SHELAH'S DIVIDING LINES IN QUANTUM MEASUREMENT: 30 VARIATIONS IN THREE PARTS

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In homage to Alexander Grothendieck, Saharon Shelah and Stevo Todorčević

Abstract. Ultrapowers give logic and operator theory a common instrument; here they lead to a common account of idealization in quantum measurement. The dividing lines which Shelah drew to classify theories become distinctions among bipartite expectation functions. Stability characterizes agreement of the two orders of idealization and uniform approximation of idealized responses by finite mixtures, with the probing system idealized first. Its uniform form controls adaptive discrimination and fixed-threshold prediction at every positive margin. NIP, the absence of the independence property, sorts physical orders, from the counting of photons to the causal order of a spacetime with one dimension of space, by the topology of their limiting responses and Todorčević's trichotomy; and simplicity, the most abstract of the three, is found at work in quantum error correction, where forking separates the errors that do no harm from those that do. We explain types and independence to the physicist and operator theorist, and formulate the physical examples for the logician. Both will find a single theorem of Grothendieck at the bottom of stability and of NIP. A reader who belongs to neither community may also read the paper, provided that ultrafilters hold no terrors, being less alarming than their name.

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The operator theorist and the logician possess a common instrument: the ultrapower. In analysis it enlarges a Banach space or an operator algebra so that bounded sequences are represented by elements of a single structure; in model theory it is an elementary extension in which those elements have types over the original structure. The two descriptions direct attention to different features of the same construction. Types and independence, on which Shelah erected his classification of theories, can therefore be studied in the operator-theoretic setting without changing the underlying instrument. The expectation of a bipartite observable supplies the connection with preparations and measurements. In this paper we develop that connection in both directions: we explain types and independence to the physicist and operator theorist, and state for the logician, formally and completely, the physics and quantum information theory used in the account.

The thesis is that the abstractions of logic, applied to quantum measurements, become instruments of physical description. If, in place of the formulas of a first-order theory, one considers bipartite observables (measurements on a pair of quantum systems, which we describe in the Aria, the first section of the paper), then the dividing lines that Shelah drew among theories become distinctions among measurements, and the theorems by which each line is characterized become assertions about what can be observed. The results on observables that we present are those of a series of papers written with Eduardo Dueñez [DI26a, DI26b, DI26c]; here we gather them into a single account.

In the Aria we state Grothendieck's double limit exchange condition and explain its physical meaning; at the end of the paper the Aria returns, da capo, and is heard against what the three parts have established. Between its two statements lie thirty variations: the first, on observables as structures, is preparatory, and the other twenty-nine form three parts of parallel design, in order that each part may be read against the others. In Part I we treat stability, in Part II the absence of the independence property (NIP), and in Part III simplicity. Each part opens with the configuration that its line forbids, the theorem that characterizes the line, a dictionary in which we render the model theory of the line in physical terms, and the Fraïssé limits in which the line is exhibited: Hilbert space for stability, the rational order for NIP, and, for simplicity, the random graph and the Pauli group, the group generated by the bit flips and the phase flips of qubits, or two-level quantum systems, taken modulo phases. Each of these structures is the Fraïssé limit of its finitely generated substructures, as the rational order is that of the finite linear orders: it is countable or, like Hilbert space, separable, and it is homogeneous, in the sense that every isomorphism between two finitely generated substructures extends to an automorphism, and these properties, together with the class of its finitely generated substructures, determine it up to isomorphism. Near its end, each part contains a variation on mixing, the operation that distinguishes quantum observables from arbitrary formulas: states can be mixed, and the expectation of a bipartite observable at a mixture of states of either system is the weighted average of its expectations at the components. Many variations close with a paragraph for the physicist or operator theorist, one for the logician, and a sentence recording what has been established. In Table 1 we set the three parts side by side, and in the box below we state the main results; each notion used in the table and in the box is defined in a later variation.

	I. Stability	**II. NIP**	**III. Simplicity**
forbidden configuration	a ladder (order property)	an independent sequence	a tree whose branches are consistent and whose siblings are k-inconsistent, for some fixed integer $k \geq 2$
characterization	the two iterated limits agree (Grothendieck); an independence relation with stationarity over models (Harnik, Harrington)	Baire class one (Rosenthal; Bourgain, Fremlin, Talagrand)	an independence relation with the independence theorem over models (Kim, Pillay)
nonforking extensions of a type over a model	unique	invariant over the model, possibly several	possibly several, amalgamated by the independence theorem
Fraïssé limit	Hilbert space	the rational order	the random graph; the Pauli group
dictionary	orders of idealization	physical orders	quantum error correction
what mixing does	uniformly approximates idealized responses, with the probing system idealized first	makes hereditary separability imply metrizability	makes simplicity imply NIP for diagonal observables

TABLE 1. The three parts side by side.

Main results.

(I) *Stability* [DI26a]. An observable is stable exactly when the two orders of idealization agree. This condition has seven equivalent forms. They include uniform approximation of idealized responses by finite mixtures, with the probing system idealized first, and a symmetric independent product on the space of stable observables. At each stable observable, the expectation of this product is the unique separately weak* continuous extension of its values on pairs of actual states. A strictly stronger, uniform condition is equivalent to polynomial growth of adaptive expectation-response transcripts at every positive margin, with the threshold fixed within each protocol (see Theorems 5.1 and 7.1).

(II) *NIP* [DI26b]. An observable is NIP exactly when its idealized responses are of the first Baire class. For NIP observables, hereditary separability of the full response compactum implies metrizability, so only three of the four classes of Todorčević's trichotomy occur, and they contain, respectively, the order of photon numbers, the order of positions, and the causal order of spacetime, the last joined in its class by the comparison of phases on a circle. Each of the seven minimal families of the Argyros–Dodos–Kanellopoulos heptachotomy is realized by a tree of preparations of photons, of a particle in a box, or of a spin chain (see Theorems 11.1, 19.1, 15.1 and 16.1).

(III) *Simplicity* [DI26c]. The Pauli group on countably many qubits is the Fraïssé limit of finite commutation patterns, and the notions of simple theories translate into quantum error correction. For diagonal observables, simplicity implies NIP: if the pure responses have the independence property, then the mixtures carry the tree property of the second kind, and the observable is not simple (see Variations 23–26 and Theorem 28.1).

In Figure 1 we place the observables of Parts I and II in the classes of stable and of NIP observables, together with the minimal families of trees that their preparations realize.

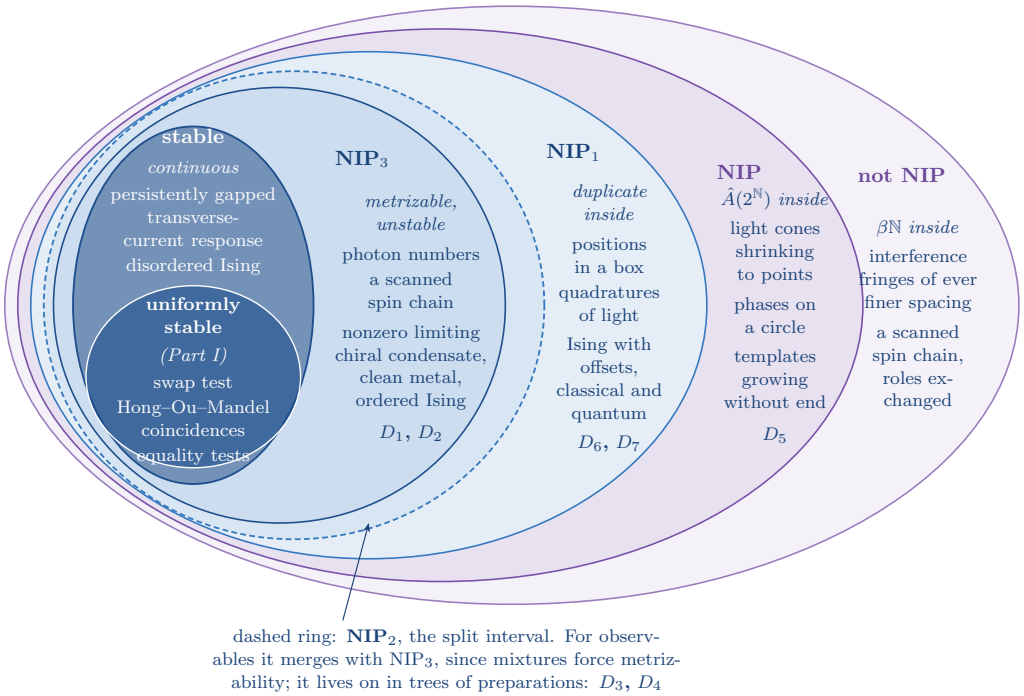


FIGURE 1. The classes of stable and NIP observables, from the innermost (continuous idealized responses) outward, with the physical observables that inhabit them and the minimal families $D_1, \ldots, D_7$ that their trees of preparations realize.

ARIA

Physics lives on idealized limits. A physicist speaks freely of systems that no laboratory has ever prepared: a particle with a sharp momentum, a particle localized at a point, an oscillator excited to infinite energy. None of them is produced by an actual preparation, but each is approached by sequences of actual preparations: wave functions concentrated ever more sharply in momentum or in position, states of ever higher energy. In the framework used here, an idealized preparation is modeled by a limiting state, obtained from a sequence of actual preparations once an ultrafilter is chosen; different sequences, or different ultrafilters, can give different limits that fit the same physical description.

To make this precise, let a quantum system be described by a Hilbert space H; its actual preparations are described by *density operators*, positive operators on H of trace 1 (see Variatio 1). The bounded operators on H form an algebra $B(H)$, and a *state* of $B(H)$ is a positive linear functional on $B(H)$ with value 1 at the identity; each density operator ρ gives the state $A \mapsto \mathrm{Tr}(\rho A)$, where Tr is the trace. Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$, and let $(\rho_n)_{n\in\mathbb{N}}$ be a sequence of density operators on H. Recall that every bounded sequence $(a_n)_{n\in\mathbb{N}}$ of reals has a unique ultralimit $\lim_{n,\mathcal{U}} a_n$, the real a such that $\{n : |a_n - a| < \varepsilon\} \in \mathcal{U}$ for every $\varepsilon > 0$; it is the ordinary limit whenever the latter exists. The *limit* of $(\rho_n)_{n\in\mathbb{N}}$ along $\mathcal{U}$ is the state $\lim_{n,\mathcal{U}} \rho_n$ of $B(H)$ that assigns to each $A \in B(H)$ the value $\lim_{n,\mathcal{U}} \mathrm{Tr}(\rho_n A)$; this state need not be given by any density operator. The energy eigenstates of an oscillator, its states of definite energy, form a sequence whose energies grow without bound (we describe the oscillator in the second paragraph below); along every nonprincipal ultrafilter on $\mathbb{N}$, the limit of this sequence is a state that is not given by any density operator and assigns probability

zero to every finite set of energy levels. For a single system, idealization commutes with taking bounded expectations: the expected value of an observable, a bounded self-adjoint operator A on H, in the state $\lim_{n,\mathcal{U}} \rho_n$ is the ultralimit along $\mathcal{U}$ of its expected values $\mathrm{Tr}(\rho_n A)$ in the actual preparations.

Difficulties begin with two systems. Quantum information theory describes its procedures, called protocols, in terms of parties, each of whom controls a system in a separate laboratory; here there are two, by custom Alice and Bob. Alice's system is described by a Hilbert space H_1 and Bob's by a Hilbert space H_2; Alice and Bob prepare their systems independently and submit the pair to a joint measurement. The two systems together are described by the tensor product $H_1 \otimes H_2$, and a *bipartite observable* is a bounded self-adjoint operator C on $H_1 \otimes H_2$, which describes a measurement performed on the pair, with an outcome that depends jointly on both systems. When Alice prepares her system in the state given by a density operator ρ on H_1 and Bob, independently, prepares his in the state given by a density operator σ on H_2, the joint state is $\rho \otimes \sigma$, and the expected outcome of C is $\mathrm{Tr}\big((\rho \otimes \sigma)C\big)$. The two roles are symmetric in principle; we speak of Alice's preparations and Bob's probes because Alice's state is the object under study and Bob's states are the tests applied to it, much as a specimen is distinguished from the instruments that examine it. A logician will recognize the division: a bipartite observable C defines a formula $\varphi(x, y)$ whose variables are partitioned, as Shelah partitions them, with Alice holding x and Bob holding y, and the number $\mathrm{Tr}\big((\rho \otimes \sigma)C\big)$ is the value of this formula at the pair (ρ, σ). In the same language, a sequence $(\rho_n)_{n\in\mathbb{N}}$ of Alice's density operators gives an element of the ultrapower along $\mathcal{U}$ of the structure that C defines (see Variatio 1), namely the class $[\rho_n]_{\mathcal{U}}$ of the sequence; the state $\lim_{n,\mathcal{U}} \rho_n$ of $B(H_1)$ determines the *φ-type* of that element, the function that records the value of φ at the pair formed by the element and each density operator of Bob. When both systems are idealized, Alice's along the sequence $(\rho_n)_{n\in\mathbb{N}}$ and Bob's along a sequence $(\sigma_m)_{m\in\mathbb{N}}$ of his probes, the expected outcome of C becomes one of two iterated limits,

$$\lim_n \lim_m \mathrm{Tr}\big((\rho_n \otimes \sigma_m)C\big) \quad \text{or} \quad \lim_m \lim_n \mathrm{Tr}\big((\rho_n \otimes \sigma_m)C\big),$$

the first with Bob's system idealized first and the second with Alice's, whenever these limits exist; nothing guarantees that the two agree.

A simple example is already decisive. An oscillator is described by the Hilbert space $L^2(\mathbb{R})$ of square-integrable functions of its position q and by its energy operator, or *Hamiltonian*, the unbounded self-adjoint operator $\frac{1}{2}(-d^2/dq^2+q^2)$, written in units in which the reduced Planck constant, the mass and the angular frequency equal 1; the Hamiltonian has an orthonormal basis of eigenvectors $e_0, e_1, \ldots$, with e_k of eigenvalue $k + \frac{1}{2}$ for $k = 0, 1, 2, \ldots$, and in the state given by e_k a measurement of energy gives $k + \frac{1}{2}$ with certainty. Only the basis and the order of the energies are used below: the measurements are built from the orthogonal projections onto the lines spanned by the eigenvectors, and these projections are bounded operators. Suppose that two identical oscillators, Alice's and Bob's, are prepared in energy eigenstates, Alice's in e_k and Bob's in f_l, where $f_0, f_1, \ldots$ is the basis of eigenvectors of the Hamiltonian of Bob's oscillator, with f_l of eigenvalue $l + \frac{1}{2}$, and that a measurement asks whether Bob's oscillator has the higher energy. The answer a_{kl} is 1 when $k < l$, and 0 when $k \geq l$. Then we have

$$\lim_l \lim_k a_{kl} = 0, \qquad \lim_k \lim_l a_{kl} = 1.$$

The matrix $(a_{kl})_{k,l\geq 0}$ has ones above the diagonal and zeros on and below it; we call this configuration the *staircase*. If Bob's oscillator is driven to infinite energy first, it outruns Alice's, and the answer is 1; if Alice's is driven there first, it outruns Bob's, and the answer

is 0. Once both oscillators are idealized, these two procedures assign different probabilities to the event "Bob's oscillator has the higher energy": the value depends on which system is idealized first.

The agreement of the two orders has a finite approximation consequence. Fix a bipartite observable C, and suppose that, for all sequences of Alice's preparations and of Bob's probes, the two iterated limits of the expectation of C agree whenever both exist. We show that every state of Alice's algebra is then simulated, to arbitrary accuracy in its responses and uniformly over all of Bob's probes, by finite mixtures of actual preparations, with Bob's system idealized first (see Theorem 5.1). The common values also define the independent product on the space of stable observables; at each such observable its expectation is the unique extension from pairs of actual states that is weak* continuous in each marginal state separately. A strictly stronger condition, uniform stability, governs finite discrimination: at every positive margin, the number of binary transcripts realized by N adaptive expectation queries is bounded by a polynomial in the positive integer N, and thresholded responses can be predicted online with a finite mistake bound, uniform in the threshold (see Theorem 7.1). The threshold is fixed within each protocol; estimating an expectation from repeated preparations is a separate statistical task. Failure of uniform stability gives all 2^N transcripts for every positive integer N at some margin. Qualitative stability alone need not prevent this exponential behavior.

The theme of this paper is the theorem that says exactly when the two orders agree, and it was proved long before anyone asked the question in these terms. Grothendieck proved it in 1952, in his work on compactness in function spaces [Gro52]: let K be a compact space, and let F be a bounded set of continuous functions on K. Then every function in the pointwise closure of F is continuous if and only if, for all sequences $(f_n)_{n\in\mathbb{N}}$ in F and $(x_m)_{m\in\mathbb{N}}$ in K, we have

$$\lim_n \lim_m f_n(x_m) = \lim_m \lim_n f_n(x_m)$$

whenever both iterated limits exist. Since F is bounded, the first condition is equivalent to the relative weak compactness of F in the Banach space $C(K)$ of continuous functions on K with the supremum norm; a topological property of the whole set is thus decided by the arrays $\big(f_n(x_m)\big)_{n,m\in\mathbb{N}}$ of its values along sequences. For K, take the set of all states of $B(H_2)$, actual or idealized, which is compact in the weak* topology, the topology of convergence at every operator; for F, take the responses of Alice's actual preparations, where the *response* of a density operator ρ on H_1 is the function $\omega \mapsto \omega(T)$ on K for the bounded operator T on H_2 by which C acts on Bob's system once Alice has prepared ρ, the operator with $\mathrm{Tr}(\sigma T) = \mathrm{Tr}\big((\rho\otimes\sigma)C\big)$ for every density operator σ on H_2. We call the functions in the pointwise closure of these responses the *idealized responses* of Alice, and the theorem says that a bipartite observable is insensitive to the order of idealization exactly when the idealized responses of Alice are continuous on K.

A logician recognizes the staircase at once: it is an instance of the order property, the configuration whose absence defines the first of the dividing lines that Shelah drew across the first-order theories; read for a formula $\varphi(x,y)$, Grothendieck's criterion becomes the theorem that φ is stable in a theory exactly when every φ-type over every model of the theory is *definable*, that is, given at each parameter b of the model by the value at b of a single formula, or of a uniform limit of formulas, with parameters from the model [BY14] (theories, models and types are defined in Variatio 1). The second line, NIP, separates the failures of the double limit condition in which idealized responses remain limits of sequences of responses of actual preparations from those in which they do not (see Theorem 11.1), and

the third, simplicity, is characterized not by limits but by the amalgamation of independent types, through the Kim–Pillay independence theorem.

In Part I we study the observables that satisfy Grothendieck's criterion, in Part II the observables that are NIP, and in Part III the amalgamation of independent types, which in simple theories takes the place of the criterion.

Variatio 1. a 1 Clav.

Observables as structures

The two iterated limits of the staircase of the Aria, the matrix of answers to the question whether Bob's oscillator has the higher energy, idealize two systems at once, and so far they have been computed only along sequences of energy eigenstates. To compare the two orders of idealization for an arbitrary observable, the idealized systems must become objects in their own right, and the value that each order assigns to the observable must depend only on these objects, not on the sequences that approach them. In this variation we define idealized states, as states of the algebra of bounded operators on a Hilbert space, and the two products that correspond to the two orders of idealization, in which Alice's system or Bob's is idealized first. We then explain, for physicists and operator theorists, how ultrapowers lead to the model-theoretic notions of elementary extension, type and independence of types, how, once an observable is fixed, an idealized state determines the restriction to a single formula of the type of an element of an ultrapower, and how the two products arise from two ways of placing two such elements in a common elementary extension.

States and observables. This subsection is addressed to logicians. A quantum system is described by a complex separable Hilbert space H. Inner products are linear in the first variable and conjugate-linear in the second, so the physicist's bracket $\langle\chi|\xi\rangle$ is $\langle\xi,\chi\rangle$; for vectors $\psi,\chi\in H$, we write $|\psi\rangle\langle\chi|$ for the operator $\xi\mapsto\langle\xi,\chi\rangle\psi$. A unit vector $\psi\in H$ describes a *pure* preparation. For a positive operator T on H, the sum $\sum_n\langle T\zeta_n,\zeta_n\rangle$ over an orthonormal basis $(\zeta_n)_{n<\dim H}$ of H does not depend on the basis; when it is finite, the operator T is of *trace class*, and the sum is its *trace* $\operatorname{Tr}T$. The trace-class operators are the linear combinations of positive operators of trace class, and the trace extends linearly to them. A general preparation is described by a *density operator*, a positive trace-class operator ρ of trace 1; the pure preparations correspond to the rank-one orthogonal projections $|\psi\rangle\langle\psi|$, and a unit vector ψ also stands for the state $|\psi\rangle\langle\psi|$. The density operators form a convex set $\mathfrak{D}(H)$, and a *mixture* $\sum_{i=1}^k t_i\rho_i$ of density operators $\rho_1,\dots,\rho_k$, with $t_i\geq 0$ and $\sum_{i=1}^k t_i=1$, describes a preparation that produces ρ_i with probability t_i. A measurement with finitely many outcomes, labeled $1,\dots,r$, is described by a family of orthogonal projections $E_1,\dots,E_r$ with sum I, the identity operator, and in the state ρ the outcome j occurs with probability $\operatorname{Tr}(\rho E_j)$, by the Born rule. If each outcome j is recorded as a real number λ_j, the measurement is represented by the operator $\sum_j\lambda_jE_j$, whose expectation $\operatorname{Tr}\bigl(\rho\sum_j\lambda_jE_j\bigr)$ in the state ρ is the expected value of the recorded number; more generally, we call every bounded self-adjoint operator A on H an *observable*: by the spectral theorem, it is a norm limit of real linear combinations of commuting orthogonal projections, and the number $\operatorname{Tr}(\rho A)$ is the expected value of its outcome in the state ρ. The expectation is affine in ρ: for all density operators ρ and ρ' and every t with $0\leq t\leq 1$, we have $\operatorname{Tr}\bigl((t\rho+(1-t)\rho')A\bigr)=t\operatorname{Tr}(\rho A)+(1-t)\operatorname{Tr}(\rho' A)$, that is, the expected outcome of a mixture is the weighted average of the expected outcomes of its components. This identity expresses in formulas the description of a mixture as a preparation that chooses its components at random.

The systems of Alice and of Bob, the two experimenters of the Aria, are described by Hilbert spaces H_1 and H_2, and together by the Hilbert tensor product $H_1 \otimes H_2$. When they are prepared independently, in the states given by density operators ρ on H_1 and σ on H_2, the joint state is $\rho \otimes \sigma$, and a bounded self-adjoint operator C on $H_1 \otimes H_2$, a *bipartite observable*, has expectation

$$\varphi_C(\rho, \sigma) = \mathrm{Tr}\big((\rho \otimes \sigma)C\big).$$

For Alice's preparation ρ, the *effective observable* $T_C(\rho)$ is the unique bounded operator on H_2 with $\mathrm{Tr}(\sigma\, T_C(\rho)) = \varphi_C(\rho, \sigma)$ for every density operator σ on H_2; it is the *partial trace* $T_C(\rho) = \mathrm{Tr}_1\big((\rho \otimes I)C\big)$ over Alice's system, the operator on H_2 with $\langle T_C(\rho)\xi, \eta\rangle = \sum_n \big\langle (\rho \otimes I)C(\zeta_n \otimes \xi), \zeta_n \otimes \eta\big\rangle$ for all $\xi, \eta \in H_2$ and any orthonormal basis $(\zeta_n)_{n<\dim H_1}$ of H_1. This identity of expectations does not specify a local implementation of the joint measurement. The function $\sigma \mapsto \varphi_C(\rho, \sigma)$ on Bob's density operators is the *response* of Alice's preparation ρ to Bob's probes; in the Aria it was extended to all states ω of $B(H_2)$, as the function $\omega \mapsto \omega(T_C(\rho))$. For the energy comparison of the Aria, the measurement that asks whether Bob's oscillator has the higher energy, let $e_0, e_1, \dots$ and $f_0, f_1, \dots$ be the energy eigenvectors of Alice's oscillator and of Bob's; the observable is the orthogonal projection P below, and its effective observable at e_k, for $k = 0, 1, 2, \dots$, is the orthogonal projection $T_P(e_k)$:

$$P = \sum_{i<j} |e_i\rangle\langle e_i| \otimes |f_j\rangle\langle f_j|, \qquad T_P(e_k) = \sum_{l>k} |f_l\rangle\langle f_l|.$$

Idealized states. An *idealized state* of a system described by a Hilbert space H is a positive linear functional ω on the algebra $B(H)$ of bounded operators on H with $\omega(I) = 1$, that is, a state of $B(H)$ in the sense of the Aria. Each density operator ρ defines the idealized state $A \mapsto \mathrm{Tr}(\rho A)$; we call such states *actual*, and physicists call them *normal*. The idealized states form a compact convex set in the weak* topology, the topology of pointwise convergence on operators, and the actual states are dense in it. In this topology the energy eigenvectors $e_0, e_1, \dots$ of the oscillator of the Aria form a sequence with no limit, and the cluster points of the sequence are idealized states that are not actual. For a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the state $\omega = \lim_{k,\mathcal{U}} e_k$, given by $\omega(A) = \lim_{k,\mathcal{U}} \langle Ae_k, e_k\rangle$, vanishes on every compact operator, that is, on every norm limit of operators of finite rank, such as the orthogonal projection onto the span of finitely many of the vectors e_k; it therefore assigns probability zero to every finite set of energy levels. Such a state represents escape beyond every finite energy cutoff, and different ultrafilters give different such states.

Two orders of idealization. To compare the two orders of idealization, let ω_1 be an idealized state of Alice's system, that is, a state of $B(H_1)$, and ω_2 an idealized state of Bob's, a state of $B(H_2)$, and let $(\rho_i)_{i\in\mathcal{I}}$ and $(\sigma_j)_{j\in\mathcal{J}}$ be nets of density operators, families indexed by directed sets $\mathcal{I}$ and $\mathcal{J}$, that converge in the weak* topology to ω_1 and to ω_2, respectively; nets are used because a sequence of density operators that converges in this topology has a density operator as its limit, and therefore no idealized state that is not actual is the limit of a sequence of density operators. The expectation of C can be idealized in two orders, which define two *products* of ω_1 and ω_2 at C:

$$(\omega_1 \otimes_{\mathsf{A}} \omega_2)(C) = \lim_j \lim_i \varphi_C(\rho_i, \sigma_j), \qquad (\omega_1 \otimes_{\mathsf{B}} \omega_2)(C) = \lim_i \lim_j \varphi_C(\rho_i, \sigma_j),$$

the first with Alice's system idealized first and the second with Bob's. Both limits exist and do not depend on the nets: for the first, the inner limit exists because, for each fixed density operator σ on H_2, the value $\varphi_C(\rho, \sigma)$ is $\mathrm{Tr}(\rho S)$ for the bounded operator $S = \mathrm{Tr}_2\big((I \otimes \sigma)C\big)$

on H_1, the partial trace over Bob's system, defined as Tr_1 above with the roles of the two factors exchanged, and the net $(\rho_i)_{i\in\mathcal{J}}$ converges to ω_1; the map $\sigma \mapsto \lim_i \varphi_C(\rho_i, \sigma)$ extends to a bounded linear functional on the trace-class operators on H_2, and it therefore has the form $\sigma \mapsto \mathrm{Tr}(\sigma Z)$ for a unique bounded operator $Z = Z_{\omega_1}$ on H_2, whence the first product is $(\omega_1 \otimes_{\mathsf{A}} \omega_2)(C) = \omega_2(Z_{\omega_1})$. The argument for the second product is symmetric. The two products are the two Arens extensions [Are51] of the bounded bilinear form $(X, Y) \mapsto \mathrm{Tr}\big((X \otimes Y)C\big)$ on pairs of trace-class operators X on H_1 and Y on H_2, which extends φ_C, that is, the two extensions of this form to the biduals of the two spaces of trace-class operators, which contain the states of $B(H_1)$ and of $B(H_2)$, obtained by taking weak* limits first in one variable and then in the other.

If one of the two states is actual, the two products agree: if ω_1 is given by a density operator ρ, then the operator Z_{ω_1} is $T_C(\rho)$, and both products equal $\omega_2(T_C(\rho))$; the case of ω_2 is symmetric. Disagreement needs two idealized systems.

For the energy comparison P, fix a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, and let $\omega_1 = \lim_{k,\mathcal{U}} e_k$ and $\omega_2 = \lim_{l,\mathcal{U}} f_l$. For each fixed l, the value $\varphi_P(e_k, f_l)$ is 0 for all $k \geq l$; for each fixed k, it is 1 for all $l > k$. Since the products do not depend on the nets, we may compute them along subnets of $(e_k)_{k\in\mathbb{N}}$ and of $(f_l)_{l\in\mathbb{N}}$ that converge to ω_1 and to ω_2, and along such subnets the indices tend to infinity; hence we have

$$(\omega_1 \otimes_{\mathsf{A}} \omega_2)(P) = 0, \qquad (\omega_1 \otimes_{\mathsf{B}} \omega_2)(P) = 1.$$

The two orders of idealization, with Alice's system and with Bob's system idealized first, assign to the event "Bob's oscillator has the higher energy" the probabilities 0 and 1, respectively.

Ultrapowers, elementary extensions and types. This subsection is addressed to physicists and operator theorists. The expectation of a bipartite observable is a bounded real-valued function of a preparation of Alice and a probe of Bob; a logic in which it is to be a formula must therefore allow formulas to take real values. Two frameworks for real-valued model theory are in use, and they differ in the place they give to the metric. In the continuous logic of Ben Yaacov, Berenstein, Henson and Usvyatsov [BYBHU08], a structure consists of complete metric spaces, its *sorts*, together with uniformly continuous real-valued functions on finite products of the sorts, its *predicates*; the metric of each sort is a distinguished predicate, and all predicates take values in a fixed bounded interval. In Keisler's real-valued logic [Kei23], which is the setting of this paper, a *vocabulary* specifies sorts and predicate symbols, among other symbols that we do not need, each predicate symbol with the sorts of its arguments and with a bounded interval that depends on the symbol; a *general structure* for the vocabulary consists of a nonempty set, its universe, for each sort and, for each predicate symbol, of a function on the product of the universes of the sorts of its arguments, the *interpretation* of the symbol, with values in the interval of the symbol. A general structure determines a vocabulary but no metric signature: no distinguished distance predicate and no moduli of uniform continuity are prescribed. A bipartite observable C determines a general structure $\mathcal{M}_C$ with two sorts, the sets $\mathfrak{D}(H_1)$ and $\mathfrak{D}(H_2)$ of density operators of Alice and of Bob, and three predicates: the expectation φ_C, with values in $[-\|C\|, \|C\|]$, and, on each sort, the trace distance $(\rho, \rho') \mapsto \|\rho - \rho'\|_1$, with values in $[0, 2]$, where $\|T\|_1 = \mathrm{Tr}\,|T|$, with $|T| = (T^*T)^{1/2}$, is the trace norm of a trace-class operator T. We write φ for the predicate symbol of the expectation and φ_C for its interpretation in $\mathcal{M}_C$. Formulas are built from the *atomic* formulas, the predicate symbols applied to variables of the appropriate sorts, by applying continuous functions $u\colon \mathbb{R}^n \to \mathbb{R}$, for every $n \geq 1$, and the quantifiers $\sup_x$ and $\inf_x$,

where x is a variable of either sort; they take real values rather than the values true and false. For instance, the value of the formula $\sup_y |\varphi(x,y) - \varphi(x',y)|$ at a pair (ρ, ρ') of Alice's density operators is the supremum of the differences that a probe of Bob can detect between the expectations of C under ρ and under ρ'. When the vocabulary is countable, Keisler's expansion theorem shows that every theory, in the sense defined below, has an expansion by a distance predicate, a pseudometric that is a uniform limit of formulas, and that this predicate is unique up to uniform equivalence; the structure $\mathcal{M}_C$ needs no such expansion, since its trace distances are already predicates.

An ultrapower of $\mathcal{M}_C$ turns sequences of density operators, from which the idealized states of the Aria were obtained as limits along an ultrafilter, into elements of a single structure, at which formulas can be evaluated. Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$. The ultrapower $\mathcal{M}_C^{\mathcal{U}}$ has as sorts the powers $\mathfrak{D}(H_1)^{\mathbb{N}}$ and $\mathfrak{D}(H_2)^{\mathbb{N}}$ modulo *Leibniz equality*, under which two sequences are identified when every atomic formula, evaluated at either of them together with any further sequences, has the same ultralimit along $\mathcal{U}$; we write $[\rho_n]_{\mathcal{U}}$ for the class of a sequence $(\rho_n)_{n\in\mathbb{N}}$. Since the trace distance is a predicate and the expectation is Lipschitz in each argument with respect to it, with constant $\|C\|$, two sequences $(\rho_n)_{n\in\mathbb{N}}$ and $(\rho'_n)_{n\in\mathbb{N}}$ are identified exactly when $\lim_{n,\mathcal{U}} \|\rho_n - \rho'_n\|_1 = 0$, and the sorts are therefore the metric ultrapowers of $\mathfrak{D}(H_1)$ and $\mathfrak{D}(H_2)$. The trace distance is also the distance that measurements detect: the largest difference between the probabilities that a single measurement assigns to an outcome under ρ and under ρ' is $\frac{1}{2}\|\rho - \rho'\|_1$, and two sequences of preparations are therefore identified in $\mathcal{M}_C^{\mathcal{U}}$ exactly when no sequence of measurements tells them apart along $\mathcal{U}$. The operator norm would identify too much: for each $n \geq 1$, let G_n and G'_n be orthogonal projections of rank n with orthogonal ranges; the operator norm of $G_n/n - G'_n/n$ is $1/n$, whereas its trace norm is 2, and the measurement given by G_n and $I - G_n$ tells the preparations G_n/n and G'_n/n apart with certainty. The operator norm would thus identify the sequences $(G_n/n)_{n\geq 1}$ and $(G'_n/n)_{n\geq 1}$, which the trace distance keeps apart. Each predicate symbol is interpreted in the ultrapower by the ultralimit of its interpretation in $\mathcal{M}_C$; for instance, the interpretation of φ at the pair $([\rho_n]_{\mathcal{U}}, [\sigma_n]_{\mathcal{U}})$, formed by the classes of a sequence of Alice's density operators and a sequence of Bob's, is $\lim_{n,\mathcal{U}} \varphi_C(\rho_n, \sigma_n)$. Each sort is thus a closed convex subset of the Banach ultrapower, in the sense of Heinrich [Hei80], of the space of trace-class operators. Operator theorists take ultrapowers on both sides of the duality between states and observables. The Banach ultrapower of $B(H)$ is taken in the operator norm. In the standard representation, $B(H)$ acts by left multiplication on the Hilbert space of Hilbert–Schmidt operators on H, and the von Neumann algebra generated by the Banach ultrapower of $B(H)$ on the ultrapower of this Hilbert space is the Groh–Raynaud ultrapower [Gro84, Ray02]; for tracial von Neumann algebras one uses instead the 2-norm of the trace, and for σ-finite ones the strong* topology of Ocneanu (see [AH14]). The actual states lie in the predual of $B(H)$, the space of trace-class operators, whose Banach ultrapower is taken in the trace norm, and Raynaud showed that this ultrapower is the predual of the Groh–Raynaud ultrapower [Ray02].

Łoś's theorem, in its real-valued form, states that, for every formula $\theta(x_1, \ldots, x_k)$ and all elements $[a_n^1]_{\mathcal{U}}, \ldots, [a_n^k]_{\mathcal{U}}$ of the ultrapower, we have

$$\theta^{\mathcal{M}_C^{\mathcal{U}}}\big([a_n^1]_{\mathcal{U}}, \ldots, [a_n^k]_{\mathcal{U}}\big) = \lim_{n,\mathcal{U}} \theta^{\mathcal{M}_C}\big(a_n^1, \ldots, a_n^k\big),$$

where $\theta^{\mathcal{N}}$ denotes the interpretation of θ in a structure $\mathcal{N}$; thus $\varphi^{\mathcal{M}_C}$ is φ_C. In particular, the diagonal map, which sends each element a of $\mathcal{M}_C$ to the class of the constant sequence $(a, a, \ldots)$, preserves the value of every formula: it is an *elementary embedding*. In general,

let $\mathcal{M}$ and $\mathcal{N}$ be structures for the same vocabulary, with each universe of $\mathcal{M}$ contained in the corresponding universe of $\mathcal{N}$ and each interpretation in $\mathcal{M}$ the restriction of the corresponding one in $\mathcal{N}$; the structure $\mathcal{N}$ is an *elementary extension* of $\mathcal{M}$ if every formula with parameters from $\mathcal{M}$ takes the same value in $\mathcal{N}$ as in $\mathcal{M}$. The *theory* of a structure is the function that assigns to each formula without free variables, that is, to each *sentence*, the value of that sentence in the structure; an elementary extension of a structure has the same theory as the structure, and a *model* of a theory is a structure whose theory it is. A formula $\varphi(x, y)$ is *stable in a theory* if no model of the theory has an infinite *ladder* for it, that is, elements $a_1, a_2, \ldots$ and $b_1, b_2, \ldots$ of the sorts of x and of y, together with reals t and $\delta > 0$, such that the value of φ at (a_i, b_j) is at most $t - \delta/2$ for $i \leq j$ and at least $t + \delta/2$ for $i > j$, or the same with the two inequalities exchanged (see Variatio 2). Ultrapowers of $\mathcal{M}_C$, and ultrapowers of these, are the most concrete elementary extensions, and they suffice for the arguments of Parts I and II. Since $\mathcal{U}$ is nonprincipal, and hence countably incomplete (some countable family of its members has empty intersection), and the vocabulary is countable, the ultrapower $\mathcal{M}_C^{\mathcal{U}}$ is $\aleph_1$-saturated: it *realizes* every type over a countable set of its elements, that is, it contains an element with that type, for the notion of type defined in the next paragraph; since the sorts of $\mathcal{M}_C$ are, moreover, separable in the trace distance, every elementary extension of $\mathcal{M}_C$ with separable sorts embeds elementarily in $\mathcal{M}_C^{\mathcal{U}}$ by an embedding that extends the diagonal map [BYBHU08]. Every state of $B(H)$ is the limit, along some ultrafilter on $\mathbb{N}$, of a sequence of density operators, because the density operators form a subset of the states that is separable in the trace norm and dense in the weak* topology. The ultrafilter must depend on the state: for a fixed $\mathcal{U}$ there are only $2^{\aleph_0}$ sequences of density operators, whereas $B(H)$ has $2^{2^{\aleph_0}}$ states when H is infinite-dimensional. Sequences thus reach every idealized state once the ultrafilter is allowed to vary. Part III needs a larger elementary extension, the monster model of Variatio 20, an elementary extension that realizes every type over every set of its elements that is small compared with the model.

Elementary extensions are where types live. Let $\mathcal{N}$ be an elementary extension of a structure $\mathcal{M}$, and let a be an element of $\mathcal{N}$. The *type* of a over $\mathcal{M}$ is the function that assigns to every formula $\theta(x, y)$ and every tuple b from $\mathcal{M}$ the value $\theta^{\mathcal{N}}(a, b)$; it records everything that formulas with parameters from $\mathcal{M}$ say about a. The type of a over a set A of elements of $\mathcal{N}$ is defined in the same way, with tuples b from A, and a *type* over A is the type over A of an element of some elementary extension of $\mathcal{N}$. For a formula $\varphi(x, y)$, the *φ-type* of a over $\mathcal{M}$ is the restriction of the type of a to φ, the function $b \mapsto \varphi^{\mathcal{N}}(a, b)$ on $\mathcal{M}$. Now let $(\rho_n)_{n\in\mathbb{N}}$ be a sequence of Alice's density operators, and let $\omega = \lim_{n,\mathcal{U}} \rho_n$ be its limit along $\mathcal{U}$, the state of $B(H_1)$ defined in the Aria. The φ-type over $\mathcal{M}_C$ of the element $a = [\rho_n]_{\mathcal{U}}$ of Alice's sort in $\mathcal{M}_C^{\mathcal{U}}$ is the function

$$\sigma \longmapsto \lim_{n,\mathcal{U}} \operatorname{Tr}\big((\rho_n \otimes \sigma)C\big)$$

on Bob's density operators; this function is $\sigma \mapsto \omega\big(\operatorname{Tr}_2((I \otimes \sigma)C)\big)$, and we call it the *response* of ω. Thus, for a fixed observable C, the state ω determines the φ-type over $\mathcal{M}_C$ of every element $[\rho_n]_{\mathcal{U}}$ with $\lim_{n,\mathcal{U}} \rho_n = \omega$; different states may determine the same φ-type, and the full type of $[\rho_n]_{\mathcal{U}}$ is not determined by ω, since that type also records the trace distances from $[\rho_n]_{\mathcal{U}}$ to the elements of $\mathcal{M}_C$. Because formulas are uniformly continuous in the trace distances, a type over $\mathcal{M}_C$ is determined by its values at the tuples from a countable dense subset of $\mathcal{M}_C$; since $\mathcal{M}_C^{\mathcal{U}}$ is $\aleph_1$-saturated, every type over $\mathcal{M}_C$ is therefore realized in $\mathcal{M}_C^{\mathcal{U}}$.

Independence enters with two systems. Let $\mathcal{V}$ be a nonprincipal ultrafilter on $\mathbb{N}$, possibly equal to $\mathcal{U}$, and let $(\rho_n)_{n\in\mathbb{N}}$ and $(\sigma_m)_{m\in\mathbb{N}}$ be sequences of Alice's and of Bob's density operators; their classes $a = [\rho_n]_{\mathcal{U}}$ and $b = [\sigma_m]_{\mathcal{V}}$ are elements of Alice's sort in $\mathcal{M}_C^{\mathcal{U}}$ and of

Bob's sort in $\mathcal{M}_C^{\mathcal{V}}$, respectively. Before φ can be evaluated at the pair (a, b), the two elements must be placed in a common elementary extension of $\mathcal{M}_C$. When $\mathcal{U} = \mathcal{V}$, placing both in $\mathcal{M}_C^{\mathcal{U}}$ yields the joint limit $\lim_{n,\mathcal{U}} \varphi_C(\rho_n, \sigma_n)$, which need not equal either iterated limit and need not make a and b independent (see Variatio 6); for the energy comparison P it is 0, the value with Alice's system idealized first, when $\rho_n = e_n$ and $\sigma_n = f_n$, and 1, the value with Bob's system idealized first, when $\rho_n = e_n$ and $\sigma_n = f_{n+1}$. An iterated ultrapower offers two other ways of placing the two elements together. If a is taken in $\mathcal{M}_C^{\mathcal{U}}$ and the type of b over $\mathcal{M}_C$ is then realized in $(\mathcal{M}_C^{\mathcal{U}})^{\mathcal{V}}$ by the class of $(\sigma_m)_{m\in\mathbb{N}}$, the formula φ takes at the pair the value $\lim_{m,\mathcal{V}} \lim_{n,\mathcal{U}} \varphi_C(\rho_n, \sigma_m)$; in the other order it takes the value $\lim_{n,\mathcal{U}} \lim_{m,\mathcal{V}} \varphi_C(\rho_n, \sigma_m)$. These two values are the two products defined above: if ω_1 and ω_2 are the limits of $(\rho_n)_{n\in\mathbb{N}}$ along $\mathcal{U}$ and of $(\sigma_m)_{m\in\mathbb{N}}$ along $\mathcal{V}$, then the first value, in which Alice's element is placed first and her system is thus idealized first, is $(\omega_1 \otimes_{\mathsf{A}} \omega_2)(C)$, and the second is $(\omega_1 \otimes_{\mathsf{B}} \omega_2)(C)$. In the first order, the type over $\mathcal{M}_C^{\mathcal{U}}$ of the class of $(\sigma_m)_{m\in\mathbb{N}}$ is the limit along $\mathcal{V}$ of the types over $\mathcal{M}_C^{\mathcal{U}}$ of the elements σ_m of $\mathcal{M}_C$: it assigns to each formula $\theta(y, c)$ with parameters c from $\mathcal{M}_C^{\mathcal{U}}$ the value $\lim_{m,\mathcal{V}} \theta^{\mathcal{M}_C^{\mathcal{U}}}(\sigma_m, c)$. In general, a type over a set A of elements of an elementary extension of a structure $\mathcal{M}$, with A containing $\mathcal{M}$, is a *coheir* of its restriction to $\mathcal{M}$ if it is approximately finitely satisfiable in $\mathcal{M}$: for all formulas $\theta_1(x, c_1), \ldots, \theta_r(x, c_r)$ with parameters from A and every $\varepsilon > 0$, some element of $\mathcal{M}$ gives each $\theta_i(x, c_i)$ a value within ε of the value that the type assigns to it; equivalently, the type is the limit, along an ultrafilter, of the types over A of elements of $\mathcal{M}$. The type over $\mathcal{M}_C^{\mathcal{U}}$ of the class of $(\sigma_m)_{m\in\mathbb{N}}$ is thus a coheir of the type of b over $\mathcal{M}_C$.

For the physicist or operator theorist. The ultrapower $\mathcal{M}_C^{\mathcal{U}}$ is built from the metric ultrapowers of $\mathfrak{D}(H_1)$ and $\mathfrak{D}(H_2)$, as in Banach space theory, and Łoś's theorem makes the diagonal map an elementary embedding of $\mathcal{M}_C$ into it: this map preserves the value of every formula, not only the values of the predicates. The φ-type over $\mathcal{M}_C$ of an element $[\rho_n]_{\mathcal{U}}$ of Alice's sort is given by the response of the state $\lim_{n,\mathcal{U}} \rho_n$, whereas the full type of the element records more, such as its trace distances to the elements of $\mathcal{M}_C$. The two orders of realization in an iterated ultrapower give the two products $\otimes_{\mathsf{A}}$ and $\otimes_{\mathsf{B}}$, and the types of the pairs realized in these two orders are the independent amalgams of the types of a and b that we study in Variatio 6.

For the logician. A preparation of a quantum system is described by a density operator ρ, and a measurement with outcomes $1, \ldots, r$ by a family of orthogonal projections $E_1, \ldots, E_r$ with sum I, the outcome j occurring with probability $\operatorname{Tr}(\rho E_j)$; the number $\varphi_C(\rho, \sigma)$ is the expected outcome of the observable C when Alice's system is prepared in ρ and Bob's, independently, in σ. An idealized state is a state of $B(H)$ that need not be given by a density operator: for a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the state $\lim_{k,\mathcal{U}} e_k$ of the oscillator vanishes on every compact operator, whence every finite set of energy levels has probability zero in it; no density operator gives such a state, and no laboratory prepares it. The two products $\otimes_{\mathsf{A}}$ and $\otimes_{\mathsf{B}}$, one for each order of idealization, are given by the two Arens extensions of the bilinear form $(X, Y) \mapsto \operatorname{Tr}\big((X \otimes Y)C\big)$ on trace-class operators, which extends φ_C; for the energy comparison they assign to the event "Bob's oscillator has the higher energy" the probabilities 0 and 1, respectively. What distinguishes observables from arbitrary formulas is that states can be mixed, and that the expectation of C at a mixture, whether of Alice's preparations or of Bob's, is the weighted average of its expectations at the components.

In Theorem 3.1 below, we show that Grothendieck's double limit condition, read for the expectation of C, is the statement that the two products agree for every pair of states. The staircase of the oscillators fails it with the largest possible gap.

Part I. Stability

Variatio 2. A 1 Clav.

The order property

Two quantum systems whose Hamiltonians have orthonormal bases of eigenvectors, such as the two oscillators of the Aria, one in Alice's laboratory and the other in Bob's, can be compared by two measurements: the first asks whether the two systems are in levels of the same energy, and the second asks whether Bob's system has the higher energy. Both measurements are classical, in the sense that they commute with each other and with every measurement of energy on either system; nevertheless, they behave in opposite ways once the systems are excited without bound. For the two oscillators, the second measurement is the energy comparison of the Aria, whose staircase is the prototype of the configuration forbidden by stability, the first of Shelah's dividing lines and the subject of Part I. In this variation we define ladders and stability, show that a bipartite observable is unstable exactly when it has an infinite ladder (see Theorem 2.1), and explain the opposite behavior of the two measurements (see Proposition 2.2).

To write the two measurements as operators, let A and B be the Hamiltonians of Alice's system and of Bob's, and, for each real λ, let Π^A_λ and Π^B_λ be the orthogonal projections onto the eigenspaces of A and of B for λ, which are zero unless λ is an eigenvalue. The two measurements are then the orthogonal projections

$$[A = B] = \sum_\lambda \Pi^A_\lambda \otimes \Pi^B_\lambda, \qquad [A < B] = \sum_{\lambda<\lambda'} \Pi^A_\lambda \otimes \Pi^B_{\lambda'},$$

in which only countably many terms are nonzero; we call them the *equality test* and the *order test*.

Ladders. The definition of a ladder copies the staircase of the Aria up to a margin: the values of the expectation of an observable need not be 0 and 1, but they lie on one side of a threshold or on the other according to the position of the entry relative to the diagonal; a fixed margin keeps the two sides apart. Let C be a bipartite observable with expectation $\varphi_C(\rho,\sigma) = \mathrm{Tr}((\rho\otimes\sigma)C)$, and let $\delta > 0$ be given. A *ladder at margin* δ consists of a real threshold t and of density operators ρ_i of Alice and σ_j of Bob, for $1 \le i, j \le N$, where N is a positive integer, or for all positive integers i and j, such that

$$\varphi_C(\rho_i,\sigma_j) \le t - \tfrac{\delta}{2} \ \text{ for } i \le j, \qquad \varphi_C(\rho_i,\sigma_j) \ge t + \tfrac{\delta}{2} \ \text{ for } i > j,$$

or the same with the two inequalities exchanged; in the first case the ladder is finite, of *length* N, and in the second it is infinite. Ladders for a formula $\varphi(x,y)$ in a structure are defined in the same way, with elements of the two sorts in place of density operators; the ladders of C are the ladders of the formula φ in the structure $\mathcal{M}_C$ of Variatio 1, whose two sorts are the sets of density operators of Alice and of Bob. A formula has the *order property* of Shelah [She71] in a theory if, at some margin, it has ladders of every finite length in some model of the theory; by the argument with ultrapowers at the end of this variation, a formula is stable in a theory, in the sense of Variatio 1, exactly when it does not have the order property in

that theory. The observable C is *stable* if it satisfies Grothendieck's double limit condition: for all sequences $(\rho_k)_{k\in\mathbb{N}}$ of Alice's density operators and $(\sigma_l)_{l\in\mathbb{N}}$ of Bob's, we have

$$\lim_l \lim_k \varphi_C(\rho_k, \sigma_l) = \lim_k \lim_l \varphi_C(\rho_k, \sigma_l)$$

whenever both iterated limits exist.

In Figure 2 we draw as a ladder the staircase of the Aria, the matrix of values of the energy comparison of two oscillators. Of the two implications in Theorem 2.1 below, the one from instability to an infinite ladder carries the content: a disagreement between two iterated limits is a statement about limits alone, and the proof must extract from it a combinatorial configuration.

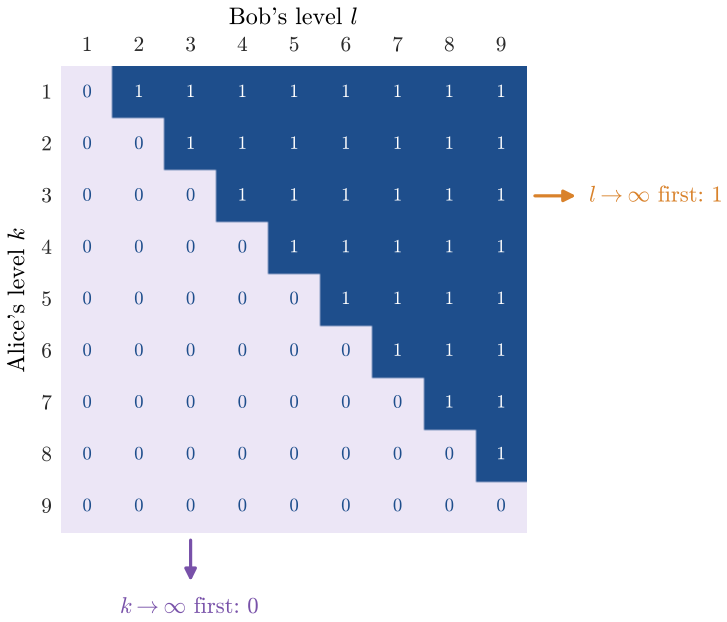


FIGURE 2. The energy comparison P of two oscillators, with energy eigenvectors $e_0, e_1, \ldots$ for Alice's and $f_0, f_1, \ldots$ for Bob's: the value $\varphi_P(e_k, f_l)$ in row k and column l is 1 when Bob's level l exceeds Alice's level k, and 0 otherwise. Every row tends to 1 as $l \to \infty$, and every column tends to 0 as $k \to \infty$; the iterated limit taken first in l is therefore 1, and the one taken first in k is 0.

Theorem 2.1. *An observable is unstable if and only if it has an infinite ladder at some margin.*

Proof. Given an infinite ladder $(\rho_i)_{i\geq 1}$, $(\sigma_j)_{j\geq 1}$ at margin δ with threshold t, consider the bounded matrix $\big(\varphi_C(\rho_i, \sigma_j)\big)_{i,j\geq 1}$. A diagonal argument yields subsequences of its rows and of its columns along which both iterated limits exist; these limits lie on opposite sides of t, at distance at least δ from each other, and the observable C is therefore unstable.

Conversely, suppose that C is unstable: there are sequences $(\rho_k)_{k\in\mathbb{N}}$ of Alice's density operators and $(\sigma_l)_{l\in\mathbb{N}}$ of Bob's for which the matrix of values $a_{kl} = \varphi_C(\rho_k, \sigma_l)$ has iterated limits $L_1 = \lim_k \lim_l a_{kl}$ and $L_2 = \lim_l \lim_k a_{kl}$ with $L_1 \neq L_2$. The idea is to choose rows and columns alternately, so that each entry of the resulting grid is close to the limit of its row or to that of its column, according to its position relative to the diagonal. Put $\alpha_k = \lim_l a_{kl}$ and $\beta_l = \lim_k a_{kl}$, whence $\alpha_k \to L_1$ and $\beta_l \to L_2$, and fix $\varepsilon > 0$ with $4\varepsilon < |L_1 - L_2|$. Having

chosen $k_1 < \cdots < k_{m-1}$ and $l_1 < \cdots < l_{m-1}$, choose $k_m > k_{m-1}$ with $|\alpha_{k_m} - L_1| < \varepsilon$ and $|a_{k_m l_p} - \beta_{l_p}| < \varepsilon$ for all $p < m$, which holds for all large k; then choose $l_m > l_{m-1}$ with $|\beta_{l_m} - L_2| < \varepsilon$ and $|a_{k_p l_m} - \alpha_{k_p}| < \varepsilon$ for all $p \leq m$, which holds for all large l. For $p \leq m$ the value $a_{k_p l_m}$ is then within 2ε of L_1, and for $p > m$ it is within 2ε of L_2: the preparations ρ_{k_p} and the probes σ_{l_m} form an infinite ladder at margin $|L_1 - L_2| - 4\varepsilon$, with threshold $(L_1 + L_2)/2$. □

Proposition 2.2. *The equality test $[A = B]$ is stable. If the spectra of A and B are both unbounded above, the order test $[A < B]$ is unstable.*

Proof. For the equality test, the expectation is $\varphi_{[A=B]}(\rho, \sigma) = \sum_\lambda \operatorname{Tr}(\rho \Pi^A_\lambda) \operatorname{Tr}(\sigma \Pi^B_\lambda) = \langle u(\rho), v(\sigma) \rangle$, where λ ranges over the eigenvalues of A, and $u(\rho) = (\operatorname{Tr}(\rho \Pi^A_\lambda))_\lambda$ and $v(\sigma) = (\operatorname{Tr}(\sigma \Pi^B_\lambda))_\lambda$ are vectors of norm at most 1 in the space ℓ^2 of square-summable families indexed by these eigenvalues. Let $(\rho_k)_{k \in \mathbb{N}}$ and $(\sigma_l)_{l \in \mathbb{N}}$ be sequences of Alice's and of Bob's density operators along which both iterated limits of $\varphi_{[A=B]}(\rho_k, \sigma_l)$ exist. Bounded sequences in a Hilbert space have weakly convergent subsequences, and passing to subsequences does not change iterated limits that exist; we may therefore assume that $u(\rho_k) \to \xi$ and $v(\sigma_l) \to \xi'$ weakly, and then both iterated limits equal $\langle \xi, \xi' \rangle$. For the order test, the spectra of A and B, being unbounded above, allow us to choose eigenvalues $\lambda_1 < \lambda'_1 < \lambda_2 < \lambda'_2 < \cdots$, with λ_k an eigenvalue of A and λ'_l one of B, and unit eigenvectors ψ_k of A for λ_k and χ_l of B for λ'_l, for all $k, l \geq 1$. By this choice, the inequality $\lambda_k < \lambda'_l$ holds exactly when $k \leq l$, and the value $\varphi_{[A<B]}(\psi_k, \chi_l)$ is therefore 1 for $k \leq l$ and 0 for $k > l$; the vectors ψ_k and χ_l thus form an infinite ladder at margin 1 with threshold $\frac{1}{2}$, with the two inequalities of the definition exchanged, and the order test is unstable, by Theorem 2.1. □

The instability of $[A < B]$ is the classical failure of the double limit condition for the order of $\mathbb{N}$, carried into a Hilbert space by two orthonormal sequences; it owes nothing to noncommutativity. The equality test is stable for the opposite reason: it is an inner product, and the inner product has no infinite ladders.

For the physicist or operator theorist. The existence of ladders of every finite length at some margin is Shelah's order property [She71], in the Krivine–Maurey real-valued form [KM81], which Iovino carried into real-valued model theory [Iov99a]. Recall from Variatio 1 the structure $\mathcal{M}_C$, whose two sorts are the sets of density operators of Alice and of Bob and whose predicates are the expectation φ_C and the trace distances, and fix a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$. Ultrapowers turn finite ladders into infinite ones: if, for every $n \geq 1$, there is a ladder $(\rho_{n,i}, \sigma_{n,j})_{1 \leq i,j \leq n}$ of C at margin δ with threshold t_n, then, by Łoś's theorem, which computes the value of a formula in the ultrapower $\mathcal{M}_C^{\mathcal{U}}$ as the ultralimit along $\mathcal{U}$ of its values in $\mathcal{M}_C$, the elements $a_i = [\rho_{n,i}]_{\mathcal{U}}$ and $b_j = [\sigma_{n,j}]_{\mathcal{U}}$, with arbitrary density operators $\rho_{n,i}$ for $n < i$ and $\sigma_{n,j}$ for $n < j$, form an infinite ladder at margin δ with threshold $\lim_{n,\mathcal{U}} t_n$ in $\mathcal{M}_C^{\mathcal{U}}$ (one of the two orientations occurs for $\mathcal{U}$-almost every n, and the ladder has that orientation). Hence the formula φ is unstable in the theory of $\mathcal{M}_C$, in the sense of Variatio 1, when the ladders of C have unbounded length at some margin. Conversely, an infinite ladder at margin δ in a model of that theory yields ladders of C of every finite length at every margin smaller than δ, since the model has the theory of $\mathcal{M}_C$. The formula φ is therefore stable in the theory of $\mathcal{M}_C$ exactly when C is *uniformly stable*, that is, when, at every margin, the ladders of C have bounded length (see Variatio 7). Stability of C is weaker: by Theorem 2.1, the double limit condition that defines it forbids infinite ladders in $\mathcal{M}_C$ only.

For the logician. The Hamiltonian of a system with quantized energy is the self-adjoint operator whose spectral projection Π_λ, for each energy λ, is the orthogonal projection onto the space of vectors of definite energy λ; in the state given by such a unit vector, a measurement of energy returns λ with certainty. The tests $[A = B]$ and $[A < B]$ are yes-no measurements on the pair of systems, and, for each of them, the value $\varphi_C(\rho, \sigma)$, with C the test, is the probability of the answer yes when Alice's system is prepared in ρ and Bob's in σ. For the equality test this probability is an inner product of two vectors of ℓ^2, namely $\sum_\lambda p_\lambda q_\lambda$, where p_λ and q_λ are the probabilities that Alice's system and Bob's system, respectively, have energy λ; for the order test, eigenstates of increasing energy, taken alternately from Alice's system and from Bob's, produce answers that are certain: yes when Alice's eigenstate comes before Bob's in the alternation, and no otherwise. Both tests commute with every measurement of energy on either system.

A ladder is the combinatorial configuration behind a failure of the double limit condition: two disagreeing iterated limits yield an infinite ladder, and Grothendieck's criterion is exactly the absence of infinite ladders.

Variatio 3. a 1 Clav. Canone all'Unisuono

Formulas and observables

The ladders of a bipartite observable are, by definition, the ladders of a formula in the structure that the observable defines, and the correspondence does not stop at configurations: types, elementary extensions and definitions of types have physical counterparts as well. In this variation we set out this dictionary of stability (see Table 2). Each theorem is stated once and read twice, by the logician and by the physicist, and the same words carry a different meaning in each reading.

The logician's reading. Let $\varphi(x, y)$ be a formula, let $\mathcal{M}$ be a structure, and let p be a type in x over $\mathcal{M}$; it is the limit, along an ultrafilter $\mathcal{W}$ on a set I, of the types of elements a_i of $\mathcal{M}$, for $i \in I$, and we write $\varphi(p, b) = \lim_{i,\mathcal{W}} \varphi^{\mathcal{M}}(a_i, b)$ for the value that it assigns to the formula $\varphi(x, b)$. The φ-type of p is *definable* if the function $b \mapsto \varphi(p, b)$ on $\mathcal{M}$ extends to a continuous function on the space $S_y(\mathcal{M})$ of types in y over $\mathcal{M}$, which is then called its *definition*. If the formula has no infinite ladder in $\mathcal{M}$, then every φ-type over $\mathcal{M}$ is definable, and the formula is stable in the theory of $\mathcal{M}$ exactly when every φ-type over every model of the theory is definable [BY14]. In the notation of Variatio 1, let $(\rho_n)_{n\in\mathbb{N}}$ be a sequence of Alice's density operators, let $a = [\rho_n]_{\mathcal{U}}$ be its class in the ultrapower $\mathcal{M}_C^{\mathcal{U}}$ of the structure $\mathcal{M}_C$, and write $\mathrm{tp}(a/\mathcal{M}_C)$ for the type of a over $\mathcal{M}_C$. Let b be an element of Bob's sort in an elementary extension $\mathcal{N}$ of $\mathcal{M}_C$, and let a' be the class of $(\rho_n)_{n\in\mathbb{N}}$ in $\mathcal{N}^{\mathcal{U}}$; its type over $\mathcal{N}$ is the coheir of $\mathrm{tp}(a/\mathcal{M}_C)$ given by the limit along $\mathcal{U}$ of the types of the elements ρ_n. The value $\varphi(a', b)$, which is the value with Bob's system idealized first when b is the class of a sequence of Bob's density operators, depends only on the type of b over $\mathcal{M}_C$; this function of the type is continuous exactly when the φ-type of a is definable and its definition, evaluated at the type of b, gives $\varphi(a', b)$ for every such b, that is, when the coheir follows the definition. Definability alone is not enough: for the energy comparison, as we show in the paragraph "Both readings" below, the φ-type of the class of $(e_k)_{k\in\mathbb{N}}$ is definable, but its coheir does not follow its definition.

The physicist's reading. Let C be a bipartite observable on the tensor product $H_1 \otimes H_2$ of the Hilbert spaces of Alice's and Bob's systems, and let ω_1 be an idealized state of Alice, the weak* limit of a net $(\rho_i)_{i\in\mathcal{I}}$ of her density operators. Its response, the function $\sigma \mapsto \lim_i \varphi_C(\rho_i, \sigma)$ on Bob's density operators, is given by an operator: as we saw in Variatio 1,

The logician	**The physicist**
a structure with two sorts	the preparation spaces of Alice's and Bob's systems
an element of a sort	an actual preparation, a density operator
a formula $\varphi(x, y)$	the expectation $\varphi_C(\rho, \sigma)$ of an observable
an elementary extension, such as an ultrapower	an enlargement containing families of preparations
the φ-type of an element of an ultrapower	the response of an idealized state, the ultrafilter limit of preparations
the φ-type of an element a of the structure, a function of the elements b of the other sort	the response of a preparation to the probes
a definable φ-type whose coheirs follow its definition	a Bob-first idealized response represented on all probes by one observable
realizing one coheir before the other	idealizing one system before the other
an independent amalgam, read at φ	a candidate product value at the observable C
the order property in the theory	ladders of every finite length at some fixed positive margin
a formula stable in the given model	a stable observable: its two orders of idealization agree
a formula stable in the theory of the model	a uniformly stable observable: bounded ladder lengths at every positive margin

TABLE 2. The dictionary of stability, read for the expectation predicate φ in the preparation structure $\mathcal{M}_C$. An idealized state determines a φ-type, not generally a full type or a unique element of an elementary extension. Stability in the given model and stability in its theory occupy different rows; the latter is uniform stability (see Variatio 7). Definability alone does not ensure representation on idealized probes: the coheirs must follow the definition, as explained below.

there is a unique bounded operator Z_{ω_1} on H_2 with $\lim_i \varphi_C(\rho_i, \sigma) = \mathrm{Tr}(\sigma Z_{\omega_1})$ for every density operator σ on H_2. The response is *represented by an observable* if the same operator also gives the value at idealized probes, in the order with Bob's system idealized first: we have $(\omega_1 \otimes_{\mathsf{B}} \omega_2)(C) = \omega_2(Z_{\omega_1})$ for every state ω_2 of $B(H_2)$, where $\otimes_{\mathsf{B}}$ is the product of Variatio 1 with Bob's system idealized first. In Theorem 3.1 below, we show that the observable C is stable exactly when every idealized response of C is represented by an observable.

Theorem 3.1 (Grothendieck [Gro52]**;** [DI26a, Theorem 5.1]**).** *For a bipartite observable C on $H_1 \otimes H_2$, the following are equivalent:*

(1) *the observable C is stable;*

(2) *the two orders of idealization agree: for every state ω_1 of $B(H_1)$ and every state ω_2 of $B(H_2)$, we have $(\omega_1 \otimes_{\mathsf{A}} \omega_2)(C) = (\omega_1 \otimes_{\mathsf{B}} \omega_2)(C)$, where $\otimes_{\mathsf{A}}$ and $\otimes_{\mathsf{B}}$ are the products of Variatio 1, with Alice's and with Bob's system idealized first, respectively;*
(3) *every idealized response of C is represented by an observable.*

Proof. One argument serves both readings. As we saw in Variatio 1, the product with Alice's system idealized first is always $(\omega_1 \otimes_{\mathsf{A}} \omega_2)(C) = \omega_2(Z_{\omega_1})$; conditions (2) and (3) therefore say the same thing. If (2) fails at a pair of states ω_1 and ω_2, take nets $(\rho_i)_{i\in\mathcal{I}}$ and $(\sigma_j)_{j\in\mathcal{J}}$ of density operators converging to ω_1 and to ω_2, respectively; the iterated limits $L_1 = \lim_i \lim_j \varphi_C(\rho_i, \sigma_j)$ and $L_2 = \lim_j \lim_i \varphi_C(\rho_i, \sigma_j)$ are the two products at C, and they differ. Arguing as in the proof of Theorem 2.1, with "for all large k" and "for all large l" replaced by "eventually along the net", we choose indices alternately: each step imposes finitely many conditions, all of which hold eventually along a net, and the choice yields sequences $(\rho_{i_p})_{p\geq 1}$ and $(\sigma_{j_m})_{m\geq 1}$ that form an infinite ladder; by Theorem 2.1, the observable C is unstable. Conversely, if C is unstable, there are sequences $(\rho_k)_{k\in\mathbb{N}}$ of Alice's density operators and $(\sigma_l)_{l\in\mathbb{N}}$ of Bob's whose iterated limits $L_1 = \lim_k \lim_l \varphi_C(\rho_k, \sigma_l)$ and $L_2 = \lim_l \lim_k \varphi_C(\rho_k, \sigma_l)$ exist and differ; their limits ω_1 and ω_2 along a nonprincipal ultrafilter on $\mathbb{N}$ are limits of subnets, along which ordinary limits persist, and therefore $(\omega_1 \otimes_{\mathsf{B}} \omega_2)(C) = L_1$ and $(\omega_1 \otimes_{\mathsf{A}} \omega_2)(C) = L_2$. □

The logician's reading. Whenever the φ-type over $\mathcal{M}_C$ determined by ω_1 is definable, its definition is given by the operator Z_{ω_1}: for a nonprincipal ultrafilter $\mathcal{V}$ on $\mathbb{N}$ and a sequence $(\sigma_m)_{m\in\mathbb{N}}$ of Bob's density operators, the definition sends the type over $\mathcal{M}_C$ of the element $[\sigma_m]_{\mathcal{V}}$ of $\mathcal{M}_C^{\mathcal{V}}$ to $\omega_2(Z_{\omega_1})$, where $\omega_2 = \lim_{m,\mathcal{V}} \sigma_m$. Ben Yaacov observed that Grothendieck's theorem (Theorem 3.1), read for formulas, yields the definability of types in stable theories [BY14].

The physicist's reading. For the energy comparison P of Variatio 1, the projection that asks whether Bob's oscillator has the higher energy, let $e_0, e_1, \ldots$ and $f_0, f_1, \ldots$ be the energy eigenvectors of Alice's and of Bob's oscillator, let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$, and take $\omega_1 = \lim_{k,\mathcal{U}} e_k$. The effective observables $T_P(e_k) = \sum_{l>k} |f_l\rangle\langle f_l|$, by which P acts on Bob's system once Alice has prepared e_k, decrease to 0 strongly, that is, the vector $T_P(e_k)\xi$ tends to 0 for every $\xi \in H_2$, although each of these operators has norm 1; hence the value $\mathrm{Tr}(\sigma\, T_P(e_k))$ tends to 0 for every density operator σ, and $Z_{\omega_1} = 0$: on every actual probe, the response of ω_1 takes the value 0. On the idealized probe $\omega_2 = \lim_{l,\mathcal{U}} f_l$, with Bob's system idealized first, it takes the value 1, while $\omega_2(Z_{\omega_1}) = 0$. No operator reproduces this response.

Both readings. By Theorem 3.1, Grothendieck's condition, that the iterated limits agree, holds exactly when every idealized response is represented by an observable; in the logician's reading, it holds exactly when, for every nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$ and every element a of Alice's sort in $\mathcal{M}_C^{\mathcal{U}}$, the φ-type of a is definable and the coheir of the type of a follows its definition. For the energy comparison P, the φ-type over $\mathcal{M}_P$ of the class of $(e_k)_{k\in\mathbb{N}}$, which describes Alice's oscillator excited to infinite energy, is definable, with the constant 0 as its definition, but its coheir takes the value 1 at the class of Bob's sequence $(f_l)_{l\in\mathbb{N}}$; accordingly, the response of $\lim_{k,\mathcal{U}} e_k$ is not represented by an observable.

Variatio 4. A 1 Clav.

Hilbert space

Two experiments, one with an auxiliary qubit, a two-level quantum system, and one with a beam splitter, compare two unknown preparations of the same system, and both measure their overlap, the Hilbert–Schmidt inner product of the two density operators. In this variation we

show that every observable whose expectation is an inner product is stable, and we recall that the separable infinite-dimensional Hilbert space is the Fraïssé limit of the finite-dimensional Hilbert spaces.

To describe the two experiments, recall that a *qubit* is a quantum system described by the Hilbert space $\mathbb{C}^2$; physicists write $|0\rangle$ and $|1\rangle$ for its standard orthonormal basis, and $|\psi\rangle$ for a vector ψ. In the *swap test*, two *registers*, quantum systems described by the same Hilbert space K, are prepared in the states given by density operators ρ and σ on K, respectively, and are coupled to an auxiliary qubit, the *ancilla*, prepared in $(|0\rangle + |1\rangle)/\sqrt{2}$. A unitary operator, that is, an invertible linear isometry, then acts on the registers and the ancilla together: it exchanges the two registers on the subspace in which the ancilla is in $|1\rangle$, and it acts as the identity on the subspace in which the ancilla is in $|0\rangle$; finally, the ancilla is measured in the basis $(|0\rangle + |1\rangle)/\sqrt{2}$, $(|0\rangle - |1\rangle)/\sqrt{2}$, and the outcome $(|0\rangle + |1\rangle)/\sqrt{2}$ counts as acceptance. The test accepts with probability $\frac{1}{2}(1 + \mathrm{Tr}(\rho\sigma))$ [BCWdW01]. In optics the quantity $\mathrm{Tr}(\rho\sigma)$ is measured without an ancilla. A photon is described by $\mathbb{C}^2 \otimes K$, where the factor $\mathbb{C}^2$ records which of two paths, or *ports*, it occupies and K now describes its internal degrees of freedom, such as polarization or spatial mode; an ideal lossless balanced beam splitter acts on the port factor, leaving the internal factor unchanged, by the unitary operator $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, and the joint state of two photons lies in the symmetric subspace of $(\mathbb{C}^2 \otimes K)^{\otimes 2}$, because photons are identical bosons, particles whose joint states are unchanged when two of them are exchanged. Two independently prepared single photons whose internal states are given by ρ and σ enter such a beam splitter, one in each input port. The internal states include all degrees of freedom relevant to distinguishability. With ideal output counting that does not resolve these degrees of freedom, they leave by different ports, an event called a *coincidence*, with probability $\frac{1}{2}(1 - \mathrm{Tr}(\rho\sigma))$, which vanishes for equal pure states. This is the Hong–Ou–Mandel effect [HOM87]. Both experiments thus measure the overlap $\mathrm{Tr}(\rho\sigma)$, the Hilbert–Schmidt inner product of ρ and σ.

Lemma 4.1 (The swap trick). *Let K be a Hilbert space, and let F be the unitary operator on $K \otimes K$ with $F(\psi \otimes \chi) = \chi \otimes \psi$ for all $\psi, \chi \in K$. Then, for all density operators ρ and σ on K, we have $\mathrm{Tr}((\rho \otimes \sigma)F) = \mathrm{Tr}(\rho\sigma)$, the Hilbert–Schmidt inner product of ρ and σ. The probability of acceptance in the swap test and the probability of a coincidence in the Hong–Ou–Mandel experiment are, respectively, the expectations at (ρ, σ) of the orthogonal projections $\frac{1}{2}(I + F)$ and $\frac{1}{2}(I - F)$.*

Proof. For $\rho = |\psi\rangle\langle\psi'|$ and $\sigma = |\chi\rangle\langle\chi'|$, the operator $(\rho \otimes \sigma)F$ is $|\psi \otimes \chi\rangle\langle\chi' \otimes \psi'|$, whose trace is $\langle\psi, \chi'\rangle\langle\chi, \psi'\rangle = \mathrm{Tr}(\rho\sigma)$; the general case follows by linearity and continuity. □

Theorem 4.2. *Let C be a bipartite observable. If there are bounded maps u and v, from the sets $\mathfrak{D}(H_1)$ and $\mathfrak{D}(H_2)$ of density operators of Alice and of Bob into a Hilbert space, with $\varphi_C(\rho, \sigma) = \langle u(\rho), v(\sigma)\rangle$ for all $\rho \in \mathfrak{D}(H_1)$ and $\sigma \in \mathfrak{D}(H_2)$, then the observable C is stable. In particular, the swap F and the two projections $\frac{1}{2}(I \pm F)$ of Lemma 4.1 are stable.*

Proof. Let $(\rho_k)_{k\in\mathbb{N}}$ and $(\sigma_l)_{l\in\mathbb{N}}$ be sequences of Alice's and of Bob's density operators along which both iterated limits of $\varphi_C(\rho_k, \sigma_l)$ exist. Arguing as in the proof of Proposition 2.2, we may assume that $u(\rho_k) \to \xi$ and $v(\sigma_l) \to \xi'$ weakly. For each k, the inner limit over l is then $\langle u(\rho_k), \xi'\rangle$, which tends to $\langle\xi, \xi'\rangle$; by symmetry, the other order yields the same value. □

A natural guess is that the swap owes its stability to approximation by *local* observables, the finite sums of products of an operator on the first factor with an operator on the second,

which the two laboratories can measure separately and combine afterwards. If K is infinite-dimensional, the guess fails, for the swap F is not a norm limit of local observables: if a bipartite observable C is such a limit, its effective map $\rho \mapsto T_C(\rho)$, which assigns to each preparation of Alice the observable that C induces on Bob's system, is compact, that is, it maps bounded sets to sets with compact closure, whereas for the swap we have $T_F(\rho) = \rho$, and this map sends the pure states of an orthonormal sequence to operators at mutual distance 1, a sequence with no convergent subsequence. The stability of the swap comes instead from Theorem 4.2, since its expectation is an inner product.

The Fraïssé limit. By Fraïssé's theorem, a class of finitely generated structures with countably many isomorphism types that is closed under substructures and has the *joint embedding property*, any two members embed in a third, and the *amalgamation property*, any two embeddings of one member into two others can be followed by embeddings of these two into a fourth member so that the two composites agree, is, up to isomorphism, the class of finitely generated substructures of a countable *homogeneous* structure, one in which every isomorphism between finitely generated substructures extends to an automorphism; this structure, the *limit* of the class, is unique up to isomorphism [Fra54]. Ben Yaacov extended the theory to metric structures [BY15]. The finite-dimensional Hilbert spaces form a class of this kind in the metric sense: the class is closed under subspaces, and two isometric embeddings of a space E into spaces E_1 and E_2 amalgamate in the space $E_1 \oplus_E E_2$, in which the orthogonal complements of E in E_1 and in E_2 are orthogonal to each other; joint embedding is the case $E = \{0\}$. The limit is the separable infinite-dimensional Hilbert space, and its homogeneity is the statement that every isometry between finite-dimensional subspaces extends to a unitary operator on the whole space.

For the physicist or operator theorist. Let H be the separable infinite-dimensional Hilbert space, and let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$. The ultrapower $H^{\mathcal{U}}$ is again a Hilbert space and, by Łoś's theorem, which computes the values of formulas in the ultrapower as ultralimits of their values in H, an elementary extension of H. For the formula $\mathrm{Re}\langle x, y\rangle$ and a bounded sequence $(\xi_n)_{n\in\mathbb{N}}$ in H, the type of the class $[\xi_n]_{\mathcal{U}}$ over H, restricted to this formula, is the function $y \mapsto \mathrm{Re}\langle \xi, y\rangle$, where ξ is the weak limit of $(\xi_n)_{n\in\mathbb{N}}$ along $\mathcal{U}$: this function is given by the element ξ of H, and the type is therefore definable (see Variatio 3). The expectation of an overlap measurement in an ultrapower is again an inner product, of vectors of an ultrapower of a Hilbert space, and the argument of Theorem 4.2 shows that it has no infinite ladder; by Łoś's theorem, the ladders of overlap measurements therefore have bounded length at every margin, and in Variatio 8 we give an explicit bound. Over a closed subspace E of an elementary extension of H, vectors a and b of that extension are independent in the sense of nonforking (see Variatio 6) exactly when $a - P_E a$ and $b - P_E b$ are orthogonal, where P_E is the orthogonal projection onto E [BYBHU08]: in the amalgam $E_1 \oplus_E E_2$, the spaces E_1 and E_2 are independent over E.

For the logician. The swap test and the Hong–Ou–Mandel experiment both perform the two-outcome measurement given by the orthogonal projections $\frac{1}{2}(I + F)$ and $\frac{1}{2}(I - F)$ of Lemma 4.1. For pure states ψ and χ, the overlap $\mathrm{Tr}(\rho\sigma)$ of $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\chi\rangle\langle\chi|$ is $|\langle\psi, \chi\rangle|^2$, the probability that a system prepared in ψ passes the test for χ, the measurement given by the projections $|\chi\rangle\langle\chi|$ and $I - |\chi\rangle\langle\chi|$. Since the two experiments measure the pair and never either system alone, they compare two unknown states without determining either of them.

For an inner product, the weak limits of Alice's and of Bob's vectors are again vectors of the Hilbert space, and both iterated limits equal their inner product. Idealization never leaves the Hilbert space, and this is why Hilbert space is the Fraïssé limit associated with stability.

Variatio 5. a 1 ô vero 2 Clav.

Stability in seven languages

The double limit condition has been found, under different names, in analysis, Banach space theory, the theory of Banach algebras, probability and model theory. For a bipartite observable, each of these forms becomes a statement about idealized preparations. The theorem below collects seven of them, and a single example is then read in all seven. Two of these forms, the agreement of the two orders of idealization and the representation of idealized responses by observables, are already in Theorem 3.1 of Variatio 3; three others describe operations of a laboratory, and each of them is developed in a later variation: the independent product in Variatio 6, the simulation of idealized preparations by finite mixtures in Variatio 9, and randomization in Variatio 10. Between the first two of these variations, in Variations 7 and 8, we treat the uniform version of the condition, which the ultrapower turns into a statement about every elementary extension.

Throughout this variation, the letter C denotes a bipartite observable on the tensor product $H_1 \otimes H_2$ of Alice's and Bob's Hilbert spaces, with expectation $\varphi_C(\rho, \sigma) = \mathrm{Tr}((\rho \otimes \sigma)C)$ and with effective observable $T_C(\rho)$ for each density operator ρ of Alice, the observable that C induces on Bob's system once Alice has prepared ρ; the symbols $\otimes_{\mathsf{A}}$ and $\otimes_{\mathsf{B}}$ denote the products of Variatio 1, given by the two orders of idealization, with Alice's system and with Bob's system idealized first, respectively.

Theorem 5.1 ([DI26a, Theorem A])**.** *For every bipartite observable C on $H_1 \otimes H_2$, the following are equivalent:*

(A1) *the observable C is stable: for all sequences $(\rho_k)_{k\in\mathbb{N}}$ of Alice's density operators and $(\sigma_l)_{l\in\mathbb{N}}$ of Bob's, the two iterated limits of $\varphi_C(\rho_k, \sigma_l)$ agree whenever both exist;*

(A2) (Grothendieck) *the effective observables $T_C(\rho)$, for ρ in the set $\mathfrak{D}(H_1)$ of Alice's density operators, form a relatively weakly compact subset of the Banach space $B(H_2)$, for the weak topology of that Banach space, which is finer than the weak operator topology, and every idealized response is therefore represented by an observable;*

(A3) (Iovino; Davis, Figiel, Johnson, Pełczyński) *the expectation factors through the pairing of a reflexive Banach space X with its dual X^*: there are bounded maps u from $\mathfrak{D}(H_1)$ into X and v from $\mathfrak{D}(H_2)$ into X^* with $\varphi_C(\rho, \sigma) = v(\sigma)(u(\rho))$ for all ρ and σ;*

(A4) (Arens) *the two orders of idealization agree on all pairs of idealized states: the two products satisfy $(\omega_1 \otimes_{\mathsf{A}} \omega_2)(C) = (\omega_1 \otimes_{\mathsf{B}} \omega_2)(C)$ for every state ω_1 of $B(H_1)$ and every state ω_2 of $B(H_2)$;*

(A5) (Mazur) *every idealized preparation of Alice is simulated, to any accuracy and uniformly over all of Bob's probes, by finite mixtures of actual preparations, with Bob's system idealized first;*

(A6) (Pták, Simons) *for any two finitely additive probability measures, one on $\mathfrak{D}(H_1)$ and one on $\mathfrak{D}(H_2)$, the two iterated integrals of the expectation against them agree;*

(A7) (Harrington) *for every idealized preparation of Alice and every idealized probe of Bob, the two orders of realizing the pair give the same value, the value at C of the unique, symmetric independent product of the two states (see Theorem 6.2).*

The equivalence of the seven conditions. The argument for the equivalence is organized around (A4), the agreement of the two products: each of the other conditions is joined to it, or to a condition already joined to it; most of the links are classical theorems of the fields from which the conditions come. The equivalence of (A1) and (A4) is Theorem 3.1, whose proof extracts a ladder from two disagreeing iterated limits. For (A2), the operator Z_{ω_1} that represents, on actual probes, the response of an idealized state ω_1 of Alice is a weak* cluster point of the effective observables; under (A4) it is a weak cluster point, and the Eberlein–Šmulian theorem, by which a subset of a Banach space is relatively weakly compact as soon as every sequence in it has a weak cluster point, yields relative weak compactness [Ebe47]. Condition (A3) follows from (A2): the map $\rho \mapsto T_C(\rho)$, extended linearly to the trace-class operators on H_1, is then a weakly compact operator into $B(H_2)$, and, by the Davis–Figiel–Johnson–Pełczyński theorem, it factors through a reflexive space [DFJP74]; the factorization, by bounded maps, of every bounded function with the double limit property through such a pairing is due to Iovino [Iov99b], and, conversely, reflexivity yields (A1) from (A3), because bounded subsets of a reflexive space are relatively weakly compact. Condition (A5) rests on Mazur's theorem [Maz33]: a convex set has the same weak and norm closures, and a weak approximation of the operator Z_{ω_1} by effective observables of mixtures thus becomes a norm approximation, uniform over probes. In (A6), an iterated integral of a function that is affine in each variable depends only on the barycenters of the two measures, which are states, and the two iterated integrals are therefore the two products of (A4); the equivalence of the double limit condition with the agreement of iterated integrals, for arbitrary bounded functions, is the Pták–Simons theorem [Ptá63, Sim68]. Finally, for (A7), the two orders of

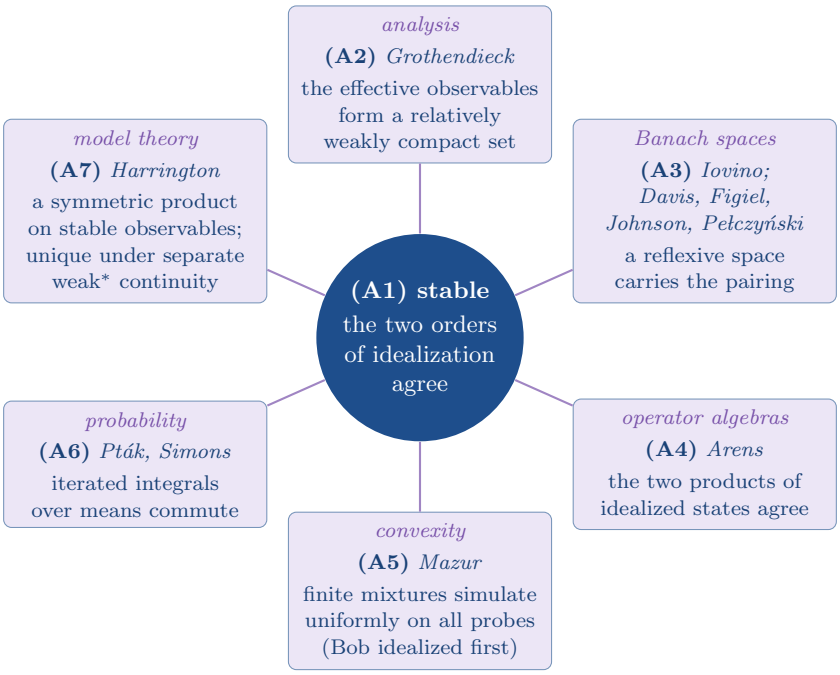


FIGURE 3. Seven readings of one condition. Each form of stability in Theorem 5.1 comes from a different field and says something different about idealized preparations.

realizing a pair of types are the two iterated ultrafilter limits, and their agreement is (A4) in the language of types; for a formula stable in the theory, it is Harrington's lemma [HH84].

One example in seven languages. Let P be the energy comparison of two oscillators, with $\varphi_P(e_k, f_l) = 1$ for $k < l$ and 0 otherwise, where $e_0, e_1, \dots$ and $f_0, f_1, \dots$ are the energy eigenvectors of Alice's oscillator and of Bob's, and let $\omega_1 = \lim_{k,\mathcal{U}} e_k$ and $\omega_2 = \lim_{l,\mathcal{U}} f_l$ for a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$.

(A1) The iterated limit taken first over Alice's index is 0, and the one taken first over Bob's is 1.

(A2) The effective observables $T_P(e_k) = \sum_{l>k} |f_l\rangle\langle f_l|$ decrease to 0 in the strong operator topology, but the state ω_2 assigns every one of them the value 1 and their strong limit the value 0; no subsequence converges weakly in $B(H_2)$.

(A3) The comparison does not factor, by bounded maps, through the pairing of a reflexive space with its dual: for the vectors $u(e_k)$ and $v(f_l)$ of such a factorization, weak limits of subsequences would force the two iterated limits to agree.

(A4) The two products of ω_1 and ω_2 at P are 0 and 1.

(A5) No mixture ρ of Alice's preparations simulates ω_1 within any error less than 1: the response of ω_1 is 0 on every f_l, while $\mathrm{Tr}(f_l\, T_P(\rho)) = \sum_{k<l}\langle e_k, \rho e_k\rangle$ tends to 1 as l grows.

(A6) Let μ be the finitely additive probability measure that assigns to a set of Alice's density operators the value 1 if the set of indices k with e_k in it belongs to $\mathcal{U}$, and the value 0 otherwise, and let ν be defined in the same way from $(f_l)_{l\in\mathbb{N}}$; one iterated integral against μ and ν is 0, and the other is 1.

(A7) The φ-types determined by ω_1 and ω_2 are realized in two orders that assign the comparison the values 0 and 1; the two states have no independent product at P.

For the physicist or operator theorist. Read in the structure $\mathcal{M}_C$ of Variatio 1, whose sorts are the sets of density operators of Alice and of Bob, condition (A2) becomes the statement that the coheirs of the φ-types over $\mathcal{M}_C$ of the elements of Alice's sort of ultrapowers of $\mathcal{M}_C$ follow their definitions: for an element $a = [\rho_n]_{\mathcal{U}}$, for a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, and the class $b = [\sigma_m]_{\mathcal{V}}$ of a sequence of Bob's density operators along a nonprincipal ultrafilter $\mathcal{V}$ on $\mathbb{N}$, the coheir value $\varphi(a', b)$ of Variatio 3, the value with Bob's system idealized first, is $\omega_2(Z_{\omega_1})$, where $\omega_1 = \lim_{n,\mathcal{U}} \rho_n$ and $\omega_2 = \lim_{m,\mathcal{V}} \sigma_m$, and it depends continuously on the type of b over $\mathcal{M}_C$. Condition (A6) concerns *Keisler measures*, the averaged analogue of types: a Keisler measure over a structure $\mathcal{M}$ in the variable x is a regular Borel probability measure on the compact space of types in x over $\mathcal{M}$, equivalently a positive normalized linear functional on the formulas $\theta(x, b)$ with b in $\mathcal{M}$, and each type is identified with the Dirac measure at it. Ultrapowers produce Keisler measures as they produce types: for a sequence $(\mu_n)_{n\in\mathbb{N}}$ of finitely supported probability measures on Alice's sort of $\mathcal{M}_C$, the functional $\theta(x, b) \mapsto \lim_{n,\mathcal{U}} \int \theta(x, b)\, d\mu_n(x)$ is a Keisler measure, and, read for these measures, condition (A6) says that the two iterated integrals of φ_C against such a measure on Alice's sort and such a measure on Bob's agree. Condition (A7) is the counterpart, for the single structure $\mathcal{M}_C$, of Harrington's lemma (see Theorem 6.1): the two independent amalgams obtained in an iterated ultrapower coincide.

For the logician. The conditions that describe operations of a laboratory are (A5), (A6) and (A7). A finite mixture $\sum_{i=1}^{r} t_i \rho_i$ of preparations $\rho_1, \ldots, \rho_r$, with weights $t_i \geq 0$ summing to 1, is prepared by choosing the preparation ρ_i at random with probability t_i and forgetting the choice, and condition (A5) says that, for every $\varepsilon > 0$, such a procedure reproduces an idealized preparation within ε, simultaneously for all of Bob's probes, in the order with Bob's system idealized first; by Theorem 5.1, this happens exactly for stable observables. Condition (A6) concerns preparations drawn at random from infinitely many (see Variatio 10), and condition (A7) concerns the joint state of two systems prepared independently, which is $\rho \otimes \sigma$ for actual preparations ρ and σ and must be defined by a limit for idealized ones. For the energy comparison, all three fail with the largest possible gap, and for one reason: the event "Bob's oscillator has the higher energy" has probability 0 or 1 according to the order in which the two oscillators, excited to infinite energy, are approached.

The seven conditions are seven readings of Grothendieck's double limit criterion, and the staircase of the Aria, the matrix of the energy comparison, violates all seven for one reason: its two iterated limits are 0 and 1.

Variatio 6. a 1 Clav. Canone alla Seconda

The independent product

Two systems are prepared independently. If the preparations are actual, the joint state is the tensor product of the two density operators. If they are idealized, there are two candidates for the joint state, the two products of Variatio 1, obtained by placing the two systems, in either order, in an iterated ultrapower. In this variation we state when the two candidates coincide, in two readings: the logician speaks of two-valued types over a model, the physicist of states of an algebra of observables.

Let C be a bipartite observable, and recall from Variatio 1 the structure $\mathcal{M}_C$, whose two sorts are the sets of density operators of Alice and of Bob and whose predicate symbol φ stands for the expectation of C. Let p and q be types over $\mathcal{M}_C$ in Alice's sort and in Bob's, respectively, realized by elements a and b of ultrapowers of $\mathcal{M}_C$, as in Variatio 1. We call a joint type of (a, b) obtained in either order of the iterated ultrapower of Variatio 1 an *independent amalgam* of p and q over $\mathcal{M}_C$; in general it depends on the ultrafilters, and not only on p and q. The uniqueness of independent amalgams separates stable theories from unstable ones. If φ is stable in the theory of $\mathcal{M}_C$, then over a model the coheir of a φ-type does not depend on the ultrafilter and is the unique nonforking extension, for the notion of forking recalled below, and independence is symmetric [HH84]. For a single structure the relevant condition is weaker: by Theorem 3.1, the two orders agree on all pairs of states exactly when C is stable, that is, when φ has no infinite ladder in $\mathcal{M}_C$ itself, whereas stability of φ in the theory of $\mathcal{M}_C$ is uniform stability (see Theorem 7.1). In physical terms, an independent amalgam of the φ-types determined by two idealized states yields a candidate value for the product of the two states at C, and in Theorem 6.2 we assemble these values, over all stable observables, into a state. Simple theories retain a symmetric independence relation without uniqueness, and in Part III we show what it becomes in quantum error correction.

The logician's reading. Let M be a model of a stable theory, let p and q be complete types over M in variables x and y, and let $\varphi(x, y)$ be a formula. Stability provides definitions: there are formulas $d_p\varphi(y)$ and $d_q\varphi(x)$ over M such that, for every tuple b from M, the formula $\varphi(x, b)$ belongs to p exactly when $M \models d_p\varphi(b)$, and, for every tuple a from M, the formula $\varphi(a, y)$ belongs to q exactly when $M \models d_q\varphi(a)$.

Theorem 6.1 (Harrington's lemma [HH84]). *Let M be a model of a stable theory, let p and q be complete types over M in the variables x and y, and let $\varphi(x, y)$ be a formula. Then the formula $d_p\varphi(y)$ belongs to q if and only if the formula $d_q\varphi(x)$ belongs to p.*

Every type over a model is finitely satisfiable in it; hence there is an ultrafilter $\mathcal{U}$ on the tuples of M such that a formula $\theta(x, b)$ belongs to p exactly when the set of tuples a with $M \models \theta(a, b)$ belongs to $\mathcal{U}$, and there is an ultrafilter $\mathcal{V}$ that gives q in the same way. If φ is regarded as a $\{0, 1\}$-valued function on pairs of tuples, then the two sides of the lemma read:

$$\lim_{b,\mathcal{V}} \lim_{a,\mathcal{U}} \varphi(a, b) = 1 \qquad \text{and} \qquad \lim_{a,\mathcal{U}} \lim_{b,\mathcal{V}} \varphi(a, b) = 1.$$

Harrington's lemma thus asserts that two iterated ultrafilter limits may be exchanged. To state its role we recall forking. A formula $\theta(x, b)$ *divides* over a set A if there are elements $b = b_0, b_1, b_2, \ldots$ of an elementary extension of M that are *indiscernible* over A (for every n, any two increasing n-tuples of them have the same type over A) and an integer k such that no k of the formulas $\theta(x, b_i)$ have a common solution, that is, the sequence $(b_i)_{i\in\mathbb{N}}$ is *k-inconsistent* for θ; in real-valued logic, where $\theta(x, b)$ is read as the condition $\theta(x, b) = 0$, k-inconsistency is required with a margin: for some $\varepsilon > 0$, no x satisfies $\theta(x, b_i) \le \varepsilon$ for k distinct indices i. A type *forks* over A if it implies a finite disjunction of formulas that divide over A, and we write $a \mathop{\smile\!\!\!\!\!\!|}_A b$ when the type $\mathrm{tp}(a/Ab)$ of a over $A \cup \{b\}$ does not fork over A. In every theory, a coheir of a type over a model does not fork over that model; hence the two orders of the iterated ultrapower of Variatio 1 always produce nonforking amalgams, and in a stable theory they produce the only one. Harrington's lemma is the step that makes nonforking independence symmetric in a stable theory, and, together with the uniqueness of nonforking extensions over models, it yields the characterization of stable theories, due to Harnik and Harrington, as the theories with an independence relation that is stationary over models [HH84].

The physicist's reading. Let $\mathcal{O}$ be the set of stable observables on the tensor product $H_1 \otimes H_2$ of Alice's and Bob's Hilbert spaces, a norm-closed real vector space of self-adjoint operators that contains I [DI26a, Proposition 10.1]. For a state ω_1 of $B(H_1)$ and a state ω_2 of $B(H_2)$, the two orders of idealization agree at every $C \in \mathcal{O}$, by Theorem 3.1, and we write $\omega_1 \otimes \omega_2$ for the functional on $\mathcal{O}$ whose value at C is their common value. We seek a state on this space of observables that extends the product of actual preparations and whose value at each $C \in \mathcal{O}$ depends weak* continuously on each marginal state separately. These requirements determine the scalar extension at each stable observable, because actual states are dense. The theorem below shows that the resulting functional is a state on $\mathcal{O}$ and is symmetric under exchange of the two systems. In this reading, uniqueness concerns the separately continuous extension on stable observables, rather than arbitrary joint states with prescribed marginals.

Theorem 6.2 ([DI26a, Theorem 8.2])**.** *The following hold:*

(a) (Existence) *for all states ω_1 of $B(H_1)$ and ω_2 of $B(H_2)$, the functional $\omega_1 \otimes \omega_2$ is a state of $\mathcal{O}$: linear, positive, with value 1 at I;*

(b) (Extension) *if ω_1 and ω_2 are given by density operators ρ and σ, then we have $(\omega_1 \otimes \omega_2)(C) = \mathrm{Tr}((\rho \otimes \sigma)C)$ for every $C \in \mathcal{O}$;*

(c) (Uniqueness) *for each $C \in \mathcal{O}$, the function $(\omega_1\omega_2) \mapsto (\omega_1 \otimes \omega_2)(C)$ is the only function on pairs of states that is weak* continuous in each variable separately and satisfies* (b)*;*

(d) (Symmetry) *if $\Sigma\colon H_1 \otimes H_2 \to H_2 \otimes H_1$ is the unitary operator with $\Sigma(\xi \otimes \eta) = \eta \otimes \xi$, then, for all states ω_1 of $B(H_1)$ and ω_2 of $B(H_2)$ and every $C \in \mathcal{O}$, the value $(\omega_2 \otimes \omega_1)(\Sigma C \Sigma^*)$ equals $(\omega_1 \otimes \omega_2)(C)$, where the product on the left is taken on the stable observables on $H_2 \otimes H_1$.*

Moreover, a bipartite observable on $H_1 \otimes H_2$ belongs to $\mathcal{O}$ if and only if the two orders of idealization assign it the same value at every pair of states.

Idea of the proof. Positivity holds for each order separately, since, at a positive observable, each order gives an iterated limit of nonnegative expectations in product states. Extension holds because the two orders agree when one factor is actual. For uniqueness, the product with Alice's system idealized first is weak* continuous in Bob's state, being evaluation at the operator Z_{ω_1} that represents the response of ω_1 on actual probes, and the product with Bob's system idealized first is continuous in Alice's; when they agree, the common value is separately continuous, and a separately continuous function is determined by its values at the pairs of actual states, which are dense. Symmetry is the exchange of the two iterated limits. □

Both readings. A type over a model is *stationary* if it has a unique nonforking extension to every larger set; in the iterated ultrapower, stationarity means that the type of the element realized second does not depend on the ultrafilter. In the language of independent amalgams, (a) corresponds to the existence of an independent amalgam, (c) to stationarity over a model, and (d) to the symmetry of independence. For the energy comparison, at the states that describe Alice's and Bob's oscillators excited to infinite energy, the two orders of realization yield the values 0 and 1, and uniqueness and symmetry are lost together.

On stable observables, the two orders of idealization have a common value, and these values form a symmetric product of the marginal states. For each stable observable, separate weak* continuity makes this extension unique; the double limit condition is what makes the two readings agree.

Variatio 7. a 1 ô vero 2 Clav. Al tempo di Giga

Discrimination and learning

Alice prepares one of many known states, and Bob distinguishes it by adaptive queries of expectation responses, each probe chosen in view of the previous answers. In the online model of learning, a learner is shown one probe at a time, predicts whether the expectation for the unknown preparation lies above or below a threshold, and is then told the correct answer. The threshold is fixed within each protocol, a positive margin separates the counted responses from it, and the bounds are uniform in the threshold. In this variation we show that, for uniformly stable observables, the number of preparations distinguished by a given number of such queries grows at most polynomially, and the learner makes boundedly many mistakes. A query concerns an expectation value; estimating that value from repeated preparations is a separate statistical task.

Trees. A ladder is a linear configuration, whereas an adaptive protocol branches, each probe depending on the outcomes of the previous ones; the configuration that records such a protocol is a tree. Let C be a bipartite observable, and fix a margin $\delta > 0$ and a threshold t. For $n \geq 1$, a *tree of height* n assigns a probe σ_s to every node s of the complete binary tree of height n and a preparation ρ_η to every branch η, in such a way that, whenever η passes through s,

$$\varphi_C(\rho_\eta, \sigma_s) \geq t + \tfrac{\delta}{2} \text{ if } \eta \text{ turns right at } s, \qquad \varphi_C(\rho_\eta, \sigma_s) \leq t - \tfrac{\delta}{2} \text{ if it turns left,}$$

or the same holds, for all branches and nodes, with the two inequalities exchanged. Thus, with n probes, each chosen in view of the previous outcomes, Bob tells apart the 2^n preparations of a tree of height n. The pairs (ρ_i, σ_i) of a ladder are its *rungs*, and, for every $h \geq 1$, a ladder with 2^h rungs contains a tree of height h: the first probe is that of the middle rung, and binary search continues in the half of the ladder consistent with the outcome. The observable C is *uniformly stable* if, at every margin, its ladders have bounded length.

Theorem 7.1 ([DI26a, Theorem B])**.** *For every bipartite observable C on $H_1 \otimes H_2$, the following are equivalent:*

(B1) *the observable C is uniformly stable;*
(B2) (Shelah) *at every margin, the trees of C have bounded height;*
(B3) *the expectation remains stable on ultrapowers, that is, on families of experiments;*
(B4) (Sauer, Shelah) *at every margin δ, the number of preparations that n adaptive probes tell apart is at most a polynomial in n, whose degree is the bound on the height of trees;*
(B5) (Littlestone) *at every margin, the response of an unknown preparation can be learned online with a bounded number of mistakes.*

If the conditions fail, then at some margin, for every n, there are 2^n preparations that n probes tell apart.

Two arguments. The counting bound of (B4) rests on an observation about the largest height $d(\mathcal{R})$ of a tree whose preparations lie in a set $\mathcal{R}$: for any probe, the preparations in $\mathcal{R}$ that answer high and those that answer low form two disjoint parts of $\mathcal{R}$, which need not exhaust it, and the two parts cannot both carry trees of height $d(\mathcal{R})$, because two such trees, placed above that probe, would form a tree of height $d(\mathcal{R})+1$. By induction on n, with Pascal's rule, the number of preparations that n probes tell apart is at most $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{d}$, a

polynomial of degree d, where $d = d(\mathcal{R})$. This is the tree form of the Sauer–Shelah lemma [Sau72, She72].

Littlestone's learner [Lit88] rests on the same observation: it maintains the set V of preparations consistent with the answers so far and predicts the side whose consistent preparations carry the higher tree. Each mistake decreases $d(V)$, so there are at most d mistakes. Conversely, along a tree of height n, choosing at each node the branch opposite to the prediction produces n mistakes.

For (B3), the idea is a count: trees of every height glue, in an ultrapower, into a tree whose $2^{\aleph_0}$ branches are told apart by countably many probes, whereas, if the expectation of the ultrapower were stable, the responses of the preparations at the branches to these probes would lie in a separable space. Fix a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$ and consider elements $[\rho_k]_{\mathcal{U}}$ and $[\sigma_k]_{\mathcal{U}}$ of the ultrapower $\mathcal{M}_C^{\mathcal{U}}$ of the structure $\mathcal{M}_C$ of Variatio 1, given by sequences $(\rho_k)_{k\in\mathbb{N}}$ of Alice's preparations and $(\sigma_k)_{k\in\mathbb{N}}$ of Bob's probes, families of experiments indexed by k, at which the expectation takes the value $\lim_{k,\mathcal{U}} \varphi_C(\rho_k, \sigma_k)$. If C has trees of every height at a fixed margin, they glue along the ultrafilter into such an infinite tree. If the expectation of the ultrapower were stable, then, by the theorem of Iovino recalled in Variatio 5, it would factor through the pairing of a reflexive space X with its dual, by bounded maps on the two sorts; the closed span in X^* of the images of the countably many probes would then be a separable reflexive space with a separable dual, and the images of the $2^{\aleph_0}$ preparations would be $2^{\aleph_0}$ points of that dual at mutual distances bounded below, which is impossible.

Example 7.2 (Stable but not uniformly stable). Let both systems carry the Hilbert space $\bigoplus_{n\geq 1} \mathbb{C}^n$, a sum of spin multiplets, the Hilbert spaces $\mathbb{C}^n$ of particles of spin $(n-1)/2$, whose summands we call *sectors*, with orthonormal bases $e_1^n, \dots, e_n^n$ of the sectors (only these bases are used), and let C_{sec} be the orthogonal projection onto the closed span of the vectors $e_i^n \otimes e_j^n$ with $i \leq j$, which describes the event "both systems lie in the same sector, and Alice's level does not exceed Bob's". In the sector $\mathbb{C}^n$ the basis vectors form a ladder of length n at margin 1, and the observable C_{sec} is therefore not uniformly stable. Nevertheless, it is stable: its effective map $\rho \mapsto T_{C_{\mathrm{sec}}}(\rho)$ factors through the ℓ^2-sum of the spaces ℓ_n^1, which is reflexive [DI26a, Example 9.8].

For the physicist or operator theorist. Let $\mathcal{M}_C$ be the structure of Variatio 1, whose sorts are the sets of density operators of Alice and of Bob, and let $\mathcal{M}_C^{\mathcal{U}}$ be its ultrapower along a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$. A tree of height n is a binary tree of Shelah for the formula φ, and the largest height of such trees is Shelah's binary tree rank; as observed by Chase and Freitag [CF19], it is the *Littlestone dimension* of the class of responses, the optimal mistake bound of (B5). By Łoś's theorem, under which formulas take in the ultrapower the ultralimits of their values in $\mathcal{M}_C$, trees of unbounded height in $\mathcal{M}_C$ glue into an infinite tree in $\mathcal{M}_C^{\mathcal{U}}$, which, by the argument for (B3) above, makes the expectation of the ultrapower unstable; conversely, a bound on the height of the trees in $\mathcal{M}_C$ holds in $\mathcal{M}_C^{\mathcal{U}}$ as well, and, since a ladder with 2^h rungs contains a tree of height h, the expectation of the ultrapower then has no infinite ladder. This is the equivalence of (B2) and (B3); hence uniform stability is stability of φ in the theory of $\mathcal{M}_C$ (see Variatio 1), a property shared by all elementary extensions, and in Example 7.2 we show that it is stronger than stability of C. The count in (B4) is a count of φ-types over finite sets of probes, and Rowbottom's original definition of stability, the condition that, for an infinite cardinal λ, there are at most λ types over every set of λ parameters, is its infinite form; here a type over a set A is the type of an element of some elementary extension, and for countable A all of them are realized in $\mathcal{M}_C^{\mathcal{U}}$.

For the logician. In the response-query model behind Theorem 7.1, Alice supplies copies of an unknown preparation ρ chosen from a known list, and Bob chooses a probe σ at each step. The answer records whether $\varphi_C(\rho,\sigma) \geq t+\delta/2$ or $\varphi_C(\rho,\sigma) \leq t-\delta/2$, with t fixed throughout the protocol. Experimentally, Bob estimates this expectation by measuring C on repeated preparations of $\rho \otimes \sigma$; the theorem counts the queries and assumes their correct threshold labels, rather than bounding the repetitions needed for statistical confidence. Each probe may depend on the previous labels, and a tree of height n distinguishes its 2^n leaf preparations with n such queries. For a uniformly stable observable with bound d on tree height at the prescribed margin, no tree of height greater than d is realized, and the dimension-based learner makes at most d mistakes when predicting the labels. In Example 7.2, the summand $\mathbb{C}^n$ carries the spin-$(n-1)/2$ representation; ever larger multiplets carry ever longer ladders, so qualitative stability does not give this uniform bound.

The ultrapower expectation is a joint limit, not an iterated one, and uniform stability is Grothendieck's double limit criterion applied to families of experiments: it holds exactly when trees have bounded height, and then adaptive discrimination grows polynomially and online learning makes boundedly many mistakes.

Variatio 8. A 2 Clav.

Bounds independent of the dimension

The swap test and the Hong–Ou–Mandel coincidence (see Variatio 4) are stable in every dimension. In this variation we show that they are uniformly stable, with bounds that do not depend on the dimension. Bob chooses probe states adaptively and queries the coincidence or acceptance probability on fresh copies of Alice's preparation, using the previous response labels to choose the next probe. The bound in Theorem 8.1, combined with the counting argument of Variatio 7, gives a polynomial bound on the number of preparations distinguished by a given number of expectation queries; its degree depends only on the margin. The same bound therefore holds in a polarization model and in a model with infinitely many spatial modes.

Decision trees. An adaptive protocol of Bob, in which each probe is chosen in view of the previous outcomes, is a binary decision tree. Fix a margin $\delta > 0$ and a bounded real function φ on pairs consisting of a preparation of Alice and a probe of Bob, such as the expectation of a bipartite observable. A *tree of height n at margin δ*, in this variation, differs from a tree of Variatio 7 in having a threshold at each node: it assigns a probe σ_s and a threshold t_s to every node s of the complete binary tree of height n, and a preparation ρ_b of Alice to every branch b, in such a way that, whenever the branch b passes through the node s,

$$\varphi(\rho_b,\sigma_s) \geq t_s + \tfrac{\delta}{2} \text{ if } b \text{ turns right at } s, \qquad \varphi(\rho_b,\sigma_s) \leq t_s - \tfrac{\delta}{2} \text{ if } b \text{ turns left.}$$

The 2^n preparations at the branches are then told apart by n probes, each answered with margin δ. Every tree of Variatio 7 becomes a tree in this sense, with all thresholds equal, once left and right are exchanged at every node if necessary; a bound on the height of the trees of this variation therefore bounds the height of the trees of Variatio 7. Since both answers occur at every node, every threshold satisfies $|t_s| \leq \sup|\varphi|$; we use this bound in the proof below.

Theorem 8.1 (Novikoff [Nov62]; [DI26a, Theorem 9.11]).
Suppose that $\varphi(\rho,\sigma) = \langle u(\rho), v(\sigma)\rangle$ for all preparations ρ and probes σ, where u and v are maps into a real Hilbert space $\mathcal{K}$, and that R_u and R_v are positive reals with $\|u(\rho)\| \leq R_u$ and $\|v(\sigma)\| \leq R_v$ for all ρ and σ. Then every tree at margin δ has height at most $16(R_uR_v/\delta)^2$.

For the swap test and for the Hong–Ou–Mandel coincidence, with the margin measured in probability, every tree has height at most $4/\delta^2$.

Proof. The idea is a comparison of two rates of growth: a vector corrected once at each step of a walk down the tree grows in length at most like the square root of the number of steps, while its inner product with a fixed vector determined by the branch followed grows at least linearly; the Cauchy–Schwarz inequality, which compares the two, bounds the height of the tree. Write $x_s = (v(\sigma_s), -t_s/R_u)$ in the Hilbert space $\mathcal{K} \oplus \mathbb{R}$; each of these vectors satisfies $\|x_s\|^2 \le 2R_v^2$. We walk down the tree from the root, keeping a vector w, initially 0. At the node s, we turn left if $\langle w, x_s\rangle \ge 0$ and right otherwise, set $y_s = +1$ for a right turn and $y_s = -1$ for a left one, and replace w by $w + y_s x_s$. Since $y_s\langle w, x_s\rangle \le 0$ at every step, each step increases $\|w\|^2$ by at most $\|x_s\|^2 \le 2R_v^2$, and after n steps $\|w\|^2 \le 2nR_v^2$. Let b be the branch that was followed and put $w^* = (u(\rho_b), R_u)$, a vector with $\|w^*\|^2 \le 2R_u^2$ and $\langle w^*, x_s\rangle = \varphi(\rho_b, \sigma_s) - t_s$. By the definition of a tree, $y_s\langle w^*, x_s\rangle \ge \delta/2$ at every node of b, whence $\langle w, w^*\rangle \ge n\delta/2$. The Cauchy–Schwarz inequality now gives $n\delta/2 \le \sqrt{2n}\, R_v \cdot \sqrt{2}\, R_u$, that is, $n \le 16(R_u R_v/\delta)^2$. For the swap test, the overlap $\operatorname{Tr}\rho\sigma$ is the Hilbert–Schmidt inner product of two operators of norm at most 1, and a margin δ in probability is a margin 2δ in $\operatorname{Tr}\rho\sigma$; the bound becomes $16/(2\delta)^2 = 4/\delta^2$. The same holds for coincidences. □

At margin $\delta = 0.1$ in probability, no protocol of 401 beam-splitter probes tells apart 2^{401} photon states, however many modes the photons have. No such bound holds for arbitrary observables: for the energy comparison of two oscillators, Bob's probe f_l, his energy eigenstate of level l, answers "yes" exactly on Alice's energy eigenstates e_k with $k < l$, and binary search on the levels builds trees of every height at margin 1.

Since a ladder with 2^h rungs contains a tree of height h, by the binary search of Variatio 7, a ladder of the swap test at margin δ in probability has fewer than $2^{1+4/\delta^2}$ rungs.

For the physicist or operator theorist. For a fixed threshold and margin, the largest height of the trees of Variatio 7 is Shelah's binary tree rank, or Littlestone dimension, of the thresholded response class; it is the optimal mistake bound in the corresponding response game. The trees of this variation allow the threshold to vary from node to node, so a bound for them also bounds the fixed-threshold rank. In Theorem 8.1 we prove this stronger bound for every expectation that factors through a Hilbert space, in terms only of the factor norms and the margin. In particular, the fixed-threshold bounds give uniform stability by Theorem 7.1, hence stability of φ in the theory of the preparation structure $\mathcal{M}_C$ and in every elementary extension, including ultrapowers. The proof runs the perceptron against the labels of a branch chosen to contradict its predictions; the bound on its mistakes is therefore a bound on the height of the tree.

For the logician. The internal space K of a photon is $\mathbb{C}^2$ in a polarization-only model; allowing spatial or temporal modes leads to infinite-dimensional models. For the Hong–Ou–Mandel case of Theorem 8.1, each query is realized by the arrangement of Variatio 4, repeated on fresh copies: Alice's photon has internal state ρ, Bob supplies a photon in a chosen state σ_s, and the ideal lossless balanced beam splitter, preserving the internal modes, gives coincidence probability $\frac{1}{2}(1-\operatorname{Tr}(\rho\sigma_s))$. At margin δ in this probability, a full tree of such expectation queries has height at most $4/\delta^2$, independently of the dimension of K. Repeated runs estimate each probability; they are not counted by the tree height. In each fixed finite dimension some bound follows from compactness of the state space; the bound proved here remains uniform as the dimension grows.

The bound on the ladders of the swap test found above, which holds for the Hong–Ou–Mandel coincidence as well, excludes in particular infinite ladders, and the two orders of idealization of these overlap measurements therefore agree, by Theorem 2.1: at margin δ in probability, the staircase of the Aria, the pattern of ones above the diagonal and zeros on and below it, can be imitated by the values of an overlap measurement only with fewer than $2^{1+4/\delta^2}$ steps.

Variatio 9. a 1 Clav. Canone alla Terza

Mixtures simulate idealized preparations

An oscillator excited to infinite energy cannot be prepared in a laboratory, but a finite mixture of actual preparations, which produces each of them with a prescribed probability, can. In this variation we show that, for stable observables, finite mixtures of actual preparations simulate idealized ones uniformly over all probes, in the order of idealization with Bob's system idealized first. The logician averages formulas and the physicist averages preparations; since the expectation of an observable is affine in the state, the two averages coincide.

The logician's reading. Let $\varphi(x, y)$ be a formula, and let M be a model of a theory in which φ is stable. In real-valued logic, the definition of a φ-type p over M, the continuous function on the space $S_y(M)$ of types in y over M that extends the map $b \mapsto \varphi(p, b)$ (see Variatio 3), is a uniform limit of continuous combinations of instances $\varphi(a, y)$ with a in M [BYU10]. Ben Yaacov observed that, by Mazur's theorem, according to which a convex subset of a normed space has the same weak and norm closures, averages suffice: the definition is a uniform limit of averages $\frac{1}{n}\sum_{i=1}^{n}\varphi(a_i, y)$, with $n \geq 1$ and $a_1, \ldots, a_n$ in M [BY14].

The physicist's reading. Let H_1 and H_2 be the Hilbert spaces of Alice's and of Bob's systems, and let C be a bipartite observable on $H_1 \otimes H_2$, with expectation $\varphi_C(\rho, \sigma) = \mathrm{Tr}((\rho \otimes \sigma)C)$ for ρ in the set $\mathfrak{D}(H_1)$ of density operators on H_1 and σ in $\mathfrak{D}(H_2)$. Recall that the effective observable of a preparation ρ of Alice is the unique bounded operator $T_C(\rho)$ on H_2 with $\mathrm{Tr}(\sigma\, T_C(\rho)) = \varphi_C(\rho, \sigma)$ for every $\sigma \in \mathfrak{D}(H_2)$ (see Variatio 1). Let ω_1 be an idealized state of Alice, and let $\otimes_{\mathsf{B}}$ denote the product of states that corresponds to the order of idealization with Bob's system idealized first.

Theorem 9.1 ([DI26a, Theorems 6.1 and 6.5]**).**

(a) *The observable C is stable if and only if, for every idealized state ω_1 of Alice and every $\varepsilon > 0$, some density operator ρ satisfies $|(\omega_1 \otimes_{\mathsf{B}} \omega_2)(C) - \omega_2(T_C(\rho))| \leq \varepsilon$ for all states ω_2 of Bob, idealized or not. The density operator ρ can be taken to be a finite mixture of terms from any tail of any net of density operators converging to ω_1.*

(b) *Let $\mathcal{K}$ be a real Hilbert space, let $u\colon \mathfrak{D}(H_1) \to \mathcal{K}$ and $v\colon \mathfrak{D}(H_2) \to \mathcal{K}$ be maps, and suppose that $\varphi_C(\rho, \sigma) = \langle u(\rho), v(\sigma)\rangle$ for all $\rho \in \mathfrak{D}(H_1)$ and $\sigma \in \mathfrak{D}(H_2)$, that some constant R_u satisfies $\|u(\rho)\| \leq R_u$ for all $\rho \in \mathfrak{D}(H_1)$, and that v extends to a bounded linear map, of norm $\|v\|$, from the self-adjoint trace-class operators on H_2, with the trace norm, to $\mathcal{K}$. Then, for every $\varepsilon > 0$ and every positive integer $N \geq (2R_u\|v\|/\varepsilon)^2$, the density operator in (a) can be taken to be an equal-weight mixture of N terms, not necessarily distinct, of any tail of any net of density operators converging to ω_1. For the swap test and the Hong–Ou–Mandel coincidence, which measure the overlap $\mathrm{Tr}(\rho\sigma)$ of two states of a common Hilbert space K (see Variatio 4), the same holds for every integer $N \geq 1/\varepsilon^2$, whatever the dimension of K.*

Idea of the proof. (a) If C is stable, the response of ω_1 is represented by an observable Z (see Theorem 3.1), which lies in the weak closure of the effective observables $T_C(\rho)$ of the

mixtures ρ of terms from any tail of any net of density operators converging to ω_1. By Mazur's theorem [Maz33], the weak and norm closures of a convex set coincide, so some mixture ρ has $\|Z - T_C(\rho)\| \leq \varepsilon$, and a norm approximation is uniform over all states ω_2. Conversely, if the approximation holds for every ε, the effective observables of the approximating mixtures converge in norm as ε tends to 0, because the states determine the norm of a self-adjoint operator, and their limit represents the idealized response of ω_1; as ω_1 is arbitrary, every idealized response is represented by an observable, and C is stable, by Theorem 3.1.

(b) The argument is Maurey's empirical method [Pis81]: Mazur's theorem provides a finite mixture, and random sampling replaces it by an equal-weight mixture of N of its terms. For $\eta \in \mathcal{K}$, the functional $\tau \mapsto \langle \eta, v(\tau) \rangle$ on the self-adjoint trace-class operators on H_2 is bounded and linear, hence of the form $\tau \mapsto \operatorname{Tr}(\tau B_\eta)$ for a unique self-adjoint operator B_η on H_2. The map $\eta \mapsto B_\eta$ is linear, satisfies $\|B_\eta\| \leq \|v\| \, \|\eta\|$, and sends $u(\rho)$ to $T_C(\rho)$ for every $\rho \in \mathfrak{D}(H_1)$, because $\operatorname{Tr}(\sigma B_{u(\rho)}) = \langle u(\rho), v(\sigma) \rangle = \operatorname{Tr}(\sigma \, T_C(\rho))$ for every $\sigma \in \mathfrak{D}(H_2)$. Let $(\rho_i)_{i \in I}$ be a net in $\mathfrak{D}(H_1)$ converging to ω_1, let $i_0 \in I$ be given, and let ξ^* be a weak cluster point of the net $(u(\rho_i))_{i \in I}$, which exists because closed balls of a Hilbert space are weakly compact. For every state ω_2 of Bob, the inner limit in the definition of $\otimes_{\mathsf{B}}$ is $\omega_2(T_C(\rho_i))$, whence $(\omega_1 \otimes_{\mathsf{B}} \omega_2)(C) = \lim_i \omega_2(B_{u(\rho_i)})$. The functional $\eta \mapsto \omega_2(B_\eta)$ on $\mathcal{K}$ is bounded and linear, hence weakly continuous, and passing to a subnet along which $u(\rho_i)$ converges weakly to ξ^* gives $(\omega_1 \otimes_{\mathsf{B}} \omega_2)(C) = \omega_2(B_{\xi^*})$. As a cluster point of the net, the vector ξ^* lies in the weak closure of the convex hull of the vectors $u(\rho_i)$ with $i \geq i_0$, hence, by Mazur's theorem, in the norm closure of that convex hull: there is a convex combination $\xi = \sum_{k=1}^{m} t_k u(\rho_{i_k})$, with indices $i_1, \ldots, i_m \geq i_0$, such that $\|v\| \, \|\xi^* - \xi\| \leq \varepsilon/2$. Let $\kappa_1, \ldots, \kappa_N$ be independent random elements of $\{1, \ldots, m\}$, each equal to k with probability t_k, and let $\bar{\xi}$ be the average of the vectors $u(\rho_{i_{\kappa_1}}), \ldots, u(\rho_{i_{\kappa_N}})$. These vectors are independent, with mean ξ and norm at most R_u; since the variance of a random vector is at most its mean square norm, we have $\mathbb{E}\|\bar{\xi} - \xi\|^2 = \frac{1}{N}\mathbb{E}\|u(\rho_{i_{\kappa_1}}) - \xi\|^2 \leq R_u^2/N$. Hence some outcome of the draw satisfies $\|\bar{\xi} - \xi\| \leq R_u/\sqrt{N}$, and the choice of N gives $\|v\| R_u/\sqrt{N} \leq \varepsilon/2$. Fix such an outcome, and let $\bar{\rho}$ be the equal-weight mixture of the N density operators $\rho_{i_{\kappa_1}}, \ldots, \rho_{i_{\kappa_N}}$, which are terms of the net with indices at least i_0. Because T_C is affine on $\mathfrak{D}(H_1)$ and the map $\eta \mapsto B_\eta$ is linear, the operator $T_C(\bar{\rho})$ equals $B_{\bar{\xi}}$. The map u is evaluated only at terms of the net, never at a mixture, and it therefore need not be affine. For every state ω_2 of Bob, it follows that $|(\omega_1 \otimes_{\mathsf{B}} \omega_2)(C) - \omega_2(T_C(\bar{\rho}))| = |\omega_2(B_{\xi^* - \bar{\xi}})| \leq \|v\| \, (\|\xi^* - \xi\| + \|\xi - \bar{\xi}\|) \leq \varepsilon$. For the swap F of Lemma 4.1, the unitary on $K \otimes K$ that exchanges the two factors, take $H_1 = H_2 = K$, and let u and v both be the inclusion of $\mathfrak{D}(K)$ into the real Hilbert space of self-adjoint Hilbert–Schmidt operators on K, with the inner product $\langle \tau, \tau' \rangle = \operatorname{Tr}(\tau\tau')$. By that lemma, these maps satisfy $\varphi_F(\rho, \sigma) = \langle u(\rho), v(\sigma) \rangle$, and since the Hilbert–Schmidt norm is at most the trace norm, the vectors $u(\rho)$ have norm at most $R_u = 1$ and the extension of v has norm at most 1; every integer $N \geq 4/\varepsilon^2$ therefore suffices for F. The swap test and the Hong–Ou–Mandel coincidence are, by the same lemma, the projections $\frac{1}{2}(I \pm F)$, for which $T_{\frac{1}{2}(I \pm F)}(\rho) = \frac{1}{2}(I \pm T_F(\rho))$ and $(\omega_1 \otimes_{\mathsf{B}} \omega_2)(\frac{1}{2}(I \pm F)) = \frac{1}{2}(1 \pm (\omega_1 \otimes_{\mathsf{B}} \omega_2)(F))$. Their errors are therefore half those of F, and the bound for F at accuracy 2ε shows that every integer $N \geq 4/(2\varepsilon)^2 = 1/\varepsilon^2$ suffices for them, whatever the dimension of K. □

The logician's reading. The order of idealization in part (a) of Theorem 9.1, with Bob's system idealized first, is the order in which a definition is evaluated: Bob's type is substituted into the definition of Alice's.

The physicist's reading. The order cannot be dropped: let $(e_k)_{k\geq 0}$ and $(f_l)_{l\geq 1}$ be orthonormal bases of H_1 and of H_2, let S_k, for $k \geq 1$, be the orthogonal projection onto the span of $f_1, \dots, f_k$, and let $C' = |e_0\rangle\langle e_0| \otimes I + \sum_{k\geq 1} |e_k\rangle\langle e_k| \otimes S_k$. Then, for $k \geq 1$, the value $\varphi_{C'}(e_k, f_l)$ is 1 exactly when $l \leq k$, whence the two iterated limits of these values differ, and the observable C' is unstable. Nevertheless, let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$, and let $\omega_1 = \lim_{k,\mathcal{U}} e_k$. With Alice idealized first, the actual state e_0 reproduces ω_1 exactly: its effective observable $T_{C'}(e_0)$ is I, and the response of ω_1 also takes the value 1 at every actual probe σ, because $\operatorname{Tr}(\sigma S_k)$ tends to $\operatorname{Tr}\sigma = 1$ as k grows. With Bob idealized first, at the idealized probe $\omega_2 = \lim_{l,\mathcal{U}} f_l$, the idealized state ω_1 answers 0 and the actual state e_0 answers 1. For the energy comparison P of Variatio 1, the failure is total: no mixture comes within any error less than 1, even on actual probes [DI26a, Example 6.3].

Both readings. In general, the definitions of types in real-valued logic are uniform limits of polynomials, or of lattice combinations, of instances. For an observable, an average of instances is again an instance: for density operators $\rho_1, \dots, \rho_m$ of Alice and weights $t_1, \dots, t_m \geq 0$ with sum 1, the expectation, being affine, satisfies $\sum_{k=1}^m t_k \varphi_C(\rho_k, \cdot) = \varphi_C(\sum_{k=1}^m t_k \rho_k, \cdot)$. The approximating formula is therefore the instance of a physical preparation, a finite mixture, and polynomials of degree one suffice.

When Grothendieck's double limit condition holds, it places the observable that represents an idealized response of Alice in the weak closure of the effective observables of her actual preparations; Mazur's theorem then turns the weak limit into a norm limit, and every idealized preparation is thereby approximated, uniformly over all probes and with Bob's system idealized first, by a finite mixture that a laboratory can prepare.

Variatio 10. A 1 Clav. Fughetta

Countable randomization does not detect instability

Although an oscillator of infinite energy cannot be prepared, its level can be drawn at random. Alice draws the level of her oscillator from a probability distribution, Bob draws the level of his from another, and the energy comparison of the two oscillators, which asks whether Bob's level is the higher, is averaged over both draws. The two averages can be taken in either order, Alice's first or Bob's first. In this variation we show that the two iterated averages agree for every observable when the distributions are countably additive, and that they agree for all finitely additive means exactly when the observable is stable.

We compare the two iterated integrals for three classes of measures. It is convenient to integrate over all density operators, not only over energy eigenstates: a *mean* on the set $\mathfrak{D}(H)$ of density operators on a Hilbert space H is a finitely additive probability measure defined on all subsets of $\mathfrak{D}(H)$, and the *barycenter* of a mean μ is the state $A \mapsto \int \operatorname{Tr}(\rho A)\, d\mu(\rho)$ of $B(H)$. Let C be a bipartite observable on $H_1 \otimes H_2$, where H_1 and H_2 are the spaces of Alice's and of Bob's systems, and let μ and ν be means on $\mathfrak{D}(H_1)$ and on $\mathfrak{D}(H_2)$, with barycenters ω_μ and ω_ν. Since the expectation $\varphi_C(\rho, \sigma) = \operatorname{Tr}((\rho \otimes \sigma)C)$ is affine in each state, an iterated integral of φ_C depends only on the barycenters, and it is the value at C of one of the two products of the barycenters, $\otimes_{\mathsf{A}}$ with Alice's system idealized first and $\otimes_{\mathsf{B}}$ with Bob's (see Variatio 1):

$$\int\Big(\int \varphi_C \, d\mu\Big) d\nu = (\omega_\mu \otimes_{\mathsf{A}} \omega_\nu)(C), \qquad \int\Big(\int \varphi_C \, d\nu\Big) d\mu = (\omega_\mu \otimes_{\mathsf{B}} \omega_\nu)(C).$$

Finite mixtures. If μ is finitely supported, its barycenter is a mixture, a density operator, and finite sums commute. The two integrals agree for every observable.

Countably additive measures. Let μ be a countably additive Borel probability measure on $\mathfrak{D}(H_1)$, with the trace norm. Since the trace-class space is separable, the identity map $\rho \mapsto \rho$ is strongly measurable and has constant trace norm one. Its Bochner integral $\bar{\rho} = \int \rho \, d\mu(\rho)$ therefore exists, is positive and has trace one, and represents the normal barycenter of μ. Both iterated integrals are defined even when ν is only a mean: averaging the functions $\rho \mapsto \varphi_C(\rho, \sigma)$ against ν preserves their common Lipschitz constant $\|C\|$, so the outer integrand for μ is Borel measurable; the outer integral against ν requires only a bounded integrand. The two products agree whenever one factor is actual, so the two iterated integrals agree for every observable and every mean ν of Bob [DI26a, Proposition 7.2]. No countably additive randomization detects instability, for any observable.

Finitely additive means. For finitely additive means, the answer is stability (Theorem 10.1 below).

Theorem 10.1 (Pták, Simons; [DI26a, Theorem 7.1]). *An observable C is stable if and only if, for all means μ on $\mathfrak{D}(H_1)$ and ν on $\mathfrak{D}(H_2)$,*

$$\int\Big(\int \varphi_C(\rho,\sigma)\, d\mu(\rho)\Big) d\nu(\sigma) = \int\Big(\int \varphi_C(\rho,\sigma)\, d\nu(\sigma)\Big) d\mu(\rho).$$

Proof. The two sides are the two products of the barycenters. Every state is a barycenter: if ω is the limit of a net (ρ_i) of density operators, and $\mathcal{W}$ is an ultrafilter on the index set containing all tails, then the mean μ given by $\mu(S) = \lim_{i,\mathcal{W}} \mathbf{1}_S(\rho_i)$, for $S \subseteq \mathfrak{D}(H)$, has barycenter ω. Hence the identity holds for all means exactly when the two orders of idealization agree for all states, that is, by Theorem 3.1, exactly when C is stable. □

For the energy comparison, let μ and ν be "uniform distributions on all levels": invariant means on $\mathbb{N}$, such as Banach limits of the sequence of uniform distributions on $\{1, \dots, n\}$, for $n \geq 1$, placed on the energy eigenstates e_k of Alice's oscillator and f_l of Bob's, for $k, l \in \mathbb{N}$. For each fixed l, almost every k in the sense of μ exceeds l, so the inner integral over Alice's levels is 0; for each fixed k, almost every l in the sense of ν exceeds k, so the inner integral over Bob's levels is 1. The two iterated integrals are 0 and 1. The same happens for ultrafilters, the $\{0,1\}$-valued means: for a nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$, the set $\{(k,l) : k < l\}$ has measure 0 under $\lim_{l,\mathcal{W}} \lim_{k,\mathcal{W}}$ and measure 1 under $\lim_{k,\mathcal{W}} \lim_{l,\mathcal{W}}$. No countably additive version exists, because a countably additive distribution on $\mathbb{N}$ cannot assign measure zero to every finite set.

For the physicist or operator theorist. A mean μ on Alice's state space $\mathfrak{D}(H_1)$ defines a *Keisler measure*, the averaged analogue of a type (see Variatio 5), over the structure $\mathcal{M}_C$ of Variatio 1, whose two sorts are the state spaces of Alice and of Bob: the functional $\theta(x, b) \mapsto \int \theta(\rho, b)\, d\mu(\rho)$ on formulas with parameters from $\mathcal{M}_C$; a mean that takes only the values 0 and 1 is given by an ultrafilter on $\mathfrak{D}(H_1)$, and its Keisler measure is the type of an element of an ultrapower. The two iterated integrals of Theorem 10.1 are the two products of Keisler measures, the averaged form of the two independent amalgams of Variatio 6, and the theorem says that these products commute for all measures exactly when they commute for all types. A countably additive measure on $\mathfrak{D}(H_1)$ with the trace norm has a density operator as barycenter, an element of $\mathcal{M}_C$ itself; for the formula φ it behaves like a realized type, and the two products agree because they agree whenever one factor is realized.

For the logician. A laboratory that draws Alice's preparation at random, by a countably additive distribution μ, and does not record the draw, prepares the density operator $\int \rho \, d\mu(\rho)$, since the expectation of every observable is affine in the state. Hence, if either system is

prepared by a countably additive randomization, that system is in an actual state, and the two iterated averages agree for every observable, because the two orders of idealization agree when one factor is actual (see Variatio 1). The disagreement found for the energy comparison needs Alice's level to be distributed uniformly over all of $\mathbb{N}$, a finitely additive mean that vanishes on every finite set of levels; its barycenter is the oscillator excited to infinite energy, an idealized state that no random procedure produces.

Grothendieck's double limit exchange condition is thus a Fubini theorem. For finite mixtures and countably additive measures it holds for every observable; for finitely additive means it holds exactly for the stable ones.

Part II. NIP

Variatio 11. A 2 Clav.

The independence property

Photons that pass a double slit are detected on Bob's screen in bright and dark fringes, whose spacing is inversely proportional to the wavenumber of the photons, 2π divided by their wavelength. If photons are prepared with wavenumbers $8, 8^2, 8^3, \dots$, then, as we show in Variatio 16, a single position of Bob's detector can be bright for any chosen finite set of them and dark for any other finite set disjoint from it: the fringes realize every pattern. The energy comparison of the two oscillators of the Aria cannot do this, because its answers are monotone. Both measurements are unstable; Part II is devoted to the difference between their two kinds of instability. In this variation we draw the second dividing line, between the instability of the energy comparison, which is NIP, and instability with the independence property, and name the configuration that NIP forbids.

A family of real functions on a set Y has the *independence property* if there are a sequence $(r_n)_{n\in\mathbb{N}}$ in it and fixed reals $a < b$ such that every finite pattern is realized: for all disjoint finite sets $E, F \subseteq \mathbb{N}$, some $y \in Y$ satisfies $r_n(y) \le a$ for $n \in E$ and $r_n(y) \ge b$ for $n \in F$. For a bipartite observable C with expectation φ_C, we call this the independence property for Alice's responses when the functions are $\sigma \mapsto \varphi_C(\rho, \sigma)$, with ρ ranging over Alice's preparations and σ over Bob's probes. Its absence is *NIP for Alice's responses*; unqualified NIP will refer to this direction. In Theorem 11.1 below, we show that stable observables are NIP; the energy comparison is NIP, since its answers are monotone, but unstable. The direction matters: a spin chain read site by site is NIP for Alice's responses but not for Bob's (see Variatio 14 and [DI26b, Corollary 8.7]). For a formula in a complete theory, NIP concerns finite configurations in all models of that theory and is symmetric under exchange of the variables [She90]. For an observable, the symmetric uniform condition bounds, at each positive gap, the sizes of finite sets of responses on which probes realize every pattern; the condition forbidding only an infinite independent sequence in the given structure need not be symmetric. In the theorem below, a function is of the *first Baire class* if it is a pointwise limit of a sequence of continuous functions, a *Rosenthal compactum* is a compact set of such functions on a Polish space with the topology of pointwise convergence, and $\beta\mathbb{N}$ is the Stone–Čech compactification of $\mathbb{N}$, whose points are the ultrafilters on $\mathbb{N}$.

Theorem 11.1 ([DI26b, Theorem 6.1])**.** *Let C be a bipartite observable.*

(a) *The observable C is stable if and only if every idealized response is continuous on the space of probes.*
(b) *The observable C is NIP if and only if every idealized response is of the first Baire class. The idealized responses then form a Rosenthal compactum.*
(c) *If C is not NIP, then the idealized responses contain a homeomorphic copy of $\beta\mathbb{N}$.*

The theorem is the dichotomy of Rosenthal and of Bourgain, Fremlin and Talagrand [Ros74, BFT78], applied to the space of probes. This space is the state space, with the weak* topology, of the C*-algebra generated by the effective observables $T_C(\rho)$ of Alice's preparations ρ (see Variatio 1), that is, of the smallest algebra of operators on Bob's Hilbert space that contains them and is closed under adjoints and norm limits. The responses of Alice's actual preparations are continuous functions on this space, and the idealized responses form their pointwise closure, the kind of set that the dichotomy describes.

For the physicist or operator theorist. For a formula in a theory, the independence property says that, for every n, some n parameters admit all 2^n patterns; by compactness, an infinite independent sequence then exists in an elementary extension, such as an ultrapower, whereas for the single structure $\mathcal{M}_C$, in which Alice's and Bob's density operators form the two sorts and the expectation is a predicate (see Variatio 1), the definition above asks for an infinite independent sequence in $\mathcal{M}_C$ itself. Theorem 11.1 describes the φ-types over $\mathcal{M}_C$ realized in ultrapowers, the idealized responses of Variatio 1: for an NIP observable, each is the pointwise limit of a sequence of responses of actual preparations, as Simon proved for NIP formulas [Sim15], while an independent sequence $(\rho_n)_{n\in\mathbb{N}}$ of Alice's preparations yields the copy of $\beta\mathbb{N}$, formed by the φ-types of the elements $[\rho_n]_{\mathcal{V}}$ for the ultrafilters $\mathcal{V}$ on $\mathbb{N}$. Because the condition concerns infinite sequences in $\mathcal{M}_C$ and not finite configurations in every model of its theory, it need not be symmetric in Alice and Bob, whereas NIP of a formula in a theory, like the uniform version for observables, is symmetric.

For the logician. In the double-slit experiment Bob measures the position at which the photon is detected, and the response of a preparation of wavenumber k to a detector at x is the relative intensity $\frac{1}{2}(1+\cos kx)$ of the fringes there. The independence property of the fringes is a statement about single detector positions: for the wavenumbers $8, 8^2, 8^3, \ldots$, the intensities at x are close to 1 or close to 0 according to the base-8 digits of $x/2\pi$, which can be chosen freely (see Variatio 16). For each finite pattern, continuity allows the point probe to be replaced by a normal preparation localized near x, for which the averaged relative intensities are at least $\frac{5}{8}$ on the high indices and at most $\frac{3}{8}$ on the low indices. The gap is therefore independent of the finite pattern. For the comparison of photon numbers, by contrast, which is the energy comparison of the Aria when the two oscillators are modes of light, Bob's state f_l, with l photons, answers yes exactly for Alice's states e_k with $k < l$ photons, so on any n levels only the $n+1$ initial segments occur as patterns. When a spin chain is scanned site by site (see Variatio 14), the roles are reversed: the chains of Alice realize every pattern on Bob's sites, which is why NIP for Alice's responses does not imply NIP for Bob's.

When the double limit condition fails, there are two cases. If the observable is NIP, every idealized response is the limit of a sequence of actual responses; otherwise the idealized responses contain a copy of $\beta\mathbb{N}$, whose nonprincipal points are not limits of sequences of actual responses.

Variatio 12. Canone alla Quarta

The rational order

The energy comparison of Part I compares two discrete quantities; its continuous counterpart, the comparison of two positions on a line, is governed by the rational order, in which every finite linear order embeds. The rational order has in Part II the role that Hilbert space, the Fraïssé limit of the finite-dimensional Hilbert spaces, has in Part I (see Variatio 4). In this variation we approach the rational order from two directions: the logician descends from the theory of the rational order to its types, and the physicist ascends from two particles on a line to the idealized limits of their comparison.

The finite linear orders form a *Fraïssé class*, a class of finite structures closed under substructures and with the joint embedding and amalgamation properties, and their limit in the sense of Fraïssé's theorem (see Variatio 4) is $(\mathbb{Q}, <)$, the dense linear order without endpoints. Its theory is NIP and unstable, the prototype of a theory that is ordered but tame. The types of the formula $x < y$ over the real line are the cuts, and each real number r splits into two of them, just below and just above r; in the ultrapower $(\mathbb{R}, <)^{\mathcal{U}}$ along a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the class of the sequence $(r + 1/n)_{n\geq 1}$ realizes the type just above r and the class of $(r - 1/n)_{n\geq 1}$ the type just below. The space of these types is homeomorphic to the split interval of Alexandroff, obtained from an interval by splitting each point into a left and a right copy, with the topology of the lexicographic order; it is hereditarily separable, every subspace having a countable dense subset, but not metrizable. The physical home of the rational order is the comparison of two continuous quantities, such as the positions of two particles on a line.

Counting types measures the complexity of a formula: over finite sets of parameters the count separates NIP from the independence property, and over infinite sets it also separates stability from NIP. Write $\varphi(x, y)$ for the formula $x < y$. Over a finite set of n parameters it has only $n+1$ φ-types, since a new element falls into one of the $n+1$ gaps that the parameters leave. A set of n parameters is *shattered* by a formula $\theta(x, y)$ if each of its subsets is cut out by $\theta(a, y)$ for some a, and the *Vapnik–Chervonenkis dimension* of θ is the largest n for which some n parameters are shattered; by Łoś's theorem, unbounded shattering in a structure M yields the independence property in its ultrapower $M^{\mathcal{U}}$ along a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$. This is NIP in its finite form: by the Sauer–Shelah lemma, whose tree form appears in Variatio 7, a formula of Vapnik–Chervonenkis dimension d has at most $\sum_{i\leq d} \binom{n}{i}$ types over n parameters, a polynomial of degree d in n, and the formula φ, of dimension one, attains the bound $n + 1$, whereas a formula with the independence property has, for every n, 2^n types over some set of n parameters. Over an infinite set the count changes character. The types are the cuts of the set, and their number is governed, for a set of infinite cardinality κ, by the cardinal $\operatorname{ded}\kappa$, the supremum of the numbers of cuts of linear orders of size κ; Shelah showed that, for a theory T, with $|T|$ the cardinality of its language plus $\aleph_0$, there are, over a set of size κ, at most $\kappa^{|T|}$ complete types if T is stable and at most $(\operatorname{ded}\kappa)^{|T|}$ if T is NIP [She90, CS16]. The passage from stability to NIP is the passage from points to cuts.

Independence of types changes accordingly. A type over a monster model, a very large saturated and strongly homogeneous model of a complete theory (see Variatio 20), is called *global*. In an NIP theory, a global type does not fork over a model M, in the sense recalled in Variatio 6, exactly when it is invariant over M, that is, fixed by every automorphism of the monster model that fixes M pointwise [HP11]; the coheirs of Variatio 1 are of this kind. Unlike the stable case of Variatio 6, a type over a model may have several invariant extensions. The rational order exhibits two of them for the type just below a real number r.

When the type just below r is realized for Alice and for Bob in an iterated ultrapower of the structure $\mathcal{M}_C$ of the comparison C of the positions of two particles on a line (see Variatio 1), the two orders yield two independent amalgams: if Alice's particle, spread uniformly over $(r-1/n, r)$ for $n \geq 1$, is idealized first, it lies closer to r than Bob's, spread uniformly over $(r-1/m, r)$ for $m \geq 1$, and the comparison "Alice's particle lies to the left of Bob's" takes the value 0; in the other order it takes the value 1. This is the staircase of the Aria, compressed into a single point.

For the physicist or operator theorist. For the formula $x < y$, the ultrapower $(\mathbb{R}, <)^{\mathcal{U}}$ exhibits the types: over $\mathbb{R}$, the element $[r+1/n]$ has the φ-type $y \mapsto 1_{(r,\infty)}(y)$ and the element $[r-1/n]$ the φ-type $y \mapsto 1_{[r,\infty)}(y)$, two functions of the parameter that differ only at $y = r$. Together with the two types of infinite elements, these are all the φ-types over $\mathbb{R}$, and each is a pointwise limit of a sequence of realized types, as Theorem 11.1(b) requires of an NIP formula. Since the two types of such a pair are separated only by the parameter r itself, no countable set of parameters separates all of them, and the type space, the split interval, is hereditarily separable but not metrizable.

For the logician. For two particles on a line the comparison is the relation observable of $\{(x, y) : x < y\}$, multiplication by its indicator on $L^2(\mathbb{R}^2)$, and in states ρ and σ it answers yes with probability $(\mu_\rho \otimes \mu_\sigma)(\{x < y\})$, where μ_ρ and μ_σ are the distributions of the two positions. If Alice's particle is spread over $(r, r + 1/n)$ and n grows, its responses converge to $\sigma \mapsto \mu_\sigma((r, \infty))$, and if it is spread over $(r - 1/n, r)$, to $\sigma \mapsto \mu_\sigma([r, \infty))$. On actual probes the two limits agree, since a density operator assigns probability zero to the point r; they disagree on the idealized probe obtained by spreading Bob's particle over $(r - 1/m, r + 1/m)$ and letting m grow first, where the left limit answers 1 and the right limit answers 0. These two idealizations of a particle at r form a split pair.

The rational order thus fails the double limit condition in the mildest way: at each point of the line, the limits of a response from the left and from the right may disagree, these one-sided limits form the split pairs, and every type remains of the first Baire class.

Variatio 13. a 2 Clav.

The dictionary of NIP

Simon showed that, for an NIP formula and a countable model, the global types of the formula (see Variatio 12) that are finitely satisfiable in the countable model, viewed as functions on the space of types of parameters over it, are of the first Baire class, each the limit of a sequence of types realized in it [Sim15]. Bipartite observables present the same situation, with Bob's probes as parameters and Alice's idealized responses as types. In the language of Variatio 1, such a type is the response of an idealized preparation of Alice, the limit along an ultrafilter of a sequence of her actual preparations, evaluated, with Bob idealized first, at the elements of Bob's sort in any elementary extension; these types are coheirs, hence invariant extensions in the sense of Variatio 12, which do not fork (see Variatio 6). A Rosenthal compactum may be metrizable or very far from it, and the levels of NIP record how far: an NIP observable is NIP_3, NIP_2 or NIP_1 when its idealized responses form a metrizable, a hereditarily separable or a first countable compactum, respectively. In this variation we set out the second dictionary (see Table 3) and describe each of these levels by how many probes, and which, are needed to separate the idealized responses.

Model theory and topology	**Quantum measurement**
the independence property	a sequence of Alice's preparations on which Bob's probes realize every finite pattern at fixed thresholds: fringes
NIP	comparisons of physical quantities, and the causal order
Baire class one (Rosenthal; Bourgain, Fremlin, Talagrand)	every idealized response is the limit of a sequence of actual responses
a Rosenthal compactum	the idealized responses of an NIP observable
metrizable, and for observables hereditarily separable	one countable set of probes separates all idealized responses
first countable, not hereditarily separable	each response has its own countable determining probes, but no common countable separating set exists
a split pair	the two pure-phase responses at one transition point in an offset magnetization family
a closed subspace of the Helly space	the idealized responses of a comparison of positions
$\hat{A}(2^{\mathbb{N}})$, not first countable	a light cone shrinks to a point
the seven minimal families	selected trees of preparations of photons, a particle in a box or a spin chain
symmetry of NIP in the two variables	holds for uniform NIP; fails for NIP: a spin chain read site by site

TABLE 3. The dictionary of NIP, read from Alice's preparations to Bob's probes, with Bob's system idealized first. The levels classify the pointwise closure of the response family. For a full NIP observable response compactum, mixtures make hereditary separability equivalent to metrizability; the closure of a selected tree need not be convex (see Variations 19 and 15). Probe values here are expectation values, including those of idealized probes; the table does not specify a number of measurement repetitions for estimating them. The compactum $\hat{A}(2^{\mathbb{N}})$, the seven minimal families and the position-comparison subspace of the Helly space are described in Variations 14, 15 and 16, respectively.

Two facts about Rosenthal compacta, the sets of functions of the first Baire class on a Polish space that are compact in the topology of pointwise convergence (see Variatio 11), carry the dictionary. The first is the Bourgain–Fremlin–Talagrand theorem that a Rosenthal compactum is *Fréchet–Urysohn*: every point in the closure of a set is the limit of a sequence from the set [BFT78]. Write r_ρ for the response $\sigma \mapsto \varphi_C(\rho, \sigma)$ of a preparation ρ of Alice to Bob's probes σ, where φ_C is the expectation of a bipartite observable C (see Variatio 1). If C is NIP, then, for every sequence $(\rho_n)_{n\in\mathbb{N}}$ of Alice's preparations and every nonprincipal ultrafilter

$\mathcal{U}$ on $\mathbb{N}$, the idealized response $\lim_{n,\mathcal{U}} r_{\rho_n}$ is therefore also $\lim_k r_{\rho_{n_k}}$ for some sequence of indices $(n_k)_{k\in\mathbb{N}}$; the ultrafilter, indispensable in Variatio 1, can be replaced by a sequence. For the interference fringes of photons of wavenumbers $8, 8^2, 8^3, \ldots$ behind a double slit (see Variatio 11), this fails: the closure of their responses contains a copy of $\beta\mathbb{N}$, which has no nontrivial convergent sequences, and the points of the copy that correspond to nonprincipal ultrafilters are ultrafilter limits that no sequence reaches. The second fact is that the closure is determined by the responses: its topology is pointwise convergence on Bob's probes, so each of the levels NIP_3, NIP_2 and NIP_1 is a statement about how many probes, and which, decide convergence.

A single example shows both facts at work: the comparison of the positions of two particles on a line, which answers yes when Alice's particle lies to the left of Bob's (see Variatio 12). Fix a real number r, and, for each $n \geq 1$, let Alice's particle be spread uniformly over $(r, r + 1/n)$. Its response to a probe σ is the probability $(\mu_n \otimes \mu_\sigma)(\{x < y\})$, where μ_n and μ_σ are the distributions of the positions of Alice's and of Bob's particles, respectively, and as n grows it converges, at every probe, to $\sigma \mapsto \mu_\sigma((r, \infty))$: an ultrafilter is not needed. In an ultrapower along a nonprincipal ultrafilter on $\mathbb{N}$, the class of these preparations has as its idealized response the counterpart of the cut just above r; the class of the particles spread uniformly over $(r - 1/n, r)$ has that of the cut just below. On actual probes the two idealized responses agree, because a density operator assigns probability zero to the point r; they are told apart only by idealized probes concentrated at r, with Bob idealized first. The responses are monotone in the position of Bob's particle, each with at most countably many jumps, and they are functions of the first Baire class, as Simon's theorem requires.

Metrizable compacta (NIP_3). A compact set of functions is metrizable exactly when countably many points separate it [DI26b, Proposition 4.5]. For types, this says that one fixed countable set of instances of the formula decides every limit type. For an observable, it says that countably many probes tell the idealized responses apart. For a mode of light, an oscillator whose energy eigenvectors $e_0, e_1, e_2, \ldots$ are the states with no photon, one photon, two photons, and so on, the energy comparison of the Aria is the comparison of photon numbers; it is of this kind, NIP_3, because its idealized responses can jump only at the countably many levels of the spectrum.

Hereditarily separable compacta (NIP_2). A separable Rosenthal compactum is hereditarily separable exactly when it has no discrete subset of the size of the continuum [Tod99]. For types, this says that there is no continuum of limit types each of which finitely many instances tell apart from all the others. The comparison of positions fails this condition. At each point of the line it has a split pair, the two idealized responses of a particle that approaches the point from the right and from the left (see Variatio 12), and the *midpoint* of the pair, the response of the equal mixture of the two idealized preparations, is the average of the two responses. These midpoints, one for each point of the line, form an uncountable discrete set, each isolated by the idealized probe concentrated at its own point, where it alone takes the value $\frac{1}{2}$. For observables, the condition and metrizability coincide (see Variatio 19), which is why NIP_2 and NIP_3 merge.

First countable compacta (NIP_1). A point of a compact space of functions has a countable base of neighborhoods exactly when there is a countable set of coordinates on which no other point of the compactum agrees with it [DI26b, Proposition 4.6]. For types, this says that each limit type is decided by countably many instances of its own, chosen for it. The comparison of positions is of this kind, NIP_1: a response that jumps at a point is decided by the probes at that point and at rational points around it.

Beyond first countability. The causal order of spacetime and the comparison of phases on a circle, both defined in Variatio 14, have limit types that no countable set of instances decides. For the causal order, take events of Alice closer and closer to Bob's time slice, whose light cones meet the slice in intervals shrinking to a point c: their idealized limit is distinguished from nearby idealized responses only by probes that assign positive mass to c, and a countable set of probes assigns positive mass to only countably many points.

Read in this way, the classes measure how far an idealized response may be from continuous while remaining a limit of actual ones. Continuity, which Grothendieck's theorem equates with the double limit condition, is the strongest of them: a continuous response is determined at each probe by the probes near it.

Variatio 14. A 2 Clav.

Todorčević's trichotomy

When countably many probes no longer separate the idealized responses, the failure takes one of three specific forms [Tod99]. In this variation we state Todorčević's trichotomy for separable Rosenthal compacta (see Variatio 11), such as the space of idealized responses of an NIP observable, and list, in Table 4, a physical representative of each class. The table anticipates two later results: in Theorem 16.1 of Variatio 16 we place the representatives in their classes, and in Theorem 19.1 of Variatio 19 we show that the hereditarily separable, non-metrizable class is empty for observables.

Recall that the idealized responses are metrizable exactly when countably many probes separate them [DI26b, Proposition 4.5]. A separable Rosenthal compactum that is not metrizable contains a homeomorphic copy of one of a few canonical compacta: one of the two split Cantor sets $\hat{S}_+(2^{\mathbb{N}})$ and $\hat{S}_-(2^{\mathbb{N}})$ if it is hereditarily separable; one of the duplicates $\hat{D}(2^{\mathbb{N}})$ and $\hat{D}(S(2^{\mathbb{N}}))$ if it is first countable but not hereditarily separable; and the compactum $\hat{A}(2^{\mathbb{N}})$ if it is not first countable. Each of them has a countable dense set of isolated points, which the hat records, and is built over a classical space: the split Cantor sets over the space $S(2^{\mathbb{N}})$ obtained from the Cantor set $2^{\mathbb{N}}$ by splitting each point into a left and a right copy, the two duplicates over the Alexandroff duplicate of the Cantor set, in which each point receives an isolated twin, and over that of $S(2^{\mathbb{N}})$, respectively, and $\hat{A}(2^{\mathbb{N}})$ over the one-point compactification of a discrete set of cardinality $2^{\aleph_0}$. Figure 4 draws the trichotomy as a decision tree.

The observables of Part II. For the logician, we define once the observables that represent the classes, and the other observables that we classify in Part II. A particle on a measure space X is described by $L^2(X)$, and its position is the family of the operators M_{1_E} of multiplication by the indicator functions 1_E of Borel sets $E \subseteq X$: a state ρ places the particle in E with probability $\mu_\rho(E) = \mathrm{Tr}(\rho M_{1_E})$. For particles of Alice and of Bob on X and on Y and a Borel set $R \subseteq X \times Y$, the *relation observable* C_R is multiplication by 1_R on $L^2(X) \otimes L^2(Y) = L^2(X \times Y)$; it represents the event "Alice's value and Bob's value are related by R", and its expectation is $\varphi_{C_R}(\rho, \sigma) = (\mu_\rho \otimes \mu_\sigma)(R)$ [DI26b, Lemma 7.1]. For countable sets A and B, with standard bases $(|a\rangle)_{a\in A}$ of $\ell^2(A)$ and $(|b\rangle)_{b\in B}$ of $\ell^2(B)$, and a function $\varphi\colon A \times B \to [0,1]$, the *diagonal observable* $C_\varphi = \sum_{a\in A,\, b\in B} \varphi(a,b)\, |a\rangle\langle a| \otimes |b\rangle\langle b|$ on $\ell^2(A) \otimes \ell^2(B)$ reads $\varphi(a,b)$ on the pair of labels (a,b).

- *Photon numbers.* A mode of light is an oscillator, as in the Aria, and its energy eigenvector e_n, for $n \in \mathbb{N}$, is the state with n photons; the comparison of photon numbers is the energy comparison P of Variatio 1, the relation observable of $\{(k,l) : k < l\}$ on $\mathbb{N} \times \mathbb{N}$.

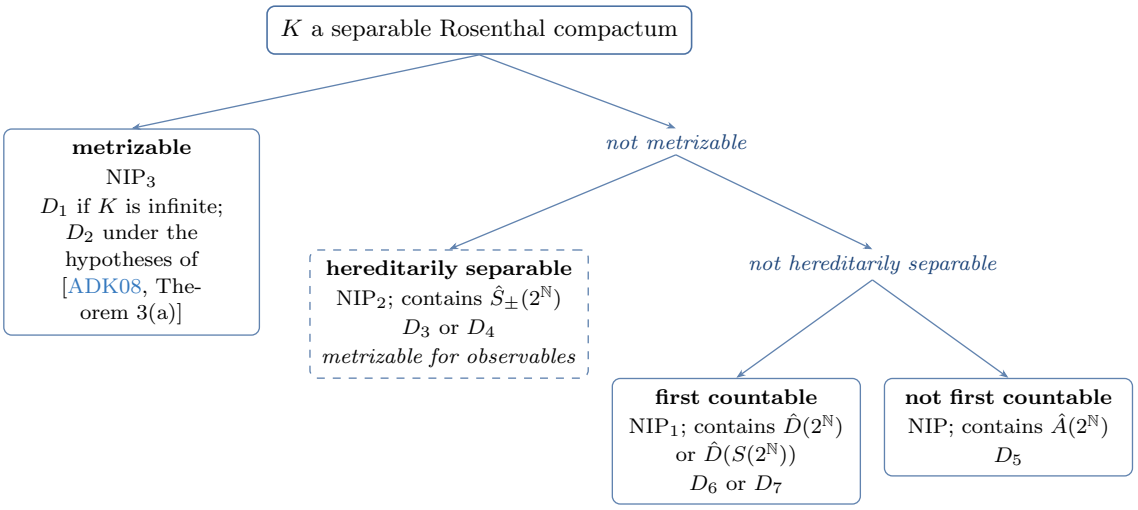


FIGURE 4. Todorčević's trichotomy as a decision tree, with the canonical compacta and the minimal families of the heptachotomy (see Variatio 15) at its leaves (compare [DIMW+26, Section 4.2]). For observables the dashed leaf merges with the metrizable one (see Theorem 19.1).

- *Positions.* The comparison of positions is the relation observable of $\{(x, y) : x < y\}$ for particles on a line, or in a box $[0, 1]$; a particle *spread over* an interval I is in the state $|I|^{-1}|1_I\rangle\langle 1_I|$, where $|I|$ is the length of I. The *quadratures* of a mode are $Q_\theta = \cos\theta\, Q - i \sin\theta\, d/dq$ on $L^2(\mathbb{R})$, for $\theta \in [0, 2\pi)$, where Q, the position, is multiplication by the variable q; each of them is unitarily equivalent to Q, so comparing quadratures is unitarily equivalent to comparing positions.
- *Light cones.* In units in which light has speed 1, an event (t, ξ) of a $(1 + 1)$-dimensional spacetime with $t \leq 0$ can influence a detection at the point y of Bob's time slice $t = 0$ exactly when $|y - \xi| \leq -t$, that is, when $a \leq y \leq b$ for the points $a = \xi + t$ and $b = \xi - t$ where the two light rays through the event cross the slice. The causal-order observable is the relation observable of $\{(a, b, y) : a \leq y \leq b\}$ on $[0, 1]^2 \times [0, 1]$.
- *Phases.* For particles on a circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, the phase comparison is the relation observable of $\{(x, y) : y - x \in (0, \frac{1}{2}) \bmod 1\}$, the event "Bob's angle is ahead of Alice's by less than half a turn".
- *Spin chains.* Alice holds a chain of spins of variable finite length, with Hilbert space $\ell^2(2^{<\mathbb{N}})$, where $2^{<\mathbb{N}}$ is the set of finite sequences of zeros and ones; for $t \in 2^{<\mathbb{N}}$, the basis vector e_t is the chain of length $|t|$ whose spin at site n, for $n < |t|$, is up exactly when $t(n) = 1$. In *scanning*, or reading site by site, Bob's label is a site n and the observable reads 1 when Alice's spin at n is up. In *template matching*, Bob's label is a finite reference configuration u and the observable reads 1 when Alice's chain is an initial segment of u. The *combined* observable offers Bob both kinds of label. All three are diagonal observables.
- *Fringes.* The *wavenumber* of monochromatic light is $k = 2\pi/\lambda$, with λ the wavelength. Behind two slits, the two paths to a point x of a distant screen differ in phase by kx, in suitable units, so the amplitudes add to a multiple of $1 + e^{ikx}$ and the probability density is proportional to $|1 + e^{ikx}|^2 = 2(1 + \cos kx)$. Alice's register holds the wavenumber and Bob's the detector position, and the fringe observable is $C = \sum_n |e_n\rangle\langle e_n| \otimes M_{f_n}$ on

Class	**Witness**	**Response interpretation**	**Realization**
stable	continuous	idealizations commute	swap test
NIP_3	metrizable	one countable set of probes separates all responses	photon numbers; spin chains
$\mathrm{NIP}_2\backslash\mathrm{NIP}_3$	split interval	mixtures add distinguishable midpoints	absent for full convex observable compacta
$\mathrm{NIP}_1\backslash\mathrm{NIP}_2$	duplicate	each response has its own countable determining probes	positions; quadratures; Ising family with offsets
$\mathrm{NIP}\backslash\mathrm{NIP}_1$	$\hat{A}(2^{\mathbb{N}})$	some response has no countable determining set of probes	the causal order; phases on a circle
not NIP	$\beta\mathbb{N}$	every finite pattern at a fixed positive gap	interference fringes

TABLE 4. The topology of response compacta and its interpretation through probes. The NIP levels are nested, and the stable row lies within NIP_3; the four middle rows distinguish the general classes associated with Todorčević's trichotomy. For full NIP observable response compacta, convexity excludes $\mathrm{NIP}_2\backslash\mathrm{NIP}_3$ (see Theorem 19.1). Selected tree closures can still occupy that class (see Variatio 15). The Ising entry concerns the family with offset fields when spontaneous magnetization is positive (see Variations 17 and 18).

$\ell^2(\mathbb{N})\otimes L^2[0,2\pi]$, where $(e_n)_{n\in\mathbb{N}}$ is the standard basis of $\ell^2(\mathbb{N})$ and each M_{f_n} is multiplication by $f_n(x) = \frac{1}{2}(1+\cos k_n x)$, for a sequence $(k_n)_{n\in\mathbb{N}}$ of wavenumbers.

- *The Ising family.* The example of Variatio 17 is a family of response functions, indexed by finite samples with offsets and by applied fields, to which the classification applies through the pointwise closure of the family.

For the physicist or operator theorist. Let C be a bipartite observable, and let $\mathcal{M}_C$ be its two-sorted structure (see Variatio 1). The compactum classified here is the space of idealized responses of C, that is, of the φ-types over a large elementary extension of $\mathcal{M}_C$ that are approximately finitely satisfiable in $\mathcal{M}_C$ (see Variatio 13), the coheirs of Variatio 1; by Theorem 11.1 it is a Rosenthal compactum when C is NIP. Its topology is that of pointwise convergence on the parameters, so metrizability says that the values at countably many parameters determine every such type, and first countability says that each type is determined, among all of them, by countably many parameters chosen for it. When metrizability fails, the trichotomy names the configuration of types that must be present: split pairs of types that differ at a single parameter, an uncountable discrete family of types, or a single type that is the limit of continuum many types isolated from one another.

For the logician. Metrizability has an experimental meaning: there is a fixed countable list of Bob's probes whose answers determine the response of every idealized preparation of Alice to every probe. For the comparison of photon numbers such a list exists (see Theorem 16.1), because a response can jump only at the countably many points of the spectrum. For the comparison of positions no countable list suffices, since an idealized response can jump at any point of the line, but each single idealized response is determined by probes at countably many points chosen for it. For the causal order even this fails: an event whose light cone has shrunk to a point c of Bob's time slice is distinguished only by probes that assign positive probability to c itself (see Variatio 16).

Each class of the trichotomy is thus a degree of failure of the double limit condition that stops short of the independence property, and for observables one of the classes is empty (see Theorem 19.1).

Variatio 15. a 1 Clav. Canone alla Quinta. Andante

The heptachotomy

A tree of preparations branches in two at every node and gathers continuum many paths into a single countable family. In the realizations below, the responses along each branch converge, and their limits belong to the pointwise closure of that selected family. The logician classifies the indexed families that lead to a given compactum; the physicist identifies the compacta reached by trees of photons, particles and spins. Todorčević's trichotomy (see Variatio 14) identifies canonical compacta, and the refinement of Argyros, Dodos and Kanellopoulos records how indexed families sit inside their closures. In this variation we realize its seven minimal families by selected trees of physical preparations. The full response compactum of the observable also contains responses of mixtures and is convex; the closure of the selected tree need not be. The observables themselves are placed in their classes by Theorem 16.1 of the next variation.

The heptachotomy describes families of pairwise distinct functions indexed by the dyadic tree $2^{<\mathbb{N}}$ of finite sequences of zeros and ones. Two families $\{f_t : t \in 2^{<\mathbb{N}}\}$ and $\{g_t : t \in 2^{<\mathbb{N}}\}$ are *equivalent* if the map $f_t \mapsto g_t$ extends to a homeomorphism of their pointwise closures, and such a family that is relatively compact in the first Baire class, the class of pointwise limits of sequences of continuous functions (see Variatio 11), is *minimal* if it is equivalent to its restriction to every dyadic subtree. There are exactly seven minimal families $D_1, \dots, D_7$ up to equivalence, and every family of pairwise distinct functions indexed by $2^{<\mathbb{N}}$ that is relatively compact in the first Baire class of a Polish space contains, along a *regular* dyadic subtree, that is, one each of whose levels lies in a single level of $2^{<\mathbb{N}}$, a family equivalent to one of them [ADK08, Theorem 2]. Their closures fall into the classes of the trichotomy (see Variatio 14): those of D_1 and D_2 are metrizable, namely the one-point compactification $A(2^{<\mathbb{N}})$ of the countable discrete set $2^{<\mathbb{N}}$ and the compactum $2^{\leq\mathbb{N}}$ of the finite and infinite sequences of zeros and ones, respectively; those of D_3 and D_4 are, respectively, the split Cantor sets $\hat{S}_+(2^{\mathbb{N}})$ and $\hat{S}_-(2^{\mathbb{N}})$, those of D_6 and D_7 are, respectively, the duplicates $\hat{D}(2^{\mathbb{N}})$ and $\hat{D}(S(2^{\mathbb{N}}))$, and that of D_5 is the compactum $\hat{A}(2^{\mathbb{N}})$.

Family	Closure	The picture	The physics
D_1	$A(2^{<\mathbb{N}})$	escape to infinity	photon number states
D_2	$2^{\leq\mathbb{N}}$	infinite configurations	a spin chain read site by site
D_3	$\hat{S}_+(2^{\mathbb{N}})$	approach from the right	a particle in a box
D_4	$\hat{S}_-(2^{\mathbb{N}})$	approach from the left	a particle in a box
D_6	$\hat{D}(2^{\mathbb{N}})$	twins over a Cantor set	a spin chain, sites and templates
D_7	$\hat{D}(S(2^{\mathbb{N}}))$	approach from both sides	a particle in a box, half on each side
D_5	$\hat{A}(2^{\mathbb{N}})$	a point-supported limit	light cones; templates

TABLE 5. The seven minimal families, grouped by class of Todorčević's trichotomy.

Theorem 15.1 ([DI26b, Theorem 8.5])**.** *Each of the seven minimal families is realized by a tree of pure preparations of a physically natural observable: the family D_1 by photon number states, compared by photon number; the family D_2 by a spin chain read site by site; the families D_3 and D_4 by a particle in a box spread over a gap of a Cantor set, compared by position; the family D_5 by a spin chain matched against templates, and also by the causal order of spacetime; the family D_6 by a spin chain read both site by site and against templates; and the family D_7 by a particle in a box spread half over a short interval on each side of a point.*

The realizations display what distinguishes the families (see Figure 5). The responses of photon number states are isolated points with a single idealized limit, their only accumulation point (D_1). The idealized limits of the spin chains are the infinite spin configurations, which countably many site probes tell apart (D_2). A particle that approaches a point from only one side produces one half of a split pair (D_3, D_4), and a particle that approaches it from both sides at once produces the midpoint (D_7). Light-cone sections that contain a fixed point and shrink to it, or spin chains that follow longer and longer prefixes of a fixed infinite configuration, yield idealized responses whose profiles on point probes are supported at that point or configuration (D_5). The full response integrates this profile against Bob's probe measure, so it reads the mass assigned to that point.

The families D_3 and D_4 appear although the hereditarily separable non-metrizable class is empty for observables. There is no contradiction: the closure of the family itself is a split Cantor set, but the idealized responses of the whole observable include the mixtures, and, by Theorem 19.1, they form a larger compactum, which is not hereditarily separable.

For the physicist or operator theorist. A tree $(\rho_t)_{t\in 2^{<\mathbb{N}}}$ of preparations is a tree of elements of Alice's sort in the structure $\mathcal{M}_C$ that a bipartite observable C defines. Its pointwise response closure is taken on Bob's probes, including idealized probes, with Bob's system idealized first (see Variatio 13). On actual probes, an ultrafilter limit along nodes $(t_n)_{n\in\mathbb{N}}$ restricts to the φ-type over $\mathcal{M}_C$ of the ultrapower element $[\rho_{t_n}]_{\mathcal{U}}$, where $\mathcal{U}$ is an ultrafilter on $\mathbb{N}$ (see Variatio 1); two idealized responses may have the same restriction while differing on idealized probes. When C is NIP and the responses of the nodes are pairwise distinct, passing

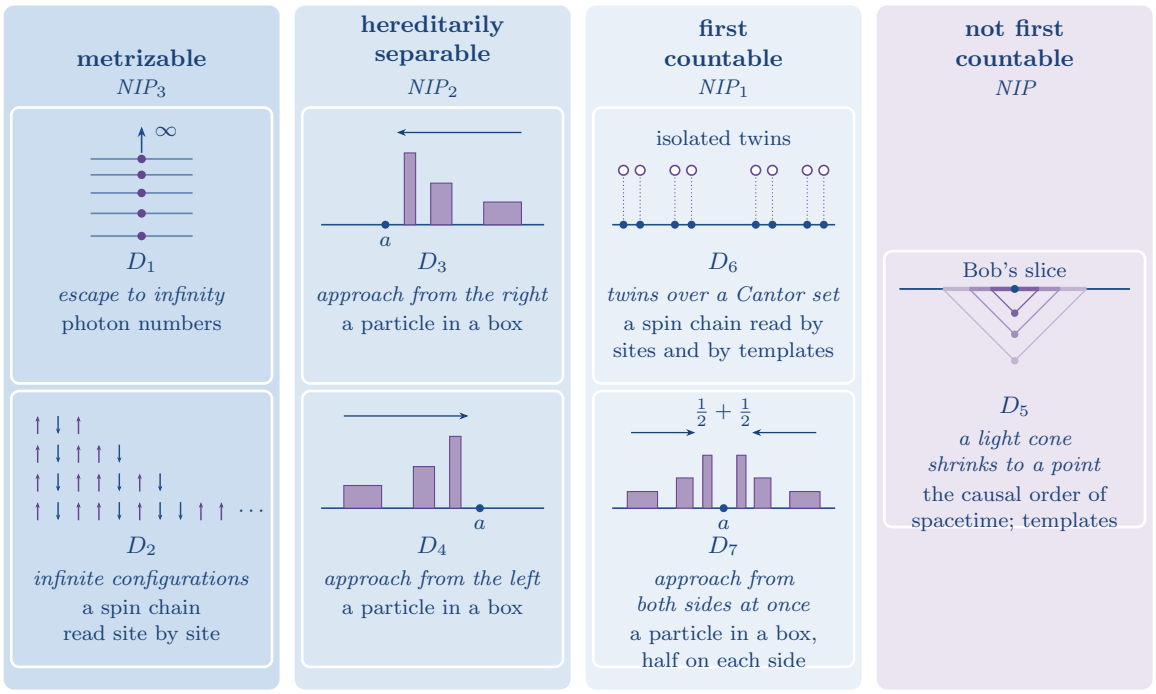


FIGURE 5. The seven minimal families of the heptachotomy, arranged by the class of Todorčević's trichotomy to which their closures belong, each with the physical picture of its realization (see Theorem 15.1): how a tree of preparations approaches its idealized limits.

to a regular dyadic subtree, by a Ramsey theorem for trees, plays the role that the extraction of indiscernible sequences plays in model theory, and the heptachotomy says that, after this thinning, the indexed family of responses is equivalent to one of the seven minimal families. The class of the trichotomy to which its closure belongs, recorded in Table 5, describes which countable collections of probes suffice to distinguish the responses.

For the logician. In each realization the nodes of the dyadic tree are pure states of Alice's system, and the idealized limits are taken along the branches. For photon numbers the nodes are states with definite, pairwise distinct numbers of photons, so every branch escapes to infinite photon number and all branches share one idealized limit (D_1); for a spin chain read site by site the node t is the chain e_t of length $|t|$ whose spin at each site $i < |t|$ is up exactly when $t(i) = 1$, the limit along a branch $x \in 2^{\mathbb{N}}$, an infinite sequence of zeros and ones, is the infinite chain x, and its response to the probe at site n is the spin $x(n)$ (D_2). For a particle in the box $[0, 1]$, compared by position with Bob's particle, the nodes are wave functions spread uniformly over short intervals that shrink, along each branch, to a point of a Cantor set, from one side (D_3, D_4) or half on each side (D_7); the idealized limit answers like a particle at that point, except on probes concentrated at the point itself, where the side of approach determines the answer.

The seven families describe the topology of the closure of a selected tree of responses, together with its indexing. All seven occur in the physical realizations above, including the two whose closures are hereditarily separable but not metrizable; mixtures place these inside

larger convex response compacta. It is not known whether the comparison of positions, which realizes D_7, also realizes D_6 (see Question 4).

Variatio 16. a 1 Clav. Ouverture

Physical orders in every class

Each of the three classes of Todorčević's trichotomy that occur for observables contains a physical order, and the interference fringes of the double slit (see Variatio 11) lie beyond NIP. In the examples below, what separates the classes is the way in which the idealized responses depart from continuity: by jumps confined to countably many points, by jumps that may occur at any point of a continuum, or by deviations concentrated at single points. In this variation we state the result and then treat the examples in turn.

Theorem 16.1 ([DI26b, Theorems 7.8, 7.10, 7.12 and Propositions 7.9, 7.13])**.**

(a) *The comparison of two photon numbers is unstable and NIP$_3$.*
(b) *The comparison of two positions on a line, or of two quadratures of light, is NIP$_1$ but not NIP$_2$; its idealized responses form the Helly space of monotone functions.*
(c) *The causal order of a $(1+1)$-dimensional spacetime, "Alice's event can influence Bob's detection", is NIP but not NIP$_1$.*
(d) *The comparison of two phases on a circle, "Bob's angle is ahead of Alice's by less than half a turn", is NIP but not NIP$_1$.*
(e) *The interference fringes of wavenumbers 8^n, for $n \geq 1$, are not NIP: the closure of their responses contains a homeomorphic copy of $\beta\mathbb{N}$.*

Photons. The idealized responses of a comparison are monotone, and a monotone function has at most countably many jumps, each with two sides. For discrete spectra, every idealized response jumps only at the countably many points of the spectrum, so countably many probes separate them. Every comparison is moreover uniformly NIP, in the sense that its finite independent sets have bounded size at every margin, and the bounds do not depend on the Hilbert spaces: replacing the distribution of the compared quantity in Bob's state by m equally weighted quantile points changes every response by at most $1/m$, and only polynomially many patterns survive the discretization [DI26b, Proposition 7.3 and Corollary 7.4].

Positions. For continuous spectra every point can be a jump, approached from either side, and the idealized responses form the Helly space.

Light cones. In $1+1$ dimensions, an event of Alice influences a detection of Bob exactly when the detection lies between the two points where the light rays through the event cross Bob's time slice. Light-cone sections that contain a fixed point and shrink to it yield an idealized response whose profile on point probes is supported at that point; on an arbitrary probe measure, the response reads the mass assigned to that point. This is the picture of the compactum $\hat{A}(2^{\mathbb{N}})$ that witnesses the failure of first countability.

Phases. An antipodal pair of angles on the circle, one approached from behind and the other from ahead, produces an idealized response whose profile on point probes equals $\frac{1}{2}$, the response of the uniform state, except at those two points; on a line, a monotone profile that is constant away from finitely many points is constant.

Fringes. For each $n \geq 1$, the response $\frac{1}{2}(1+\cos 8^n x)$ of a photon prepared with wavenumber 8^n to a detector at position x depends only on $8^n x$ modulo 2π, that is, on the base-8 digits of $x/2\pi$ from the $(n+1)$st on, and these digits can be chosen to realize every finite pattern.

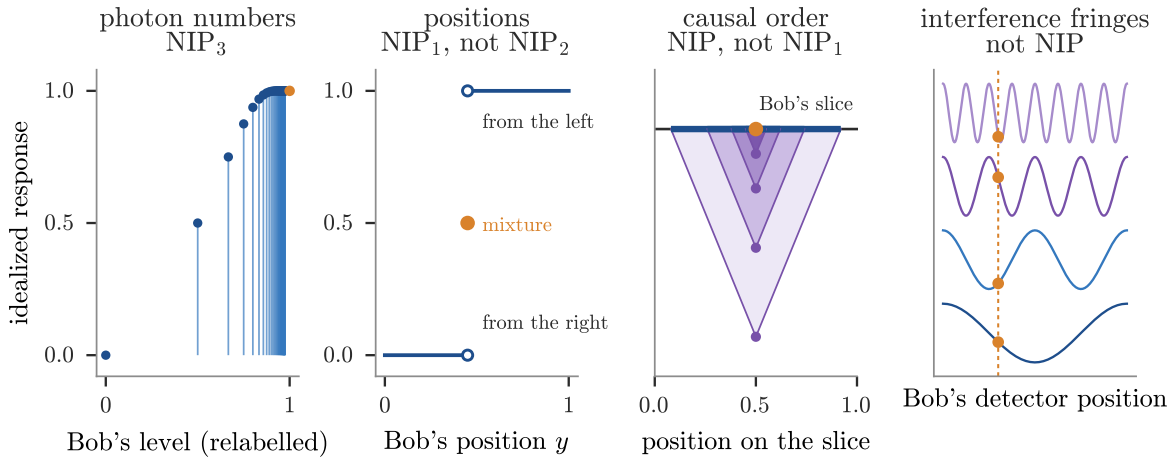


FIGURE 6. Pictures for parts (a), (b), (c) and (e) of Theorem 16.1. Photon numbers: a monotone response on the relabeled spectrum $\{1 - \frac{1}{l+1} : l \in \mathbb{N}\}$, whose points accumulate only at 1. Positions: a particle concentrating at a point from the right or from the left yields two idealized responses that differ only there, and the response of the equal mixture of the two idealized preparations is the midpoint of the pair. Causal order: events closer and closer to Bob's time slice, whose light cones meet it in intervals shrinking to a single point. Fringes: at a single detector position, fringes of increasing wavenumber realize a prescribed pattern (shown here $1, 2, 4, 8$; the proof uses 8^n).

For the physicist or operator theorist. Each observable of parts (a)–(d) of Theorem 16.1 reads a relation of a classical structure, and the types of that structure display the classes. In the structure of the causal order, the two sorts are $[0,1]^2$, the pairs (a,b) of points where the two light rays through an event of Alice cross Bob's time slice, and $[0,1]$, the detector positions y; the relation, which we write $\varphi((a,b),y)$, is $a \le y \le b$. For a point c with $0 < c < 1$ and a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the element $[(c-1/n,\ c+1/n)]_{\mathcal{U}}$ of the ultrapower, the class of a sequence indexed by the integers $n > 1/\min(c, 1-c)$, is a pair of endpoints of an interval of infinitesimal length around c, and its φ-type over the real points is $y \mapsto 1_{\{c\}}(y)$; these types, one for each c, are isolated from one another and accumulate at the zero function, which therefore has no countable base of neighborhoods, the configuration of $\hat{A}(2^{\mathbb{N}})$. For the comparison of positions the corresponding types are the split pairs of Variatio 12, and the idealized responses of the observable, which include those of mixtures, are the monotone functions of the Helly space. In the quantum structure $\mathcal{M}_C$ of each such observable C (see Variatio 1), these types appear as the restrictions of idealized responses to Bob's idealized probes concentrated at points, with Bob idealized first (see Variatio 13).

For the logician. In a $(1+1)$-dimensional spacetime with coordinates (t, ξ) and light speed 1, a signal emitted at an event reaches only the events inside its future light cone, so the event (t, ξ) with $t \le 0$ can affect a detector at the point y of the slice $t = 0$ exactly when $\xi + t \le y \le \xi - t$. The causal-order observable asks this question of an event of Alice and a detector position of Bob, both described by quantum states, and answers yes with the probability that the two positions satisfy the inequality. For events whose light-cone sections contain a fixed point c and shrink to it as the events approach the slice, the idealized limit has response $\nu(\{c\})$ at a probe represented by the probability measure ν; since no actual position

distribution assigns positive probability to a single point, this idealized response is detected only by idealized probes. Each probe measure has only countably many atoms, so continuum many such point-supported profiles cannot be monitored by countably many probes. For the phases the corresponding feature is that a circle, unlike a line, carries an idealized response whose profile on point probes equals $\frac{1}{2}$ except at two antipodal points.

Physical orders thus occupy the three classes available to full convex NIP response compacta, and the fringes lie beyond NIP. Each order considered here is unstable; the topology of its response compactum distinguishes these failures of the double limit condition. These realizations have commutative probe algebras, so their responses can be described by integration against probability distributions. The general classification applies to bounded self-adjoint bipartite observables without that commutativity assumption, while the thermodynamic families of the next two variations provide a different application of the same topology of pointwise closures.

Variatio 17. a 2 Clav.

Orders of limits in phase transitions

For spontaneous magnetization, the order of the infinite-volume and vanishing-field limits determines whether a nonzero response remains. In this variation we place that order of limits in the classification of response families. The distinction between stability and instability already appears without translating the field; offsets that move the transition point reveal the finer topological class.

The Ising model is the basic model of a ferromagnet. Fix a dimension $d \geq 1$. On a finite box Λ of the lattice $\mathbb{Z}^d$ with N sites and periodic boundary conditions, a configuration $s \in \{-1, 1\}^\Lambda$ has energy $E(s) = -J \sum_{x \sim y} s_x s_y - h \sum_x s_x$, where the first sum runs over the pairs of neighboring sites, the coupling J is positive, and the parameter h is the applied field; at inverse temperature $\beta > 0$, the *Gibbs state* assigns to s the probability $Z_N^{-1} e^{-\beta E(s)}$, where the normalizing constant $Z_N = \sum_s e^{-\beta E(s)}$ is the *partition function*, and the magnetization $m_N(h)$ is the expected mean spin in this state. The *spontaneous magnetization* is $m^* = \lim_{h \downarrow 0} \lim_N m_N(h)$, along boxes exhausting $\mathbb{Z}^d$; for $d \geq 2$ there is a critical β_c with $m^* > 0$ for $\beta > \beta_c$, the *ordered phase*, and $m^* = 0$ for $\beta < \beta_c$. The *pure phases* are the infinite-volume limits of the Gibbs states as $h \downarrow 0$ and as $h \uparrow 0$, with mean spin $\pm m^*$. Spontaneous magnetization is the standard failure of the interchange of limits: the magnetization of a ferromagnet in a field h, taken to infinite volume and then to zero field, is nonzero below the critical temperature, whereas the other order yields zero. For the classification we use a parameterized response family: Alice's labels are finite sample sizes N and field offsets a, and Bob's probes are applied fields h; the response is the magnetization $m_N(h - a)$. This is a two-variable response family; its classification does not require identifying it with the full response family of a single bounded bipartite observable. In the ordered phase the offset a moves the jump of the infinite-volume magnetization from zero field to the field $h = a$; the offsets thus supply a continuum of jumps, one for each offset, and it is this continuum that the classes of Todorčević's trichotomy detect. For the Ising model in dimension $d \geq 2$, the family with offsets is stable exactly when the spontaneous magnetization vanishes, and when it is positive this family is NIP_1 but not NIP_2, its idealized responses forming a first countable compactum that is not hereditarily separable: the plus and minus phases yield two idealized responses that differ at a single point, a *split pair*, and the midpoints of these pairs, one for each offset a, form an uncountable discrete set [DI26b, Theorem 9.2]. The family $\{m_N\}$ without offsets is also stable exactly when $m^* = 0$, but its closure is metrizable in

both regimes [DI26b, Theorem 9.2]. The split pair is illustrated in Figure 7 in the mean-field (Curie–Weiss) model, where the magnetization of N spins can be computed exactly.

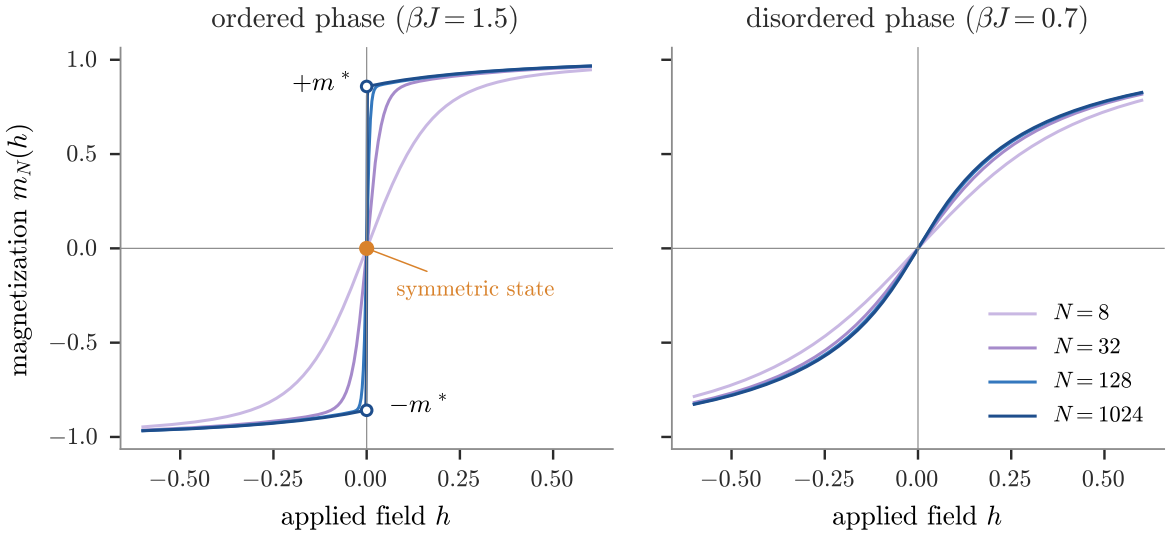


FIGURE 7. Magnetization curves of the Curie–Weiss model for increasing numbers N of spins, with the field h in units of k_BT, the product of the Boltzmann constant k_B and the temperature T. In the ordered phase the curves steepen into a jump at $h = 0$: approaching zero field from either side yields the two values $\pm m^*$, the split pair, and the symmetric state takes their midpoint 0. In the disordered phase the curves stay smooth.

For the physicist or operator theorist. The magnetizations $m_N(h - a)$ form a formula in the sense of Variatio 1, with Alice's samples, indexed by size and offset, and Bob's fields as the two sorts, and the idealized responses are its φ-types realized in ultrapowers. In the ordered phase, two of them, the plus and the minus phase at offset a, take the values m^* and $-m^*$ at the parameter $h = a$ and agree at every other parameter; they form a split pair, like the types just above and just below a real number in the rational order (see Variatio 12). Each response is monotone in h, so the family is NIP, and the types that take the midpoint value 0 at $h = a$, one for each offset a, form an uncountable discrete set, which places the family in the class $\mathrm{NIP}_1 \backslash \mathrm{NIP}_2$; in the disordered phase the idealized responses are continuous and the family is stable.

For the logician. The Ising model describes a magnet by a spin $s_x = \pm 1$ at each site of a lattice, and thermal equilibrium at inverse temperature β by the Gibbs state; the magnetization $m_N(h)$ is the expected mean spin, an odd nondecreasing smooth function of the applied field h for each finite N. The two iterated limits in question are $\lim_{h \downarrow 0} \lim_N m_N(h) = m^*$ and $\lim_N \lim_{h \downarrow 0} m_N(h) = 0$: a finite magnet in zero field has no preferred direction, while an infinite magnet in a vanishing positive field keeps a positive magnetization. In the bipartite family, Alice's preparation is a finite sample whose field is shifted by the offset a and Bob's probe is the applied field; an idealized preparation of Alice is an infinite sample, and at the field $h = a$ it responds with m^* or $-m^*$ according to the side from which the field is

approached. These two responses are the two pure phases, and their equal mixture, whose mean spin is 0, is the midpoint.

The trichotomy thus does more than detect a phase transition: the ordered phase places the magnetization in the class $\mathrm{NIP}_1\backslash\mathrm{NIP}_2$, through the split pair of the two pure phases, and the disordered phase leaves it stable.

Variatio 18. a 1 Clav. Canone alla Sesta

A quantum ferromagnet in the classification

A magnetic field perpendicular to the axis along which the spins of an Ising magnet align makes the model quantum. In this variation we carry the results of the previous one over to the quantum model.

In a transverse field the Ising Hamiltonian on a finite box of $\mathbb{Z}^d$ with N sites (see Variatio 17) is $H_N = -J\sum \sigma_x^z\sigma_y^z - \Gamma\sum \sigma_x^x$, where each site x carries a qubit, σ_x^z and σ_x^x are the Pauli matrices $\mathrm{diag}(1,-1)$ and $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$ acting on the qubit at x (see Variatio 22), the first sum runs over the pairs of neighboring sites, and the coupling J and the strength Γ of the transverse field are positive. This Hamiltonian no longer commutes with the magnetization. The classification of Variatio 17 uses the fact that the pressure, defined below, is differentiable away from zero field; for classical ferromagnets this follows from the Lee–Yang theorem [LY52]. At inverse temperature β (see Variatio 17), the *partition function* in a field h, real or complex, is $Z_N(h) = \mathrm{Tr}\exp(-\beta H_N + h\sum_x \sigma_x^z)$, and the *pressure* is $p(h) = \lim_N N^{-1}\log Z_N(h)$ for real h; it is convex, and where it is differentiable the infinite-volume magnetization is $p'(h)$. The Lee–Yang theorem states that, for a classical Ising ferromagnet on a finite set, the partition function, as a function of a complex field h, vanishes only where $\mathrm{Re}\,h = 0$. For $\Gamma = 0$ the Hamiltonian is diagonal in the basis of configurations and is the classical Ising energy at zero field; for $\Gamma > 0$ it is not, and the Gibbs state at zero field is the density operator $e^{-\beta H_N}/\mathrm{Tr}\,e^{-\beta H_N}$.

Theorem 18.1 ([DI26b, Propositions 9.4, 9.5 and Theorem 9.6])**.** *For every N, the function Z_N satisfies $Z_N(h) \neq 0$ whenever $\mathrm{Re}\,h \neq 0$; moreover, the infinite-volume pressure $\lim_N N^{-1}\log Z_N(h)$ is real analytic for real $h \neq 0$, and the magnetization with offsets, the family of responses $m_N(h-a)$ of Variatio 17, now computed in the Gibbs state of H_N, is stable when the spontaneous magnetization vanishes and NIP$_1$ but not NIP$_2$ when it is positive.*

The idea is to approximate $\exp(-\beta H_N)$ by the Lie–Trotter formula, which turns the quantum model into a classical ferromagnet in one more dimension, to apply the classical Lee–Yang theorem to the approximants, and to pass the zero-free half-planes to the limit by Hurwitz's theorem. Earlier Lee–Yang theorems for quantum spin systems exclude zeros only in a smaller region (see the historical remarks); we have not found the statement for the whole half-plane in the literature, but its proof combines standard tools, and the result may well be known to experts. The Griffiths–Hurst–Sherman inequality offers another route to the differentiability.

The Lie–Trotter formula is $e^{A+B} = \lim_n (e^{A/n}e^{B/n})^n$ for square matrices A and B of the same size, where n runs through the positive integers. Applied to $-\beta H_N + h\sum_x \sigma_x^z$, it writes $Z_N(h)$ as a limit of sums over configurations of n stacked copies of the lattice; each factor $e^{\beta\Gamma\sigma_x^x/n}$ couples neighboring copies ferromagnetically, because, with $\varepsilon = \beta\Gamma/n$ and $|s\rangle$ the eigenvector of σ^z with eigenvalue $s \in \{-1,1\}$, the matrix entries are $\langle s|e^{\varepsilon\sigma^x}|s'\rangle = c\,e^{Kss'}$ for $K = -\frac{1}{2}\log\tanh\varepsilon > 0$ and $c = (\sinh\varepsilon\cosh\varepsilon)^{1/2}$. The result is a classical Ising ferromagnet on the lattice times a cycle of length n, to which the Lee–Yang theorem applies.

Figure 8 shows the theorem on a small ring of spins.

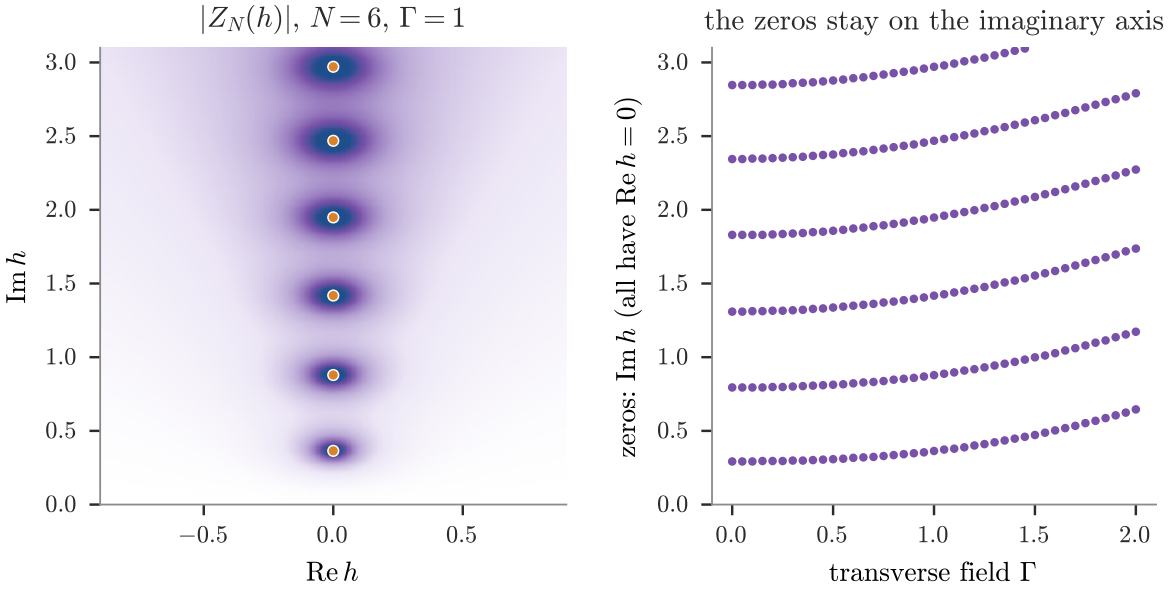


FIGURE 8. Zeros of the partition function $Z_N(h)$ of the Ising model in a transverse field on a ring of $N = 6$ spins, with $\beta = J = 1$, computed by exact diagonalization. Left: the shading is $\log\big(|Z_N(h)|/Z_N(\operatorname{Re} h)\big)$ for $\Gamma = 1$; the dark spots are the zeros, and all of them lie on the imaginary axis. Right: the zeros as the transverse field grows from the classical model, $\Gamma = 0$; they move, but none leaves the axis.

For the physicist or operator theorist. The formula being classified is the family $h \mapsto m_N(h - a)$, with sample size N and field offset a as Alice's labels and the applied field h as Bob's parameter (see Variatio 17). Each response is nondecreasing in h; at fixed thresholds $u < v$, two responses cannot realize both a low/high pattern and its high/low reversal, so the family is NIP. The finer classification uses the closure: analyticity of the pressure away from zero field supplies the regularity of the limiting magnetization (see Theorem 18.1). When spontaneous magnetization is positive, translating the field supplies split pairs at a continuum of transition points, and their midpoints form the uncountable discrete set that excludes NIP_2; the closure is first countable, so the family is NIP_1 but not NIP_2. When spontaneous magnetization vanishes, all idealized responses are continuous and the family is stable. Without offsets, the family remains NIP_3 in both cases [DI26b, Theorem 9.6].

For the logician. In the transverse-field model each site carries a qubit, the term $-J\sigma_x^z\sigma_y^z$ favors neighbors aligned along the z axis, and the term $-\Gamma\sigma_x^x$ favors at each site the equal superposition $(|0\rangle + |1\rangle)/\sqrt{2}$ of the spin-up state $|0\rangle$ and the spin-down state $|1\rangle$, a state that is not an eigenvector of σ_x^z; since the two terms do not commute, the Gibbs state is a density operator that is not diagonal in the basis of spin configurations. The magnetization is $N^{-1}\partial_h \log Z_N(h)$, the expected value of $N^{-1}\sum_x \sigma_x^z$ in the density operator $\exp(-\beta H_N + h\sum_x \sigma_x^z)/Z_N(h)$, and in infinite volume it is the derivative of the pressure. A phase transition occurs at a field h at which the pressure fails to be analytic; Theorem 18.1 excludes such points away from zero field, so the magnetization can jump only at zero field, where the two pure phases meet.

When the spontaneous magnetization is positive, the quantum magnet fails the double limit condition exactly as the classical one does, and for the same reason: a convex pressure, real analytic away from zero field.

Variatio 19. A 1 Clav.

Convexity forces metrizability

A split pair consists of two idealized responses that differ at a single point, obtained from preparations that approach the point from the left and from the right. The equal mixture of the two idealized preparations has a third response, the midpoint of the pair. In this variation we describe the second of the three effects of mixing recorded in Table 1. In Theorem 19.1 below, we show that, for NIP observables, hereditary separability of the idealized responses implies metrizability, and only three of the four classes of Todorčević's trichotomy therefore occur for observables.

Theorem 19.1 ([DI26b, Theorems 6.6 and 6.9])**.** *A separable convex Rosenthal compactum that is hereditarily separable is metrizable. Consequently, an NIP observable is NIP$_3$, or NIP$_1$ but not NIP$_2$, or not NIP$_1$: Todorčević's trichotomy has three classes for observables, witnessed, in the last two cases, by one of the minimal families D_6 and D_7 and by the minimal family D_5 of the heptachotomy (see Variatio 15), respectively.*

The idea is the following: a hereditarily separable non-metrizable Rosenthal compactum contains a copy of one of the split Cantor sets $\hat{S}_+(2^{\mathbb{N}})$ and $\hat{S}_-(2^{\mathbb{N}})$ of Todorčević (see Variatio 14), which contains uncountably many split pairs. Following limits of such pairs inside the convex set of idealized responses, the proof in our paper on NIP [DI26b] produces their midpoints and shows that the midpoints form an uncountable discrete subspace, which is not separable. For the comparison of positions, the resulting space is the Helly space.

For the physicist or operator theorist. In the structure $\mathcal{M}_C$ of a bipartite observable C (see Variatio 1), Alice's sort $\mathfrak{D}(H_1)$ is convex and the expectation φ_C is affine in Alice's variable, so every ultrapower of $\mathcal{M}_C$ along a nonprincipal ultrafilter on $\mathbb{N}$ contains, with any elements $[\rho_n]$ and $[\rho'_n]$, the element $[\frac{1}{2}(\rho_n + \rho'_n)]$, whose φ-type is the average of theirs; the idealized responses therefore form a convex compact set of φ-types. For a general NIP formula no such convexity is available, and the formula $x < y$ of the rational order has the split interval as its space of φ-types (see Variatio 12). In Theorem 19.1 we show that convexity excludes this class: the averages of the two types of each split pair are new types, each isolated among them by the parameter at which its pair splits, and they form an uncountable discrete set. In particular, the space of probability measures on the split interval is not hereditarily separable, contrary to a remark of Godefroy [God80].

For the logician. Every preparation procedure can be randomized: choosing with probability $\frac{1}{2}$ between two procedures and forgetting the choice produces the mixture $\frac{1}{2}(\rho+\rho')$, whose expectations are the averages of those of ρ and ρ'. For the comparison of positions, mixing a particle spread just to the left of a point r with one spread just to the right, and idealizing, yields a preparation that answers $\frac{1}{2}$ on Bob's particle concentrated at r, with Bob idealized first, and agrees with both one-sided limits on probes concentrated away from r; there is one such preparation for each point of the line. These preparations form an uncountable discrete set of idealized responses, so no quantum observable has the split interval as its space of

idealized responses, and the comparison of positions, the closest physical counterpart of the rational order, is NIP_1 but not NIP_2, with the Helly space as its space of idealized responses (see Theorem 16.1).

In Part I mixtures stood in for idealized preparations. Here they add idealized responses of their own, the midpoints, and the midpoints decide the class.

Part III. Simplicity

Variatio 20. a 2 Clav.

Simplicity

Each dividing line forbids a combinatorial configuration. Stability forbids the order property and NIP forbids the independence property; simplicity forbids a tree of instances of a formula in which every branch is consistent while, at every node, the infinitely many immediate successors are inconsistent in groups of a fixed finite size. Of the three prohibitions, only this one is compatible with the independence property: a simple theory may have it, as the theory of the random graph does (see Theorem 22.1 and Variatio 21), and still possess a notion of independence of types governed by axioms close to those of stability. In this variation we state the theorem that characterizes simplicity by such axioms. Simplicity is first studied in two first-order structures, the random graph and the Pauli group; observables reappear in Variatio 28, where simplicity is studied in Keisler's real-valued logic [Kei23].

A theory is *complete* if it is the theory of a structure. A *monster model* of a complete theory T is a model $\mathbb{M}$ of T that is κ-saturated and strongly κ-homogeneous for a very large cardinal κ: every type over a set of fewer than κ elements, a *small* set, is realized in $\mathbb{M}$, and two small tuples with the same type are conjugate by an automorphism of $\mathbb{M}$. An ultrapower of a structure in a countable language along a nonprincipal ultrafilter on $\mathbb{N}$, such as the ultrapower of Variatio 1, has the first property for countable sets. We write $\mathrm{tp}(a/B)$ for the type of a over B. Let $\mathbb{M}$ be a monster model of a complete first-order theory T, and let $\varphi(x,y)$ be a formula. A sequence $(b_j)_{j\in\mathbb{N}}$ is *k-inconsistent* for φ, where $k\geq 2$ is an integer, if no k of the formulas $\varphi(x,b_j)$ have a common solution. The formula has the *tree property* if there are an integer $k\geq 2$ and a tree $(b_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$, indexed by the finite sequences of natural numbers, whose branches are consistent and in which, at every node, the immediate successors of the node form a k-inconsistent sequence. The formula has the *tree property of the second kind* if there are an integer $k\geq 2$ and an array $(b_{ij})_{i,j\in\mathbb{N}}$ whose rows are all k-inconsistent and in which every choice of one entry per row is consistent. A theory is *simple* if no formula has the tree property [She80]. Stable theories are simple, and a simple theory without the independence property is stable [She90, She80], so the simple unstable theories are exactly the simple theories with the independence property.

Simplicity has a structural characterization, due to Kim and Pillay, which is the usual means of proving that a theory is simple. They proved that if a ternary relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ on small subsets of $\mathbb{M}$ is invariant under automorphisms, symmetric, monotone ($A \mathop{\smile\!\!\!\!\!\!|}_C BD$ implies $A \mathop{\smile\!\!\!\!\!\!|}_C B$, where BD denotes $B\cup D$) and transitive ($A \mathop{\smile\!\!\!\!\!\!|}_C B$ and $A \mathop{\smile\!\!\!\!\!\!|}_{CB} D$ imply $A \mathop{\smile\!\!\!\!\!\!|}_C BD$), has finite character ($A \mathop{\smile\!\!\!\!\!\!|}_C B$ if and only if $a \mathop{\smile\!\!\!\!\!\!|}_C b$ for all finite $a\subseteq A$ and $b\subseteq B$) and local character (for each finite a and each B there is $B_0\subseteq B$ with $|B_0|\leq|T|$, the cardinality of the language of T plus $\aleph_0$, and $a \mathop{\smile\!\!\!\!\!\!|}_{B_0} B$), and satisfies extension (for all a and $C\subseteq B$ there is a' with $\mathrm{tp}(a'/C)=\mathrm{tp}(a/C)$ and $a' \mathop{\smile\!\!\!\!\!\!|}_C B$) and the *independence theorem over models*, then the theory T is simple and the relation is nonforking independence [KP97]: the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds exactly when $\mathrm{tp}(A/CB)$ does not fork over C, in the sense of

Variatio 6. Stable theories have the same kind of characterization, due to Harnik and Harrington [HH84], with *stationarity over models*, the uniqueness of nonforking extensions of a type over a model, in place of the independence theorem; stationarity implies the independence theorem. Uniqueness is thus what separates stability from simplicity. Where uniqueness is lost, nonforking extensions of one type chosen separately over two sets independent over a model might conflict, and the independence theorem, the substantial axiom, ensures that they do not: if M is a model, $B \mathop{\smile\!\!\!\!\!\!|}_M D$, and two realizations a_1 and a_2 of the same type over M satisfy $a_1 \mathop{\smile\!\!\!\!\!\!|}_M B$ and $a_2 \mathop{\smile\!\!\!\!\!\!|}_M D$, then some a realizes $\mathrm{tp}(a_1/MB) \cup \mathrm{tp}(a_2/MD)$ with $a \mathop{\smile\!\!\!\!\!\!|}_M BD$. Over a model, extension is witnessed by the coheirs of Variatio 1; in every theory, the two orders of the iterated ultrapower produce nonforking amalgams; in a stable theory they produce the only one (see Variatio 6); and in a simple unstable theory other nonforking amalgams exist, whose combination the independence theorem governs.

For the physicist or operator theorist. Simplicity is a property of a theory, hence of all elementary extensions of a structure at once, and its configurations are sought in the monster model, an elementary extension, far larger than an ultrapower over $\mathbb{N}$, in which every type over a small set is realized. In the language of Variatio 1, a stable theory has over a model exactly one independent amalgam of two types, whereas a simple theory may have many nonforking amalgams, and the Kim–Pillay theorem replaces uniqueness by the independence theorem: nonforking extensions of one type over a model to two sets independent over the model can be realized together. The tree property is the obstruction: a tree with consistent branches and k-inconsistent siblings yields types that fork over every subset of size at most $|T|$ of their domain, and nonforking independence thereby loses its local character.

For the logician. The structures of Part III come from the measurement of Pauli strings (see Variatio 22), tensor products of the matrices I, X, Y and Z on finitely many qubits, which are self-adjoint unitaries with eigenvalues ± 1; each Pauli string is therefore a measurement with two outcomes. Two observables can be measured jointly, with a joint probability distribution of outcomes in every state, exactly when they commute, and any two Pauli strings either commute or anticommute; the relation between them that matters for measurement is anticommutation. In quantum error correction an unknown error, itself a Pauli string, is probed by measuring commuting Pauli strings, each outcome recording whether the error anticommutes with the string measured, and the question that the independence theorem answers, whether partial information about an unknown element can be combined, becomes a question about these outcomes.

Simplicity is not a property of the closure of a family of responses, as stability and NIP are. Its characterization is an amalgamation property, and the double limit condition reappears only when preparations are mixed.

Variatio 21. Canone alla Settima

The Rado graph

If each pair of points of a countable set is joined by an edge independently with probability $\frac{1}{2}$, the resulting graph is, with probability one, isomorphic to a fixed graph, by a theorem of Erdős and Rényi [ER63]. It is the first of the two basic structures of this part, and the example with which Shelah motivated simple theories. In this variation we show that its theory is simple and unstable, has the independence property, and has forking given by intersection.

The finite graphs form a Fraïssé class (see Variatio 12), a class of finite structures closed under substructures in which any two members can be glued along a common substructure, since two graphs that share an induced subgraph can be glued along it without new edges. The Fraïssé limit of this class is the Rado graph (V, E), with vertex set V and edge relation E, the unique countable graph, up to isomorphism, with the *extension property*: for all finite disjoint sets P and Q of vertices, some vertex outside $P \cup Q$ is adjacent to every vertex of P and to no vertex of Q. It is the random graph of the preceding paragraph. Its theory is denoted T_{ind} [She80]. The relation defined, for small sets A, B and C of vertices of a monster model of T_{ind} (see Variatio 20), by letting $A \mathop{\smile\!\!\!\!\!\!|}_C B$ hold exactly when $A \cap B \subseteq C$, satisfies the Kim–Pillay axioms [DI26c, Theorem 3.1]. The two substantive axioms can be checked by hand, and we do so, since they show how little a vertex knows.

Forking. Let $C \subseteq B$ be small sets of vertices of a monster model of T_{ind}, and let v be a vertex. The idea is that, when $v \notin B$, the type of v over B records only adjacency and distinctness, which the extension property allows one to prescribe freely, whereas, when $v \in B \setminus C$, the type names a vertex that C does not determine. If $v \in C$, the type of v over B is realized in C and does not fork over C. If $v \notin B$, the type of v over B says only which vertices of B are adjacent to v, and that v differs from each of them; every finite part of it is realized by vertices outside any given finite set, and the type does not fork over C. If $v \in B \setminus C$, the formula $x = v$ divides over C, in the sense recalled in Variatio 6: the type of v over C has infinitely many realizations, so there is a sequence $v = v_0, v_1, v_2, \ldots$ of distinct vertices, indiscernible over C, and no vertex equals two of them. Hence the type $\mathrm{tp}(v/B)$ forks over C exactly when $v \in B \setminus C$.

The independence theorem. The idea is that two types over sets independent over a model prescribe edges that can meet only over the model, where the types agree. Let M be a model, let B and D be sets with $B \cap D \subseteq M$, and let a_1 and a_2 realize the same type over M, with $a_1 \mathop{\smile\!\!\!\!\!\!|}_M B$ and $a_2 \mathop{\smile\!\!\!\!\!\!|}_M D$. If $a_1 \in M$, then a_2 equals a_1, and the element $a = a_1$ satisfies the conclusion. Otherwise $a_1 \notin MB$ and $a_2 \notin MD$, where MB denotes $M \cup B$, and the two types prescribe, for a new vertex, adjacency to the vertices of MB and of MD. The prescriptions agree on M, because a_1 and a_2 have the same type over M, and they concern disjoint sets outside M, because $B \cap D \subseteq M$. Hence their union is consistent, by the extension property, and the saturation of the monster model realizes it by a vertex a outside MBD; this vertex satisfies $a \mathop{\smile\!\!\!\!\!\!|}_M BD$.

The extension property also yields the instability of the theory. If the vertices b_1, a_1, b_2, a_2, … are chosen in this order by the extension property, one can arrange that a_i is adjacent to b_j exactly when $i < j$: a ladder, so the theory is unstable. Similarly, for distinct vertices $a_1, \ldots, a_n$ and every set $S \subseteq \{1, \ldots, n\}$, the extension property supplies a vertex adjacent to a_i exactly when $i \in S$: the independence property. Nevertheless, independent information can be combined: a vertex v is independent of B over C exactly when $v \notin B \setminus C$, so the type of a vertex forks only when the vertex itself is named.

For the physicist or operator theorist. Over a set B of vertices, the type of a vertex $v \notin B$ is determined by the set of vertices of B adjacent to v; by the extension property every subset of a finite B occurs in V, and, for a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, every subset of a countable B occurs in the ultrapower $V^{\mathcal{U}}$, which is $\aleph_1$-saturated (see Variatio 1). The independence property is this freedom to prescribe adjacency to a sequence of vertices. Nonforking independence is the relation $A \cap B \subseteq C$, so the type of a vertex forks only when the vertex is named; and, for a model M and sets B and D with $B \cap D \subseteq M$, the independence

theorem amalgamates two types over MB and over MD that agree over M, because their prescriptions of edges overlap only on M.

For the logician. The Rado graph enters measurement through its diagonal observable $C_E = \sum_{(a,b)\in E} |a\rangle\langle a| \otimes |b\rangle\langle b|$ on $\ell^2(V) \otimes \ell^2(V)$ (see Variatio 14), where $|a\rangle$ denotes the standard basis vector of $\ell^2(V)$ indexed by the vertex a: Alice's register holds a vertex, Bob's register holds a vertex, and the measurement answers yes when the two are adjacent. Since C_E is diagonal, a general preparation ρ of Alice answers every probe as the probability distribution $a \mapsto \langle a|\rho|a\rangle$ on vertices does, so superpositions of vertices behave as random vertices. The extension property says that, for any two disjoint finite sets of Alice's vertices, some actual preparation of Bob answers yes to every vertex of the first and no to every vertex of the second, and in Variatio 28 we determine what the random vertices of Alice do to the simplicity of this measurement.

Every finite pattern occurs in the random graph, so the double limit condition fails for its edge relation; nevertheless, independent information over a model can be amalgamated.

Variatio 22. a 1 Clav. Alla breve

The Pauli group

The second basic structure of this part comes from quantum computation. Its relation, the commutation of Pauli strings, decides which of them can be measured jointly, and it is all that the checks of an error-correcting code record about an unknown error (see Variatio 20). In this variation we show that the Pauli group modulo phases, with its commutation relation, is a Fraïssé limit whose theory is simple and unstable.

To make this precise, we start from a single quantum bit. A *qubit* is the Hilbert space $\mathbb{C}^2$, with the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Y = iXZ.$$

A *Pauli string* on n qubits is a tensor product of n factors, each equal to I, X, Y or Z, such as $XZZXI = X \otimes Z \otimes Z \otimes X \otimes I$. Pauli strings are the basic measurements of quantum information, and they are labeled by vectors of $\mathbb{F}_2^{2n}$, where $\mathbb{F}_2$ is the field with two elements: for $x = (x_1, \dots, x_n)$ and $z = (z_1, \dots, z_n)$ in $\mathbb{F}_2^n$, the string with label (x, z) is the one whose k-th factor, for $k = 1, \dots, n$, is I, X, Z or Y according as (x_k, z_k) is $(0,0)$, $(1,0)$, $(0,1)$ or $(1,1)$, and we write P_v for the Pauli string with label v. Up to a phase, the product $P_v P_w$ is the string P_{v+w}; hence the Pauli strings, multiplied by the phases ± 1 and $\pm i$, form a group, the *Pauli group* on n qubits, which modulo phases is the additive group of $\mathbb{F}_2^{2n}$. The crucial fact is that two Pauli strings either commute or anticommute,

$$P_v P_w = (-1)^{\omega(v,w)} P_w P_v, \qquad \omega\big((x,z),(x',z')\big) = x \cdot z' + z \cdot x',$$

where $x \cdot z'$ is the sum $x_1 z'_1 + \cdots + x_n z'_n$ in $\mathbb{F}_2$, and that ω is a nondegenerate alternating bilinear form on $\mathbb{F}_2^{2n}$. On countably many qubits, we let $V = \bigoplus_{k\in\mathbb{N}} \mathbb{F}_2^2$, with the same form; its elements label the Pauli strings that act nontrivially on finitely many qubits.

Figure 9 draws the commutation pattern of the fifteen nontrivial Pauli strings on two qubits. Three pairwise commuting strings whose labels add up to zero form a triangle; there are fifteen of them, and any two commuting strings lie on exactly one. This configuration is the generalized quadrangle of order two, known in quantum information as the *doily*.

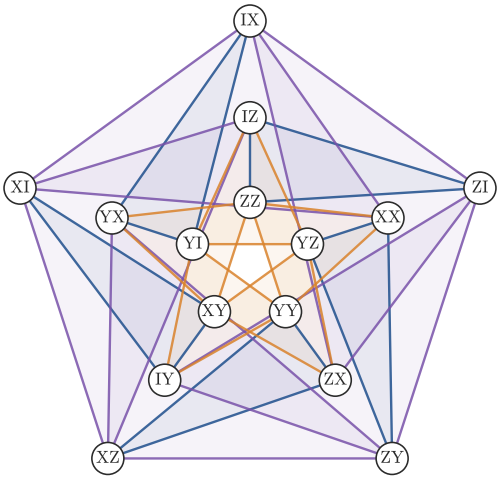


FIGURE 9. The fifteen nontrivial Pauli strings on two qubits. Two strings commute exactly when they lie on a common triangle, and each triangle is a set of three commuting strings whose product is the identity up to a phase. The picture has the five-fold symmetry of a Clifford unitary, a unitary that permutes the Pauli strings under conjugation, up to phases; this unitary has order five up to a phase, and the colors mark the three orbits of triangles under it.

Stabilizer codes in brief. Most entries of the dictionary of simplicity (see Table 6) are terms of quantum error correction; we summarize here what the logician needs of that subject. A quantum computer stores information in qubits; in the standard error model, errors are Pauli strings, and a code that corrects a set of Pauli strings corrects every error in their linear span [Got97]. A *stabilizer group* on n qubits is a subgroup $\mathcal{G}$ of the Pauli group whose elements commute pairwise and which does not contain $-I$. Its common eigenspace for the eigenvalue 1 is the *code space*; if $\mathcal{G}$ has $n-k$ independent generators, the code space has dimension 2^k and stores k qubits [Got97]. In labels, the group $\mathcal{G}$ corresponds, up to signs, to an isotropic subspace $S \subseteq \mathbb{F}_2^{2n}$, a subspace on which ω vanishes. An error is a Pauli string e, and a Pauli string that is measured in order to detect errors is a *check*; for two Pauli strings g and e we write $\omega(g,e)$ for the value of ω at their labels. Measuring independent generators $g_1,\dots,g_{n-k}$ of $\mathcal{G}$, the checks of the code, on the corrupted state reveals the *syndrome* $(\omega(g_1,e),\dots,\omega(g_{n-k},e))$ without disturbing the stored information: if ψ lies in the code space, then we have $g_i(e\psi) = (-1)^{\omega(g_i,e)}e\psi$, so measuring g_i, that is, the projections $\frac{1}{2}(I \pm g_i)$, yields $(-1)^{\omega(g_i,e)}$ with certainty and leaves $e\psi$ unchanged, and a classical procedure, the *decoder*, infers a correction from the syndrome. An error with zero syndrome is undetectable. It is harmless if it lies in $\mathcal{G}$ up to a phase, since it then acts on the code space as a scalar, and it is a *logical error* otherwise. A *logical operator* is a Pauli string that commutes with every element of $\mathcal{G}$ without lying in $\mathcal{G}$ up to a phase; it maps the code space to itself and acts on it nontrivially. The *weight* of a Pauli string is the number of its factors different from I, and the *distance* of the code is the least weight of a logical error. A *gate* is a unitary on one or two qubits, tensored with the identity on the others, and a *circuit* is a finite product of gates. A *Clifford unitary* is a unitary U that maps Pauli strings to Pauli

strings under conjugation, up to a phase; on labels, conjugation by U is a linear map that preserves ω.

The structure (V, ω) is the Fraïssé limit of the finite spaces with an alternating form over $\mathbb{F}_2$, that is, the unique countable homogeneous structure whose finite substructures are, up to isomorphism, exactly these spaces (see Variatio 4); the proof, and the model theory of (V, ω), are in Variatio 24.

Theorem 22.1 ([DI26c, Theorems 3.1 and 3.3])**.** *The theories of the Rado graph and of (V, ω) are simple and unstable. In (V, ω), nonforking independence is linear independence over the base: for small subsets A, B and C of a monster model of the theory of (V, ω), the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ of nonforking independence (see Variatio 6) holds if and only if* $\operatorname{span}(A \cup C) \cap \operatorname{span}(B \cup C) = \operatorname{span}(C)$.

The idea of the proof for (V, ω) is a gluing of linear functionals. A type over a set D records the linear relations of an element over $\operatorname{span}(D)$ and the values of ω between the element and $\operatorname{span}(D)$, which form a linear functional on $\operatorname{span}(D)$. Every finite alternating extension of a finite subspace of V embeds over that subspace: embed it in V, then use homogeneity to match the given copy of the subspace. For each pair of finite alternating spaces this is a first-order extension sentence, so it holds in the monster model as well. The conditions defining an alternating extension over a small subspace are therefore finitely satisfiable, and saturation realizes them. In the independence theorem (see Variatio 20), a model M and sets B and D with $B \mathop{\smile\!\!\!\!\!\!|}_M D$ are given, together with realizations c_1 and c_2 of one type over M such that $c_1 \mathop{\smile\!\!\!\!\!\!|}_M B$ and $c_2 \mathop{\smile\!\!\!\!\!\!|}_M D$. Writing MB for $M \cup B$, and similarly for other unions, we see that the functionals defined by c_1 on $\operatorname{span}(MB)$ and by c_2 on $\operatorname{span}(MD)$ agree on M, and that the intersection $\operatorname{span}(MB) \cap \operatorname{span}(MD)$ equals M because $B \mathop{\smile\!\!\!\!\!\!|}_M D$; hence they glue to one functional on $\operatorname{span}(MBD)$. If the common type is algebraic over M, its realization lies in M and already gives the amalgam. Otherwise, realize the glued functional by a new element linearly independent over $\operatorname{span}(MBD)$; quantifier elimination then gives $\operatorname{tp}(c_1/MB) \cup \operatorname{tp}(c_2/MD)$ and independence over M. For finite tuples, choose matching maximal subtuples linearly independent over M, prescribe their pairings with each other as well as with the parameter spans, and restore each omitted coordinate by its linear equation over M; equality of the base types makes these equations agree. For small tuples, every finite fragment of the type and independence conditions involves only finitely many coordinates, so compactness and saturation give the same conclusion. The independence property comes from nondegeneracy: for linearly independent $a_1, \ldots, a_n$, every pattern of values $\omega(a_i, b)$ is realized by some b.

For the physicist or operator theorist. The structure (V, ω) is an $\mathbb{F}_2$-vector space with an alternating form, and its ultrapower $V^{\mathcal{U}}$ along a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$ is again such a space, an elementary extension in which idealized labels live. The type over a set D of an element $e \notin \operatorname{span}(D)$ is determined by the linear functional $\omega(\cdot, e)$ on $\operatorname{span}(D)$, and every functional occurs, by nondegeneracy when D is finite and by $\aleph_1$-saturation of the ultrapower when D is countable; this freedom is the independence property. By Theorem 22.1, nonforking independence is linear independence over the base, and, as the idea of its proof shows, the independence theorem reduces to the gluing of two linear functionals that agree on a common subspace.

For the logician. On a qubit, the bit flip X exchanges the standard basis vectors $|0\rangle$ and $|1\rangle$ of $\mathbb{C}^2$, the phase flip Z multiplies $|1\rangle$ by -1, and the matrix $Y = iXZ$ does both, up to a phase; a Pauli string on n qubits applies one of these operations, or the identity, to

each qubit. Pauli strings serve both as measurements, with outcomes ± 1, and as errors, the operations that noise applies to a register; every operator on n qubits is a linear combination of them, which is why a code that corrects a set of Pauli errors corrects every error in their span. As an error, a Pauli string P acts on density operators by $\rho \mapsto P\rho P^*$, an action that does not change when P is multiplied by a phase and is therefore determined by the label of P in $\mathbb{F}_2^{2n}$; the form ω records which pairs of strings can be measured jointly: those whose labels v and w satisfy $\omega(v, w) = 0$.

The Rado graph also lives inside the Pauli group, as a pattern of commutation between bit flips and idealized Pauli strings (see Variations 27 and 28). Like the Rado graph, the structure (V, ω) is simple and unstable and has the independence property; its nonforking independence, however, is linear independence over the base, and its independence theorem becomes the gluing of linear functionals.

Variatio 23. a 2 Clav.

The dictionary of simplicity

Three notions of model theory (Fraïssé limits, forking and the independence theorem) are at work in quantum error correction under other names: in this variation we set out the third dictionary of the paper (see Table 6), after the dictionaries of stability and of NIP (see Variations 3 and 13), and present the running example; the physical side of this dictionary was introduced in Variatio 22. In each of the next three variations we state a theorem, prove it, and read it twice. A single example runs through all four: the Laflamme–Miquel–Paz–Zurek five-qubit code [LMPZ96], the smallest code that corrects an arbitrary error on one qubit, which Bennett, DiVincenzo, Smolin and Wootters found independently [BDSW96].

For the two code-specific forking rows, fix a stabilizer code on finitely many qubits and a nonzero error label on those qubits with zero syndrome; a full set of logical labels means that, together with the stabilizer labels, it spans all zero-syndrome labels on those same qubits.

Example 23.1. The five-qubit code has the stabilizer generators

$$g_1 = XZZXI, \quad g_2 = IXZZX, \quad g_3 = XIXZZ, \quad g_4 = ZXIXZ,$$

and stores one qubit, with logical operators $\bar{X} = XXXXX$ and $\bar{Z} = ZZZZZ$. Each of the fifteen errors that act as X, Y or Z on a single qubit has its own nonzero syndrome, the string of four bits that records which of the generators $g_1, \dots, g_4$ anticommute with it, and there are exactly fifteen nonzero syndromes of four bits: the code is *perfect*. For instance, the error XIIII has syndrome $(0, 0, 0, 1)$, the error $IIZII$ has syndrome $(0, 0, 1, 0)$, and the error $YIIII$ has syndrome $(1, 0, 1, 1)$.

Variatio 24. a 1 Clav. Canone all'Ottava

Fraïssé limits and Clifford circuits

Two properties make the labels of Pauli strings, with their commutation form, a Fraïssé limit: every finite pattern of commutation occurs among them, and every isomorphism between two of their finite substructures extends to an automorphism. In Proposition 24.1 below we give the first a physical form, showing that, for every symmetric matrix of zeros and ones with zeros on the diagonal, there are Pauli strings, tensor products of Pauli matrices, one for each row of the matrix, such that two of them anticommute exactly when the matrix has a one in the corresponding entry; in Theorem 24.2 we give the second, showing that any two families of Pauli strings with linearly independent labels and the same pattern of commutation are carried to each other by a Clifford circuit, a product of gates that map

Model theory	**Quantum error correction**	
the structure (V, ω)	finite-support Pauli strings modulo phases, with label addition and the alternating commutation form	Variatio 22
the age of (V, ω), its finite substructures up to isomorphism	finite alternating label spaces, including their linear relations	Proposition 24.1
ultrahomogeneity	Clifford circuits; encoding circuits of codes	Theorem 24.2
algebraic closure of a set of labels	its linear span: products of the corresponding Pauli strings modulo phases	Variatio 22
the type of an error label over a set of labels	its syndrome and all its linear relations over that set	Theorem 25.1
the type over known labels forks over base labels among them	the error is a sum of known labels but not a sum of base labels	Theorem 25.1
the type over stabilizer labels does not fork over $\emptyset$	the fixed zero-syndrome error is logical	Corollary 25.2
the type over stabilizer and logical labels forks over the stabilizer labels	the same error is logical: the full set of logical labels makes it a sum of known labels	Theorem 25.1
the independence theorem	when the check spans meet only in the common base span, syndrome functionals agreeing there have a joint extension	Proposition 26.1

TABLE 6. The dictionary of simplicity: the Pauli group as a structure.

Pauli strings to Pauli strings under conjugation, up to a phase. The two readings coincide: for the logician, the Pauli group is a Fraïssé limit; for the physicist, its symmetries are circuits.

In the language with $+$, 0 and a relation for $\omega(x, y) = 1$, where ω is the commutation form of Variatio 22, the finite spaces with an alternating form amalgamate freely, every finite one embeds in a nondegenerate one, and Witt's theorem extends every isometry between subspaces of a finite nondegenerate space, that is, every linear bijection between them that preserves the form, to an isometry of the whole space. A back-and-forth argument then shows that the structure (V, ω) of labels of Pauli strings on countably many qubits is the Fraïssé limit of the finite spaces with an alternating form over $\mathbb{F}_2$, the unique countable homogeneous structure whose finite substructures are, up to isomorphism, exactly these spaces (see Variatio 4). Its theory, in the senses recalled below, is $\aleph_0$-categorical and eliminates quantifiers; algebraic closure in it is linear span; and Cherlin and Hrushovski count (V, ω) among the basic smoothly approximable structures [CH03]. A theory is $\aleph_0$-*categorical* if all its countable models are isomorphic; by the Engeler–Ryll-Nardzewski–Svenonius theorem, this holds exactly when, for each n, the n-tuples of every model, for instance of the ultrapower $V^{\mathcal{U}}$ along a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, realize only finitely many types. *Quantifier*

elimination says that every formula is equivalent to one without quantifiers; for (V, ω), two tuples, in V or in $V^{\mathcal{U}}$, have the same type as soon as they satisfy the same linear relations and ω takes the same values on corresponding pairs of their entries. An element a of a monster model (see Variatio 20) is *algebraic* over a small set A if the type $\mathrm{tp}(a/A)$ has only finitely many realizations, and the *algebraic closure* of A is the set of such elements. An $\aleph_0$-categorical structure is *smoothly approximable* if it is the union of a chain $F_1 \subseteq F_2 \subseteq \cdots$ of finite substructures such that, for every n, any two tuples of F_n with the same type are conjugate by an automorphism that maps F_n onto itself; for (V, ω), the sets of labels of the strings on the first n qubits form such a chain. The next two results make these facts physical.

Proposition 24.1. *If M is a symmetric $k \times k$ matrix over $\mathbb{F}_2$ with zero diagonal, then there are Pauli strings $P_1, \ldots, P_k$ on at most k qubits such that P_i and P_j anticommute exactly when $M_{ij} = 1$.*

Proof. The matrix M defines an alternating form on $\mathbb{F}_2^k$. Let R be its radical and choose a complement W of R; the form is nondegenerate on W, which therefore has a symplectic basis of r pairs, where $2r = \dim W$. We send the r pairs to the labels of X and Z on the first r qubits, and a basis of R to the labels of X on $\dim R$ further qubits. This linear map preserves the form, and the images of the standard basis vectors of $\mathbb{F}_2^k$ are the labels of the required strings, on $r + \dim R \leq k$ qubits. □

Theorem 24.2. *Let n be a positive integer, let W and W' be subspaces of the space $\mathbb{F}_2^{2n}$ of labels of Pauli strings on n qubits, and let $f \colon W \to W'$ be a linear bijection with $\omega(f(v), f(w)) = \omega(v, w)$ for all $v, w \in W$. Then there is a Clifford unitary U on n qubits such that, for every $v \in W$, the Pauli string P_v with label v satisfies $UP_vU^* = \pm P_{f(v)}$. If, moreover, the vectors $v_1, \ldots, v_m$ form a basis of W, then the unitary U can be chosen with $UP_{v_i}U^* = P_{f(v_i)}$ for all $i \leq m$.*

Proof. The idea to extend f to an isometry of the whole space, to implement the extension by a Clifford unitary, and to correct the signs by a Pauli string. By Witt's theorem [Art57], the map f extends to a linear bijection T of $\mathbb{F}_2^{2n}$ that preserves ω. Every such T is induced by conjugation by a Clifford unitary [CRSS98, DDM03]; let U be one. Since UP_vU^* is a Pauli string up to a phase, Hermitian and with square I, the phase is a sign. For the second statement, let $\varepsilon_i \in \{\pm 1\}$ be the sign with $UP_{v_i}U^* = \varepsilon_i P_{f(v_i)}$. The vectors $f(v_i)$ are linearly independent, so by nondegeneracy there is $b \in \mathbb{F}_2^{2n}$ with $(-1)^{\omega(f(v_i), b)} = \varepsilon_i$ for all i. Conjugation by P_b multiplies $P_{f(v_i)}$ by $(-1)^{\omega(f(v_i), b)}$, so the Clifford unitary P_bU has the required property. □

Corollary 24.3. *Let n be a positive integer. Any two stabilizer groups on n qubits with the same number of independent generators are conjugate by a Clifford unitary. In particular, for every integer k with $0 \leq k \leq n$, every stabilizer code that stores k qubits in n has an encoding circuit made of Clifford gates, which maps onto it the trivial code, the code whose stabilizer group is generated by the $n - k$ strings that act as Z on one of the first $n - k$ qubits and as the identity on the others.*

Proof. Let $g_1, \ldots, g_m$ and $g'_1, \ldots, g'_m$ be independent generators of the two groups, with labels v_i and v'_i, respectively. Both families span isotropic subspaces, so the linear map $v_i \mapsto v'_i$ preserves ω, and Theorem 24.2 yields a Clifford unitary U with $UP_{v_i}U^* = \pm P_{v'_i}$. The signs of the generators g_i and g'_i are then matched by a Pauli correction, as in the proof of the theorem. □

On countably many qubits, every finite-dimensional subspace of V lies in the labels of the first n qubits for some n. Applying Theorem 24.2 there, and extending by the identity on the remaining qubits, shows that every isometry between finite-dimensional subspaces of (V, ω) extends to an automorphism of (V, ω) implemented by a Clifford circuit on finitely many qubits.

For the physicist or operator theorist. Proposition 24.1 identifies the age of (V, ω), the class of finite structures that embed in it, with the class of finite commutation patterns, and Theorem 24.2, extended to countably many qubits as above, yields ultrahomogeneity: every isomorphism between finite substructures extends to an automorphism, here implemented by a Clifford circuit. By quantifier elimination, the type of a tuple of labels, in V or in any ultrapower $V^{\mathcal{U}}$, is determined by its pattern of linear relations and commutations, and, since for each length there are only finitely many such patterns, the theory is $\aleph_0$-categorical by the Engeler–Ryll-Nardzewski–Svenonius theorem. Corollary 24.3 is the case of isotropic subspaces: two families of independent generators of stabilizer groups of the same size have the same type, hence are conjugate by an automorphism, up to signs that a Pauli correction adjusts, and an encoding circuit of a code is a Clifford circuit that implements such an automorphism.

For the logician. The Clifford unitaries are, up to phases, the products of three kinds of gates: the Hadamard gate $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, which exchanges the Pauli matrices X and Z under conjugation; the phase gate $\mathrm{diag}(1, i)$, which sends X to Y and fixes Z; and the controlled-NOT gate on two qubits, which flips the second qubit when the first is in the basis state $|1\rangle$. Conjugation by a Clifford unitary maps Pauli measurements to Pauli measurements and Pauli errors to Pauli errors, which is why these gates are the basic operations of stabilizer error correction. An *encoding circuit* for a code that stores k qubits in n is a unitary that maps the code space of the trivial code, in which the first $n-k$ qubits are fixed in $|0\rangle$ and the last k carry the information, onto the code space of the code; Corollary 24.3 provides one, built from these gates, for every stabilizer code.

Variatio 25. a 2 Clav. Adagio

Forking and logical errors

Some errors of a stabilizer code commute with every check, that is, with each of the generators of the stabilizer group that the code measures; every check then returns the outcome 1, and the syndrome, the list of these outcomes read as bits, is zero. Such errors are undetectable, and the decoder, the procedure that infers a correction from the syndrome, does not correct them. Some undetectable errors are harmless, others corrupt the stored qubits, and the syndrome does not distinguish the two kinds.

In this variation we separate them by forking (see Variatio 6). Forking is a strong condition: the type of an idealized error over a set of actual labels never forks over the empty set. Indeed, let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$, and let $(e_n)_{n\in\mathbb{N}}$ be a sequence of labels in the space V of labels of Pauli strings on countably many qubits (see Variatio 22) whose class $[e_n]_{\mathcal{U}}$ in the ultrapower $V^{\mathcal{U}}$ lies outside V, as it does when the labels e_n are pairwise distinct; then the class does not lie in the span of any subset B of V, and by Theorem 25.1 below its type over B does not fork over $\emptyset$. We compute forking for the Pauli group from Theorem 22.1, and then read it in a code.

Theorem 25.1. *Let e be an element of a monster model $\mathbb{M}$ of the theory of (V, ω), and let $D \subseteq B$ be small subsets of $\mathbb{M}$. Then the type $\operatorname{tp}(e/B)$ forks over D if and only if $e \in \operatorname{span}(B) \setminus \operatorname{span}(D)$.*

Proof. By Theorem 22.1, the type $\operatorname{tp}(e/B)$ does not fork over D exactly when $\operatorname{span}(eD) \cap \operatorname{span}(B) = \operatorname{span}(D)$. If $e \in \operatorname{span}(D)$, then $\operatorname{span}(eD) = \operatorname{span}(D)$, and the equality holds. If $e \notin \operatorname{span}(D)$, then we have $\operatorname{span}(eD) = \operatorname{span}(D) \oplus \{0, e\}$, and the equality fails exactly when some vector $e + d$ with $d \in \operatorname{span}(D)$ lies in $\operatorname{span}(B)$, that is, exactly when $e \in \operatorname{span}(B)$. □

For $D \subseteq B$, a type of an element outside $\operatorname{span}(D)$ can be extended without forking over D by prescribing commutation values on B consistent with linearity and with the original type over D. The nonforking realization is chosen outside $\operatorname{span}(B)$; the criterion of the theorem distinguishes this choice from a realization that becomes algebraic over B. Forking happens only when e becomes a linear combination of elements of B, that is, up to a phase, a product of the corresponding Pauli strings. In a stabilizer code this criterion separates the two kinds of undetectable error other than the identity.

Corollary 25.2. *Let n be a positive integer, identify $\mathbb{F}_2^{2n}$ with the labels in V supported on the first n qubits, let $\mathcal{G}$ be a stabilizer group on those qubits with set of labels S, and let $e \in \mathbb{F}_2^{2n}$ be a nonzero label with zero syndrome. Then the type $\operatorname{tp}(e/S)$ forks over $\emptyset$ if and only if $e \in S$, that is, if and only if e is harmless. Consequently, if the code defined by $\mathcal{G}$ encodes at least one qubit, its distance is the least number of qubits acted on by a nonzero error supported on the first n qubits with zero syndrome whose type over S does not fork over $\emptyset$.*

Proof. The first statement is Theorem 25.1 with $D = \emptyset$ and $B = S$, since $\operatorname{span}(S) = S$ and $\operatorname{span}(\emptyset) = \{0\}$. The second is the definition of distance, a logical error being a nonzero label with zero syndrome outside S. □

In the five-qubit code (see Example 23.1), consider two errors, up to phases: the product $g_1 g_2 = XYIYX$ of the first two stabilizer generators and the logical operator $\bar{X} = XXXXX$. Both have zero syndrome. The first lies in the stabilizer group, so its type over S forks over $\emptyset$, and it is harmless. The second does not, so its type does not fork over $\emptyset$, and it flips the stored qubit. The least weight of an error of the second kind is 3, as the count below shows, attained for instance by $\bar{X} g_1 = IYYIX$ up to a phase: the code has distance 3.

We now count the undetectable errors of the five-qubit code. On five qubits there are $4^5 = 1024$ Pauli strings up to phase. The undetectable ones, those with zero syndrome, are the labels on which ω vanishes against the four checks; they form a subspace of dimension $10 - 4 = 6$, sixty-four labels in all. Sixteen of them lie in the stabilizer group. One is the identity, which is no error at all, and whose type over S does not fork over $\emptyset$, since it lies in $\operatorname{span}(\emptyset)$. The other fifteen are strings of weight four; their types over S fork over $\emptyset$, and they are harmless. The remaining forty-eight are logical errors, and their types do not fork over $\emptyset$: thirty of weight three, eighteen of weight five, and none of weight four.

weight of the nonidentity undetectable error	3	4	5
harmless: the type over S forks over $\emptyset$	0	15	0
logical: the type over S does not fork over $\emptyset$	30	0	18

Every undetectable error of weight four is harmless. Every undetectable error of weight three is logical. The distance of the code, three, is the weight at which the nonforking errors begin.

Forking also tracks what is learned as operators are added to the checks. Suppose that the logical operator $\bar{Z} = ZZZZZ$ of the five-qubit code is added. The type of $\bar{X}$ over $S \cup \{\bar{Z}\}$ records that $\bar{X}$ anticommutes with $\bar{Z}$, so the error is detected. Its type still does not fork over S, because $\bar{X}$ is not a product of $\bar{Z}$ and the stabilizers. By Theorem 25.1, the type of $\bar{X}$ over a set $B \supseteq S$ forks over S exactly when $\bar{X}$ lies in $\mathrm{span}(B)$. A full set of logical operators means a set whose labels, together with the stabilizer labels, span all zero-syndrome labels on the code's qubits. After all these labels have been added, the type of every logical error over the enlarged parameter set forks over the stabilizer labels, by Theorem 25.1: the error lies in the enlarged span but outside the stabilizer span.

Forking can be observed directly in an elementary extension. Take $D = \emptyset$, $B = S$, and the harmless error $e = g_1 g_2$, whose label is $g_1 + g_2$. The formula $x = g_1 + g_2$, with parameters g_1, g_2, divides over $\emptyset$: on fresh qubits of the structure on countably many qubits there are pairs $(g_1^{(j)}, g_2^{(j)})$, $j \in \mathbb{N}$, with the same commutation pattern as (g_1, g_2) and pairwise disjoint supports, hence with pairwise distinct sums; by Ramsey's theorem and compactness they may be taken indiscernible over $\emptyset$, and no label equals two of the sums. The type $\mathrm{tp}(e/S)$ therefore forks over $\emptyset$, as Corollary 25.2 states. For a logical error e, by contrast, no formula of $\mathrm{tp}(e/S)$ divides over $\emptyset$, in accordance with Theorem 25.1: by quantifier elimination, the type says only that $e \notin \mathrm{span}(S)$ and that $\omega(s, e) = 0$ for $s \in S$, and it is realized by labels on qubits disjoint from any given finite set.

The type of a logical error over S has several nonforking extensions, and this is the decoder's situation: all labels outside $\mathrm{span}(S)$ with zero syndrome have the same type over S, since, by quantifier elimination, the type over S of such a label e is determined by the absence of linear relations between e and $\mathrm{span}(S)$ and by the linear functional $\omega(\cdot, e)$ on $\mathrm{span}(S)$, which is 0. Its nonforking extensions to $S \cup \{\bar{Z}\}$ are at least two: the extension realized by $\bar{X}$, which anticommutes with $\bar{Z}$, and the extension realized by a label supported on a sixth qubit, which commutes with $\bar{Z}$; neither label lies in $\mathrm{span}(S \cup \{\bar{Z}\})$, so neither extension forks over S. A decoder that learns the values of ω at further labels learns only conditions on commutation, which never cause forking, and on countably many qubits no finite set of such values determines the error. The type of a logical error over a set $B \supseteq S$ of known labels determines the error exactly when it forks over S, that is, when the error becomes, up to a phase, a product of strings whose labels lie in B.

For the physicist or operator theorist. Forking (see Variatio 6) can be seen directly in (V, ω). If $e = b_1 + \cdots + b_m$ with $b_i \in B$ and $e \notin \mathrm{span}(D)$, then the formula $x = b_1 + \cdots + b_m$ divides over D: since the sum is not algebraic over D, an elementary extension contains copies of $(b_1, \ldots, b_m)$ that form a sequence indiscernible over D with pairwise distinct sums, and no element equals two of them. Conditions on the values $\omega(b, x)$ for $b \in B$, by contrast, never cause forking, by Theorem 25.1. For example, an element $[e_n]_{\mathcal{U}}$ of the ultrapower with pairwise distinct labels e_n lies outside V, so its type over any subset of V does not fork over $\emptyset$, whatever commutation values it carries.

For the logician. An error on a code with stabilizer group $\mathcal{G}$ is undetectable when it commutes with every element of $\mathcal{G}$, and then it maps the code space to itself. If it lies in $\mathcal{G}$ up to a phase, it acts on the code space as a scalar and the stored information is intact; otherwise it is a logical error: it acts as a nontrivial unitary on the stored qubits, and no measurement of elements of $\mathcal{G}$ tells the two cases apart, since both yield the trivial syndrome. Corollary 25.2 reads the distinction as forking: for an undetectable error other than the identity, the type of its label over the labels of $\mathcal{G}$ forks over $\emptyset$ exactly when the error is harmless. In the five-qubit

code, every nonidentity error on at most two qubits has a nonzero syndrome, every error on one qubit has its own syndrome and can therefore be corrected (see Example 23.1), and the distance, 3, is the least weight of a logical error.

Forking thus draws exactly the distinction that the syndrome cannot: an undetectable error other than the identity is harmless if and only if the type of its label over the labels of the stabilizer group forks over the empty set. In the next variation we turn to the independence theorem, the other half of the Kim–Pillay characterization of simplicity (see Variatio 20).

Variatio 26. A 2 Clav.

The gluing of syndromes

Suppose that two sets of checks, Pauli strings measured to detect errors, are designed independently on the same qubits, in the sense that the spans of their labels meet only in the span of the labels of their common checks, and that a syndrome is prescribed for each set, the two agreeing on the common checks. The independence theorem (see Variatio 20) says that nonforking extensions of one type over a model to two sets independent over it can be amalgamated; for the Pauli group, its counterpart is that a single error produces both syndromes (see Proposition 26.1 below).

Proposition 26.1. *Let n be a positive integer, let A_0, B and D be subsets of the space $\mathbb{F}_2^{2n}$ of labels of Pauli strings on n qubits, write BA_0 for $B \cup A_0$ and DA_0 for $D \cup A_0$, suppose that $\operatorname{span}(BA_0) \cap \operatorname{span}(DA_0) = \operatorname{span}(A_0)$, and let f_1 and f_2 be linear functionals on $\operatorname{span}(BA_0)$ and on $\operatorname{span}(DA_0)$ that agree on $\operatorname{span}(A_0)$. Then some label $e \in \mathbb{F}_2^{2n}$ has the commutation values $\omega(s, e) = f_1(s)$ for $s \in \operatorname{span}(BA_0)$ and $\omega(s, e) = f_2(s)$ for $s \in \operatorname{span}(DA_0)$.*

Proof. Since f_1 and f_2 agree on the intersection of their domains, there is a linear functional f on $\operatorname{span}(BA_0) + \operatorname{span}(DA_0)$ extending both. This functional extends to $\mathbb{F}_2^{2n}$, and, because ω is nondegenerate, every linear functional on $\mathbb{F}_2^{2n}$ has the form $\omega(\cdot, e)$ for a unique e. □

In the five-qubit code, with the stabilizer generators $g_1, \dots, g_4$ of Example 23.1, let $A_0 = \emptyset$, $B = \{g_1, g_2\}$ and $D = \{g_3, g_4\}$. The spans of B and D meet only in 0, so any two bits of syndrome on g_1, g_2 and any two on g_3, g_4 are realized together by one error; by Example 23.1, even by an error on at most one qubit, such as $YIIII$ for the syndrome $(1, 0 \,|\, 1, 1)$. If D is replaced by $\{g_1 g_2\}$, the hypothesis fails, and so does the conclusion: the syndrome bit of $g_1 g_2$ is the sum of those of g_1 and g_2.

On a finite number n of qubits, the information carried by a syndrome can be measured by counting. If $s_1, \dots, s_m$ are linearly independent labels in $\mathbb{F}_2^{2n}$, then, for every syndrome $\tau \in \mathbb{F}_2^m$, exactly 2^{2n-m} labels e satisfy $\omega(s_i, e) = \tau_i$ for all $i \le m$, since the map $e \mapsto (\omega(s_i, e))_{i \le m}$ is linear and onto. A uniformly random error has a uniformly random syndrome, which reveals exactly m bits. The count records codimension, which forking ignores, and it reflects the smooth approximability of (V, ω), the structure of labels on countably many qubits, by finite structures (see Variatio 24).

For the physicist or operator theorist. The Kim–Pillay independence theorem (see Variatio 20) concerns two nonforking extensions of one type over a model M, to sets B and D independent over M, and asserts that they have a common nonforking realization. In a stable theory this is automatic, since a type over a model has only one nonforking extension to MBD (see Variatio 6); in a simple unstable theory the two extensions are chosen independently, and the theorem says that they never conflict. Proposition 26.1 is the case of (V, ω) in finite dimension, over the closed set $\operatorname{span}(A_0)$ in place of a model: the two

types are the prescribed syndromes f_1 and f_2, the independence of B and D is the hypothesis on the spans, and the common realization is the error e.

For the logician. A syndrome is the list of outcomes, each ± 1, of measuring the checks of a code on the corrupted state, and it records for each check whether the error anticommutes with it. Proposition 26.1 says that if two families of checks are independent over their common part, in the sense of the hypothesis on spans, then any two lists of outcomes that agree on the common checks are produced by a single Pauli error, and consequently no outcome of one family excludes an outcome of the other. In the five-qubit code, the outcomes of g_1, g_2 and of g_3, g_4 can be prescribed separately, whereas for the checks g_1, g_2 and $g_1 g_2$ the third outcome is the product of the first two. On n qubits, a family of m independent checks reveals exactly m bits about a uniformly random error, each syndrome being produced by the same number 2^{2n-m} of errors.

The independence theorem is thus to simplicity what uniqueness of the independent product (see Theorem 6.2) is to stability: the property of amalgams that characterizes the class.

Variatio 27. a 2 Clav. Canone alla Nona

The Rado graph inside the Pauli group

Every graph on a finite set of qubits determines a quantum state, its graph state. In Proposition 27.1 below, we show by means of graph states that every relation between two finite sets is a pattern of commutation between Pauli strings, and in Variatio 28 that, with idealized strings on one side, so is every relation between countable sets, the Rado graph among them. The Pauli group carries, in addition, the linear structure of its labels, which mirrors the multiplication of Pauli strings, and in Variatio 29 we show that this structure lets actual preparations witness the loss of simplicity under mixing (see Variatio 28).

A *stabilizer state* on n qubits is a unit vector fixed by every element of a stabilizer group with n independent generators, a subgroup of the Pauli group whose elements commute pairwise and which does not contain $-I$; the vectors fixed by such a group form a one-dimensional space, the code space of the group (see Variatio 22). Let G be a graph on a finite set W of qubits. The *graph state* of G is the stabilizer state, unique up to a phase, of the stabilizer group with the generators

$$K_v = X_v \prod_{w \sim v} Z_w \qquad (v \in W),$$

where the strings X_v and Z_w act as X on the qubit v and as Z on the qubit w, respectively, and as the identity on the other qubits, and where the product runs over the neighbors w of v in G.

Proposition 27.1.

(a) *For every finite graph G on W and all $u, v \in W$, the strings X_u and K_v anticommute exactly when u and v are adjacent.*

(b) *For every relation $\varphi \subseteq A \times B$ between finite sets, there are Pauli strings P_a, $a \in A$, and Q_b, $b \in B$, such that P_a and Q_b anticommute exactly when $(a, b) \in \varphi$.*

Proof. (a) The string X_u commutes with every X factor of K_v, and with every Z factor on a qubit other than u; it anticommutes with Z_u, which is a factor of K_v exactly when $u \sim v$. (b) We place the elements of A on distinct qubits, let $P_a = X_a$, and let Q_b be the product of Z_a over the a with $(a, b) \in \varphi$. □

For a finite graph, the strings K_v of part (a), which play the role of Bob's strings in Variatio 28 while the flips X_u play that of Alice's, are actual Pauli strings, the generators of its graph state; the passage to infinite graphs, such as the Rado graph, is made there.

For the physicist or operator theorist. Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$. In the ultrapower $V^{\mathcal{U}}$ of the space V of labels of Pauli strings on countably many qubits, the class $[b_n]_{\mathcal{U}}$ of a sequence $(b_n)_{n\in\mathbb{N}}$ of labels defines on V the linear functional $v \mapsto \lim_{n,\mathcal{U}} \omega(v, b_n)$, where ω is the commutation form; this functional records the ω-type of the class over V, and every linear functional f on V arises in this way, since, by nondegeneracy, the restriction of f to the labels of the strings on the first n qubits has the form $\omega(\cdot, b_n)$ for some label b_n of a string on these qubits. Applied to the labels of the bit flips X_a, for a in a countable set A of qubits, this shows that every relation between A and a countable set B is realized as the pattern of commutation between the flips X_a and suitable idealized labels, one for each element of B, which is how the Rado graph lives inside the Pauli group (see Variatio 28). The two theories nevertheless differ: algebraic closure is trivial in the Rado graph and is linear span in (V, ω), and accordingly the type of a vertex forks only when the vertex is named (see Variatio 21), while the type of a label forks only when the label becomes a linear combination of the given labels (see Theorem 25.1).

For the logician. A graph state is the unique state, up to a phase, left unchanged by every K_v; it is prepared by placing every qubit in $(|0\rangle + |1\rangle)/\sqrt{2}$ and applying, for each edge $\{v, w\}$, the controlled-Z gate on the qubits v and w, which multiplies by -1 the basis vector $|1\rangle_v|1\rangle_w$, in which both qubits are in the state $|1\rangle$, and fixes the other three basis vectors of the pair. A bit flip X_u applied to a graph state is detected by measuring the stabilizers K_v: by Proposition 27.1(a), the outcome is -1 exactly for the neighbors v of u, so the syndrome of a single flip is a row of the adjacency matrix. For the Rado graph on countably many qubits, each K_v would act on infinitely many qubits, which is why Bob's strings become idealized in Variatio 28, while the flips X_u remain actual.

The Pauli group thus carries, beyond the commutation patterns that it shares with the Rado graph, the linear structure of its labels; in Variatio 29 we show that this structure decides whether the loss of simplicity under mixing is already visible among actual preparations.

Variatio 28. A 2 Clav.

Mixing destroys simplicity

In each of the first two parts, mixing has a distinct effect: in Part I, for stable observables, finite mixtures of actual preparations simulate idealized ones, and in Part II mixing makes hereditary separability of the idealized responses imply their metrizability. In Theorem 28.1 below, we describe the third effect, recorded in Table 1: the simple theories of the Pauli and Rado label structures give rise to nonsimple theories of their mixture-and-probe structures. The structures, and hence the theories being compared, have changed. The idea is a comparison of two averages: the difference between the responses to the uniform mixtures on two complementary halves of a block of the labels that Alice's register can hold measures how a probe weighs the two halves, and halves meeting in a quarter of the block cannot both give a response difference greater than one half.

Before simplicity can be tested on mixtures, the mixtures must become elements of a structure; the diagonal observable of a relation provides one. Let $\varphi\colon A \times B \to \{0, 1\}$ be a

relation between countable sets of labels, and consider its diagonal observable (see Variatio 14)

$$C_\varphi = \sum_{a\in A,\, b\in B} \varphi(a,b)\, |a\rangle\langle a| \otimes |b\rangle\langle b| \quad \text{on } \ell^2(A)\otimes\ell^2(B)\,:$$

Alice's register holds a label a, Bob's register holds a label b, and the observable reads $\varphi(a,b)$. A mixture of Alice's preparations is given by a probability measure μ on A, and her response to a probe ν is the probability $R(\mu,\nu)$ that the observable reads 1. The probes form the space $\mathcal{P}(K_\varphi)$ of Radon probability measures on the closure K_φ of the functions $\varphi(\cdot,b)$ in $\{0,1\}^A$, the idealized labels of Bob [DI26c, Lemma 5.1]. Indeed, the coordinate functions $\lambda\mapsto\lambda(a)$ on K_φ separate points; their unital complex algebra is closed under conjugation and uniformly dense in $C(K_\varphi)$ by the Stone–Weierstrass theorem. Restriction to the dense set of actual labels identifies its norm closure with the probe algebra generated by the diagonal projections $b\mapsto\varphi(a,b)$, and the Riesz representation theorem identifies its states with Radon probability measures. The observable thus defines a two-sorted structure $\mathcal{S}_\varphi$ of real-valued logic, with the mixtures of Alice and the probes of Bob as sorts and the response R as predicate, and we call the observable *simple* if the theory of $\mathcal{S}_\varphi$ in Keisler's real-valued logic [Kei23] is simple in the sense of Ben Yaacov [BY03]: no formula $\theta(x,y)$ has the tree property, that is, for no real $\varepsilon>0$ and integer $k\geq 2$ is there a tree of parameters such that the conditions $\theta(x,d)=0$ along each branch are jointly satisfiable while, for any k siblings $d_1,\dots,d_k$, the conditions $\theta(x,d_1)\leq\varepsilon$, $\dots$, $\theta(x,d_k)\leq\varepsilon$ are not. A formula is *affine* if it is built from atomic formulas with the quantifiers and the connectives $u(t_1,\dots,t_n)=c_0+\sum_{i=1}^n c_i t_i$, where $n\geq 1$ and $c_0,\dots,c_n$ are real numbers. In an elementary extension, for instance in the ultrapower $\mathcal{S}_\varphi^{\mathcal{U}}$ along a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the new elements of Alice's sort are idealized preparations of Alice.

The *commutation observable* of the Pauli group is the diagonal observable of the commutation form ω on labels, and the *Rado observable* is the diagonal observable of the edge relation E of the Rado graph. Both have the independence property already on pure labels. On countably many qubits, Proposition 27.1(b) extends to countable sets A and B with the strings Q_b replaced by idealized ones: with the elements of A placed on distinct qubits, as in its proof, for each $b\in B$ some linear functional on the space V of labels takes the value 1 at the label of the bit flip X_a exactly when $(a,b)\in\varphi$, and the compact group V^* of all linear functionals on V is the space of idealized labels of Bob for the commutation observable [DI26c, Lemma 5.3]. In particular, the Rado graph and the random bipartite graph, the Fraïssé limit of the finite bipartite graphs with the two sides as two sorts, are realized as patterns of commutation between single-qubit flips and idealized Pauli strings, while for a finite graph Bob's strings are actual Pauli strings, the generators of its graph state.

The pure and mixed responses have the independence property together (see [DI26c, Lemma 6.1]). Here both properties concern Alice's responses, first on K_φ and then on the full probe space $\mathcal{P}(K_\varphi)$. To see the nontrivial direction, suppose that the pure responses are NIP. Rosenthal's dichotomy gives every sequence of pure responses a pointwise convergent subsequence; boundedness and dominated convergence make it weakly Cauchy in the real Banach space of continuous functions on K_φ, meaning that each continuous linear functional gives a Cauchy sequence. Stegall's theorem preserves this subsequence property under taking the absolutely convex hull, and approximation in norm preserves it on the norm closure, which contains every mixed preparation function. Integration against each probe therefore gives pointwise convergent subsequences of mixed responses, excluding an independent sequence. Conversely, pure preparations and pure probes are among the mixed preparations

and full probes; finite zero/one patterns on K_φ meet the dense actual labels, so the theorem's hypothesis is equivalent to failure of NIP for the observable.

Theorem 28.1 ([DI26c, Theorem 6.2])**.** *Let C_φ be a diagonal observable whose pure responses have the independence property: there are distinct labels $a_0, a_1, \dots$ such that every finite pattern of values $\varphi(a_n, b)$ is realized by some label b. Then the affine formula $R(y_1, x) - R(y_2, x)$ has the tree property of the second kind in the theory of $\mathcal{S}_\varphi$. In particular, a simple diagonal observable has no independence property.*

Proof. We first build finite arrays of every size, whose entries are pairs of uniform mixtures on complementary halves of blocks of labels, and then pass to an infinite array in an ultrapower. Let a positive integer N be given, and choose an integer t with $2^t > N$. We divide $N2^t$ of the labels a_n into N blocks $S_0, \dots, S_{N-1}$, identify each block with $\mathbb{F}_2^t$, choose distinct nonzero $w_0, \dots, w_{N-1} \in \mathbb{F}_2^t$, and, for $i, j < N$, let $T_{ij} = \{v \in S_i : w_j \cdot v = 1\}$, a half of the block S_i; two different halves in a block meet in a quarter of it. Let $\mu_{ij}^{\pm}$ be the uniform mixtures on T_{ij} and on $S_i \setminus T_{ij}$, respectively. For a probe ν with probabilities q_a of answering 1 at the label a, and with $m = 2^t$, we have

$$R(\mu_{ij}^+, \nu) - R(\mu_{ij}^-, \nu) = \frac{2}{m}\Big(\sum_{a \in T_{ij}} q_a - \sum_{a \in S_i \setminus T_{ij}} q_a\Big) \le 1,$$

and for $j \neq j'$ the sum of the two values in row i is $\frac{4}{m}\big(\sum_{T_{ij} \cap T_{ij'}} q_a - \sum_{S_i \setminus (T_{ij} \cup T_{ij'})} q_a\big) \le 1$, so the two values are never both greater than $1/2$. For any choice of one j in each row, the independence property provides a label b that answers 1 exactly on the chosen halves, and at b every chosen value is 1. These finite arrays exist for every N. For each fixed pair (i, j), use its entry in the array of size N when $N > \max(i, j)$ and fill the remaining finitely many coordinates arbitrarily. The resulting sequences define an infinite array in $\mathcal{S}_\varphi^{\mathcal{U}}$, and Łoś's theorem transfers the row bound and the satisfiability of every finite path. More explicitly, for a path $f : \mathbb{N} \to \mathbb{N}$ the countable conditions

$$R(\mu_{if(i)}^+, x) - R(\mu_{if(i)}^-, x) = 1 \qquad (i \in \mathbb{N})$$

are finitely satisfiable in the ultrapower. Its $\aleph_1$-saturation therefore realizes the entire path. For the final assertion, failure of NIP for the mixed responses gives the independent sequence of actual labels by the transfer argument above. The construction then gives TP_2, so the observable is not simple. □

The theorem is the counterpart, for independent preparations, of the theorem of Ben Yaacov that the randomization of a theory with the independence property has the tree property of the second kind [BY13]; the randomization is Keisler's real-valued theory of random elements $\xi : \Omega \to M$ of a model M of the theory, defined on an atomless probability space $(\Omega, \mathbb{P})$, with predicates $\mathbb{P}[\theta(\xi_1, \dots, \xi_n)]$ for the formulas θ of the theory. The theorem also mirrors Theorem 19.1, by which a separable convex Rosenthal compactum that is hereditarily separable is metrizable: in both cases, a class of the classical theory disappears because states can be mixed. The tree of the proof lives in an elementary extension, among idealized preparations. Whether actual preparations already carry it separates the commutation observable from the Rado observable, as we show in the next variation.

For the physicist or operator theorist. The structure $\mathcal{S}_\varphi$ has the mixtures of labels as Alice's sort, and Theorem 28.1 shows that this convex sort carries the tree property of the second kind whenever the labels have the independence property: the parameters of the finite arrays are pairs of actual mixtures, and Łoś's theorem with $\aleph_1$-saturation turns them into an infinite array in the ultrapower $\mathcal{S}_\varphi^{\mathcal{U}}$. Since a formula with the tree property of the second kind has the tree property, the real-valued theory of $\mathcal{S}_\varphi$ is not simple. The theorem parallels the result of Ben Yaacov on randomizations [BY13]: in both, a structure is replaced by probability measures on it, and the independence property of the original relation produces arrays of averages whose rows are inconsistent.

For the logician. A diagonal observable reads classical data stored in two quantum registers: Alice's register holds a label a as the basis vector $|a\rangle$, Bob's holds b as $|b\rangle$, and the measurement returns $\varphi(a,b)$; on a density operator only the probabilities $\langle \rho a, a\rangle$ of the labels matter, so Alice's preparations are, in effect, random labels. For the commutation observable, a label of Alice is read as a check, a Pauli string that a code measures, and a label of Bob as an error, and the outcome of the measurement, 1 when the two anticommute and 0 otherwise, is the syndrome bit of the error at that check. The proof of Theorem 28.1 uses only uniform mixtures over halves of blocks of labels, which a laboratory prepares by choosing a label of the half uniformly at random, and the tree property of the second kind is a statement about the probabilities with which such random checks detect a fixed error.

In Part I mixing stood in for idealization; in Part II it supplied the midpoints of split pairs, the pairs of idealized responses that differ at a single point; here it builds a tree.

Variatio 29. A 1 Ô VERO 2 CLAV.

Actual and idealized trees

In Theorem 28.1 we showed that, when the pure responses of a diagonal observable have the independence property, the mixtures of Alice's preparations carry the tree property of the second kind; the tree of its proof, an array of mixtures, was built in an elementary extension, among idealized preparations. In this variation we show that for the commutation observable, the diagonal observable of the commutation form of the Pauli group, the array parameters can be actual mixtures, and that for the Rado observable, the diagonal observable of the edge relation of the Rado graph, no such array with actual preparation parameters witnesses a tree property for a fixed quantifier-free formula in one probe variable. For the commutation observable, finite path fragments have finite-support error realizations, while entire paths are realized by pure probes already in the original compact probe space, with error labels that may have infinite support.

Theorem 29.1 ([DI26c, Theorems 7.2, 8.1 and 8.3]).

(a) *For the commutation observable, the diagonal observable of the commutation form ω on the space V of labels of Pauli strings (see Variatio 28), actual preparations witness the tree property of the second kind. Let $(e_{im})_{i,m\in\mathbb{N}}$ be linearly independent labels in V, and, for $i,j\in\mathbb{N}$, let μ_{ij} be the uniform mixture of Alice's preparations on the coset $e_{ij}+\operatorname{span}(e_{im} : m<j)$. Then, for every label $b\in V$ of Bob, the response $R(\mu_{ij},b)$, the probability that the observable reads 1, equals 1 exactly when e_{ij} is the first of the labels $e_{i0}, e_{i1}, \dots$ at which $\omega(\cdot,b)$ takes the value 1; for every choice of one entry in each of finitely many rows of the array $(\mu_{ij})_{i,j\in\mathbb{N}}$, there is a single label of Bob to which every chosen entry has response 1; and, for $j\neq j'$, no probe yields responses greater than $2/3$ to both μ_{ij} and $\mu_{ij'}$.*

(b) *For the Rado observable, the diagonal observable of the edge relation of the Rado graph, actual preparation parameters do not witness a tree property for a fixed quantifier-free formula in one probe variable: for every $n \geq 1$ and every continuous connective $u\colon [0,1]^n \to \mathbb{R}$, no infinite sequence of instances of the formula*

$$u\big(R(y_1, x), \ldots, R(y_n, x)\big),$$

with actual mixtures as parameters, is uniformly k-inconsistent, for any integer $k \geq 2$ (uniform k-inconsistency requires fixed reals $\alpha < \beta$ such that every instance takes a value at least β somewhere, but no probe gives values greater than α to any k distinct instances).

The idea of (a) is that a single mixture over a coset tests many checks at once. In the language of error correction, the labels e_{im} are checks and Bob's label b is an error. If the error commutes with the checks e_{im}, $m < j$, then no string of the subgroup $\operatorname{span}(e_{im} : m < j)$ anticommutes with it, and either every string of the coset $e_{ij} + \operatorname{span}(e_{im} : m < j)$ does or none does, according as the error anticommutes with e_{ij} or not; if it anticommutes with one of the checks e_{im}, $m < j$, then exactly half of the strings of the subgroup, and exactly half of those of the coset, anticommute with it. For $j < j'$, the pair of responses to μ_{ij} and $\mu_{ij'}$ at a pure probe is one of

$$(1, \tfrac{1}{2}), \quad (\tfrac{1}{2}, \tfrac{1}{2}), \quad (0, \tfrac{1}{2}), \quad (0, 1), \quad (0, 0).$$

These cases correspond, respectively, to the first anticommuting check occurring at j, before j, between j and j', at j', or not yet occurring. At a mixed probe the pair (x, y) lies in their convex hull, on which $x/2 + y \leq 1$; if both coordinates exceeded $2/3$, this inequality would fail. The bound is sharp: mixing a pure probe with pair $(1, \frac{1}{2})$ with weight $2/3$ and one with pair $(0, 1)$ with weight $1/3$ gives $(2/3, 2/3)$. No error has two first anticommuting checks in one row, and finitely many conditions on different rows, which concern disjoint families of linearly independent checks, can be satisfied together. For an entire path, the prescribed values extend to a linear functional on V, so the corresponding pure probe in V^* realizes it; this need not be an error of finite support. On pure probes, the difference between the responses to the uniform mixtures on the coset and on the subgroup is the indicator of a syndrome sector, the set of errors whose syndrome on the checks $e_{i0}, \ldots, e_{ij}$ is $(0, \ldots, 0, 1)$. This identity is the counterpart, for responses, of a familiar formula of error correction: for a stabilizer group $\mathcal{G}$ and a character χ of $\mathcal{G}$, the projector onto the joint eigenspace of the elements g of $\mathcal{G}$ with eigenvalues $\chi(g)$ is the average $|\mathcal{G}|^{-1} \sum_{g \in \mathcal{G}} \chi(g)\, g$. The identity, however, requires the checks only to be linearly independent, not to commute. The idea of (b) is that the responses of the Rado observable depend only on the marginals of a probe, the probabilities q_a with which it answers 1 at the vertices a, and every vector of marginals is attained. Given an infinite sequence of conditions on marginals, a gliding hump argument, familiar from Banach space theory, yields a subsequence whose parameters put their weight, up to small errors, on a common finite set of vertices and on pairwise disjoint blocks, and one probe built block by block nearly satisfies any k of these conditions.

For the physicist or operator theorist. A configuration such as an array witnessing the tree property of the second kind may exist in the structure $\mathcal{S}_\varphi$ itself, whose sorts are the mixtures of Alice's labels and the probes of Bob, or only in an elementary extension, such as the ultrapower $\mathcal{S}_\varphi^{\mathcal{U}}$ along a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, whose new elements in Alice's sort are idealized preparations; in Theorem 29.1 actual parameters suffice for the

commutation observable, whereas for the Rado observable they do not suffice for a fixed quantifier-free formula in one probe variable. An arbitrary-support error already belongs to the original pure-probe space; it is not thereby a finite-support label of the original symplectic structure. For the commutation observable, Bob's idealized labels form the compact group V^* of linear functionals on the space V of labels of Pauli strings (see Variatio 28), each label a of Alice defines the character $f \mapsto (-1)^{f(a)}$ of V^*, and averages over cosets recover, by Fourier inversion on V^*, indicators of cylinder sets, the sets of functionals with prescribed values at finitely many labels, which build the array from actual mixtures. For the Rado observable the space of Bob's idealized labels is the whole Cantor cube of functions from the vertices of the Rado graph to $\{0,1\}$, responses depend only on the marginals of a probe, and, for each fixed quantifier-free formula in one probe variable, no infinite sequence of its instances with actual preparation parameters is uniformly k-inconsistent, for any integer $k \geq 2$.

For the logician. Alice's preparation μ_{ij} is the random check obtained by multiplying e_{ij} by a uniformly random product of the earlier checks e_{im}, $m < j$, of row i; against a fixed error, a label of Bob, it answers 1 with certainty if e_{ij} is the first of the checks $e_{i0}, \dots, e_{ij}$ that anticommutes with the error, with probability $\frac{1}{2}$ if an earlier check anticommutes with it, and with probability 0 otherwise. The uniform mixture on the subgroup $\operatorname{span}(e_{im} : m < j)$ answers 1 with probability $\frac{1}{2}$ if an earlier check anticommutes with the error and with probability 0 otherwise; consequently, the difference between the response to μ_{ij} and the response to this mixture is the indicator of a syndrome sector, the set of errors whose first anticommuting check in row i is e_{ij}. For the Rado observable, the fixed quantifier-free formulas of part (b) admit no witnessing array all of whose preparation parameters are actual mixtures. The construction of Theorem 28.1 supplies such a witness in an elementary extension by taking limits of finite uniform block mixtures.

The distinction between actual and idealized configurations, which we drew for ladders in Part I, thus reappears for trees.

Variatio 30. a 1 Clav. Quodlibet

The commutation relation

A single relation of quantum mechanics runs through the three parts: a momentum kick and a translation commute up to a phase. Read with bounded kicks, with unbounded kicks and with binary kicks, it yields an example for each part (Proposition 30.1 below).

For a particle on a line, with wave functions $\psi \in L^2(\mathbb{R})$ of the position q, let T_x be the translation by a real number x, given by $(T_x\psi)(q) = \psi(q - x)$, and let K_k be the momentum kick by a real number k, given by $(K_k\psi)(q) = e^{ikq}\psi(q)$. With the position operator Q, multiplication by q, and the momentum operator $P = -i\, d/dq$, in units in which the reduced Planck constant $\hbar$ equals 1, one has $T_x = e^{-ixP}$ and $K_k = e^{ikQ}$, and $K_k^* P K_k = P + k$: the kick shifts momentum by k. The two families satisfy

$$K_k T_x = e^{ikx}\, T_x K_k,$$

Weyl's commutation relation, the bounded form of the canonical commutation relation $QP - PQ = iI$; by the Stone–von Neumann theorem, it determines (Q, P) up to unitary equivalence for irreducible families. On a qubit, the bit flip X, which exchanges the basis vectors $|0\rangle$ and $|1\rangle$, is a translation, the phase flip Z, which multiplies $|s\rangle$ by $(-1)^s$ for $s \in \{0,1\}$, is a kick, and the two anticommute, $ZX = -XZ$: the Pauli group, generated by these translations and kicks on each qubit, is the Weyl–Heisenberg group over the two-element field, with the phase $(-1)^{\omega(v,w)}$ of the commutation relation of two Pauli strings with labels v and w (see

Variatio 22) in place of e^{ikx}. The phase e^{ikx} also determines the intensity that Bob observes when Alice's photon, prepared with wavenumber k, reaches his detector at position x: the interference fringe of Variatio 16 is $\frac{1}{2}(1 + \operatorname{Re} e^{ikx})$.

Proposition 30.1. *The commutation phase e^{ikx}, read as a bipartite observable in which Alice chooses a kick k and Bob a translation x, the position of his detector, yields three observables according to the range of its parameters, and they lie in three different classes.*

(a) Bounded parameters: stable. *If Alice's wavenumbers k_n, for $n \in \mathbb{N}$, lie in a bounded set, then the fringe observable $C = \sum_n |e_n\rangle\langle e_n| \otimes M_{f_n}$ on $\ell^2(\mathbb{N}) \otimes L^2[0, 2\pi]$ (see Variatio 14), where $(e_n)_{n\in\mathbb{N}}$ is the standard basis of $\ell^2(\mathbb{N})$ and M_{f_n} is the operator of multiplication by $f_n(x) = \frac{1}{2}(1 + \cos k_n x)$, is stable.*

(b) Lacunary parameters: not NIP. *If $k_n = 8^n$, the same observable is not NIP, and its idealized responses contain a copy of $\beta\mathbb{N}$.*

(c) Binary parameters: simple, and not simple once mixed. *Over the two-element field, on countably many qubits, the commutation relation becomes the commutation form of the structure (V, ω) of labels of Pauli strings, whose theory is simple and unstable; the observable that reads it, the commutation observable, has the tree property of the second kind, witnessed by actual mixtures of Alice's preparations.*

Proof. (a) Because C is diagonal in Alice's factor, the off-diagonal parts of her states do not contribute, and the effective observable of a state ρ is multiplication by $\sum_n \langle \rho e_n, e_n\rangle f_n$. The functions f_n are uniformly bounded and, since $|f_n'(x)| \le \frac{1}{2}|k_n|$, equicontinuous; by the Arzelà–Ascoli theorem they form a relatively compact set in $C[0, 2\pi]$, and so does its closed convex hull. A norm-compact set of effective observables is weakly compact, and the observable is stable, by Theorem 5.1(A2). (b) This is [DI26b, Proposition 7.13]. (c) This is Theorem 22.1 together with Theorem 29.1(a). □

The proposition concerns three restrictions of one relation, not one observable in three classes: the bounded and the unbounded fringes are two observables, and the qubit version is a third. With bounded parameters, the relation satisfies Grothendieck's double limit criterion; with the wavenumbers 8^n it is not NIP. Over the two-element field it is the arithmetic of error correction, simple in itself and not simple once preparations are mixed.

For the physicist or operator theorist. The three regimes give three formulas. With bounded wavenumbers, the responses f_n form a relatively norm-compact set of continuous functions, so every idealized response is a continuous function of the detector position, and every type of the formula φ of the observable realized in an ultrapower of its structure is definable, as in Theorem 3.1. With the wavenumbers 8^n the formula has the independence property, since every finite pattern is realized by the responses at a suitable single position, and the φ-types realized in ultrapowers contain a copy of $\beta\mathbb{N}$ (see Theorem 11.1). Over $\mathbb{F}_2$, the relation is the alternating form of (V, ω), whose theory is a standard example of a simple unstable theory, and the mixtures of Alice's preparations carry the tree property of the second kind already in the structure itself (see Theorem 29.1).

For the logician. A translation T_x moves a particle by x and a kick K_k adds k to its momentum; applied in the two orders, they produce the same state up to the factor e^{ikx}, which is invisible for a single state but observable in interference, where two amplitudes are added. In the fringe observable this phase is measured directly: a photon of wavenumber k is detected at the position x with relative intensity $\frac{1}{2}(1 + \cos kx)$, and Bob's choice of x is the

choice of a translation. On a qubit the bit flip X is the translation of $\{0,1\}$ by 1 and the phase flip Z is a kick, so the relation $ZX = -XZ$ is the same relation with phases ± 1; on n qubits it becomes the commutation relation ω of Pauli strings, which a stabilizer measurement reads. The proposition follows one physical relation through the three dividing lines, according to whether the parameters range over a bounded set of reals, over the lacunary set $\{8^n\}$, or over the two-element field.

Aria da capo e fine

Physics lives on idealized limits. We modeled idealized preparations by limiting states of actual preparations, and represented their sequences by elements of ultrapowers. For a single system, every bounded expectation passed to the limit; in the variations, the question was what happened when two systems were combined.

Difficulties begin with two systems, and there they stayed. A simple example, the staircase, was decisive from the start: whichever of the two oscillators, Alice's and Bob's, was driven first to infinite energy outran the other, and whether Bob's had the higher energy depended on which was idealized first. The comparison of the two energies is therefore unstable, and in Part II it proved to be NIP.

The agreement of the two orders had a finite approximation consequence. Where the orders agreed, every state of Alice's algebra was simulated in its responses by finite mixtures of actual preparations, uniformly over Bob's probes and with his system idealized first. Their common values gave a symmetric independent product on the space of stable observables; at each such observable, its expectation was the unique separately weak* continuous extension from pairs of actual states. For uniformly stable observables, the number of transcripts of adaptive expectation queries was bounded by a polynomial at each positive margin, and their labels could be predicted in a fixed-threshold game with a finite mistake bound uniform in the threshold. Where uniform stability failed, the number of transcripts grew exponentially at some margin, even though qualitative stability could remain.

The theme of this paper is Grothendieck's theorem, which says exactly when the two orders agree, and it lay beneath the variations as their bass. In Part I it was heard plainly. In Part II it was broken, and, for the observables that are NIP, what remained of it was sequential: every idealized response was still a limit of a sequence of responses of actual preparations. In Part III it fell silent.

A logician recognizes the staircase at once: it is an instance of the order property, and in the variations the three dividing lines that Shelah drew became three distinctions among measurements. The first, stability, is drawn by Grothendieck's criterion itself; the second, NIP, separates the failures of the criterion in which idealized responses remain such limits from those in which they do not; and the third, simplicity, is drawn not by limits but by the amalgamation of independent types, through the Kim–Pillay theorem. Mixing, the operation that distinguishes quantum observables from arbitrary formulas, did more than simulate idealized preparations: for the observables that are NIP, it made hereditary separability of the idealized responses imply their metrizability, and, for the observables that, like the comparison of energies, read a relation between the vectors of two orthonormal bases, it made simplicity imply NIP.

In Part I we studied the observables that satisfy Grothendieck's criterion, in Part II the observables that are NIP, and in Part III the amalgamation of independent types, which in simple theories takes the place of the criterion.

Questions

Question 1. In classical model theory, a simple theory without the independence property is stable. Is a simple observable stable? By Theorem 28.1, a simple diagonal observable has no independence property, so the question is whether the unstable NIP observables of Part II, such as the energy comparison of the Aria, acquire a tree property once preparations are mixed.

Question 2. Which diagonal observables with the independence property have the tree property of the second kind witnessed by actual preparations? Theorem 29.1 suggests a criterion: whether the closed linear span of the responses of Alice's labels, as functions on the space K_φ of Bob's idealized labels (see Variatio 28), contains, with bounded norm, the indicators of an infinite independent sequence of nested cylinders, that is, of sets of idealized labels with prescribed values at finitely many labels of Alice.

Question 3. The causal order of a $(1+1)$-dimensional spacetime is NIP but not NIP_1: its idealized responses form a compactum that is not first countable (see Theorem 16.1). In which class of Todorčević's trichotomy does the causal order of a spacetime of higher dimension lie?

Question 4. The minimal family D_6 of the Argyros–Dodos–Kanellopoulos heptachotomy is realized in Part II by a spin chain read both site by site and against templates, and the family D_7 by the comparison of positions of a particle in a box (see Theorem 15.1). Does the comparison of positions also realize D_6?

Question 5. For qudits of prime dimension p, quantum systems with Hilbert space $\mathbb{C}^p$, the generalized Pauli strings, built from the shift $|s\rangle \mapsto |s+1\rangle$ and the clock $|s\rangle \mapsto e^{2\pi i s/p}|s\rangle$, for s modulo p, in place of X and Z, are labeled by a symplectic space over the field with p elements, and the dictionary of Variatio 23 carries over. For continuous variables, the commutation phase of a translation by x and a momentum kick by k is e^{ikx}, whose real part determines the interference fringe of Part II; for unbounded wavenumbers the fringe is not NIP. Where does the unbounded fringe lie with respect to simplicity?

Historical remarks

Shelah introduced stable theories in 1969 [She69], and the order property and the independence property of formulas in 1971 [She71]; simple theories followed in 1980, in a paper in which he named as his motivation the theory T_{ind} of the random graph [She80]. The double limit condition also has an analytic history: it is Grothendieck's criterion of 1952 for weak compactness [Gro52], it is closely related to the regularity of bilinear forms studied by Arens in 1951 [Are51], and it defines the stable Banach spaces introduced by Krivine and Maurey in 1981 [KM81]; Iovino connected it with Shelah's stability and developed the real-valued theory, including the stability spectrum for real-valued structures [Iov99a, Iov99b]. Rosenthal characterized the Banach spaces containing ℓ^1 in 1974 [Ros74], Bourgain, Fremlin and Talagrand studied pointwise compact sets of Baire class one in 1978 [BFT78], and Todorčević proved his trichotomy in 1999 [Tod99]; Argyros, Dodos and Kanellopoulos extracted the heptachotomy from the methods of his paper in 2008 [ADK08]. Theorem 19.1 is a special case of a known result: as Todorčević pointed out to us, hereditarily separable locally connected Rosenthal compacta are metrizable, by his Theorems 5 and 7 [Tod99] and by Lemma 2.4 of Gruenhage [Gru88]. Kim proved the symmetry of forking in simple theories in 1998 [Kim98], and Kim and Pillay the independence theorem and the characterization of simplicity in 1997 [KP97]; Ben Yaacov extended simplicity to continuous logic [BY03]. The random graph was described by Ackermann in 1937 [Ack37] and by Rado in 1964 [Rad64], and Erdős and Rényi proved in 1963 that it is the almost sure countable random graph [ER63]. Fraïssé introduced his limits

in 1954 [Fra54], and Ben Yaacov extended them to metric structures [BY15]. Smoothly approximable structures are due to Cherlin and Hrushovski [CH03]. The symplectic description of the Pauli group and of stabilizer codes is due to Gottesman [Got97] and to Calderbank, Rains, Shor and Sloane [CRSS98], and the five-qubit code to Laflamme, Miquel, Paz and Zurek [LMPZ96] and to Bennett, DiVincenzo, Smolin and Wootters [BDSW96]. Keisler introduced randomizations in 1999 [Kei99], and Ben Yaacov proved in 2013 that randomization destroys simplicity in the presence of the independence property [BY13]. Lee and Yang proved their theorem for the Ising ferromagnet in 1952 [LY52]. Asano [Asa70] and Suzuki and Fisher [SF71] extended it to quantum spin systems, the anisotropic Heisenberg ferromagnets whose coupling along the field dominates the transverse couplings; with a transverse field Γ, these results, in the form given by Fröhlich and Rodríguez [FR12, Theorem 5], exclude zeros of the partition function, as a function of the complex field h at inverse temperature β, only for $\operatorname{Re} h > \beta\Gamma$. The approximation of a quantum spin system by a classical Ising system in one more dimension, used in Variatio 18 to reach the whole half-plane $\operatorname{Re} h > 0$, is due to Suzuki [Suz76]. Hong, Ou and Mandel observed their two-photon interference in 1987 [HOM87]. Model theory has met operator algebras and quantum theory along other routes: Farah, Hart and Sherman studied stability in the model theory of operator algebras [FHS13], where it is a property of the theory of an algebra rather than of a single observable; Zilber gave a model-theoretic semantics of the canonical commutation relation [Zil16]; Ben Yaacov showed that randomization preserves NIP [BY09], a result of which the uniform NIP of comparisons of physical quantities is an elementary quantitative instance; and Hrushovski and Pillay studied invariant measures in NIP theories [HP11].

Acknowledgments

We thank Stevo Todorčević for pointing out to us that Theorem 19.1 is a special case of a known result, and for the references that establish it.

The form of this paper is that of J. S. Bach's Goldberg Variations, published in 1741: an aria, thirty variations in which every third variation up to the twenty-seventh is a canon, at intervals rising from the unison to the ninth, a quodlibet as the thirtieth, and the aria again. The section titles follow J. S. Bach's markings. Figure 10 measures this survey against the music: the share of each variation in six performances of the Goldberg Variations, four on the piano and two on the harpsichord, and its share of the words here.

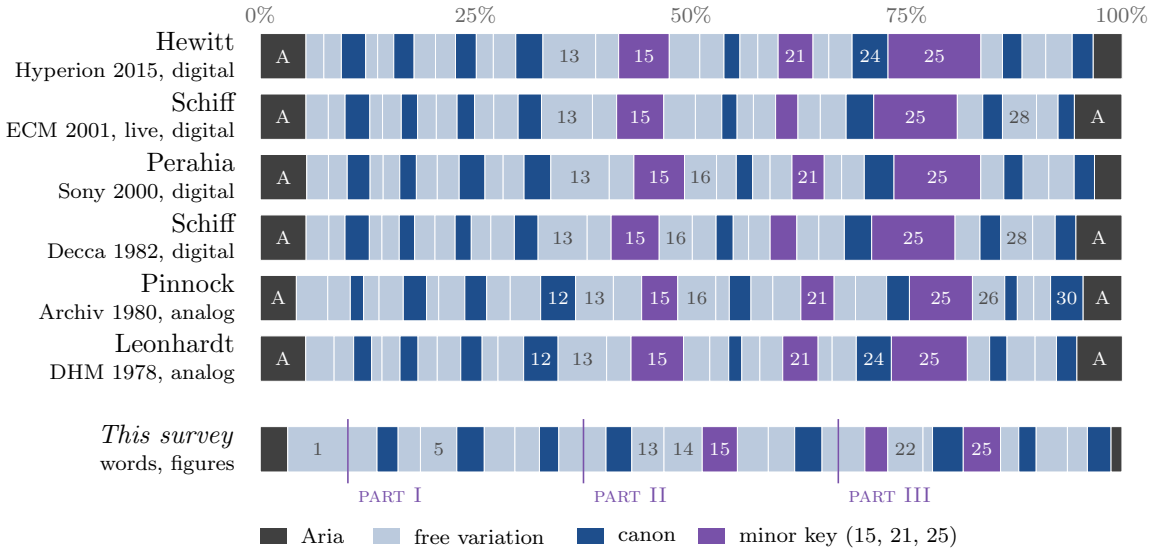


FIGURE 10. The proportions of the Goldberg Variations in performances by Angela Hewitt (piano; Hyperion, 2015), András Schiff (piano; ECM, 2001, live), Murray Perahia (piano; Sony, 2000), András Schiff (piano; Decca, 1982), Trevor Pinnock (harpsichord; Archiv, 1980) and Gustav Leonhardt (harpsichord; Deutsche Harmonia Mundi, 1978), most recent first; the harpsichord recordings are analog and the piano recordings digital, Schiff's of 1982 among the early digital ones. The last bar gives the proportions of this survey by word count, each figure counting as a fixed number of words. Each bar has the same width; the pieces are the Aria (A), Variations 1–30 and the Aria da capo.

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